

fmocdmac — FM's OCD L^AT_EX Macro*

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Abstract

This package contains almost all the latex macros I heavily use in my tcs research activity and, in particular, in the writing of conference and journal articles. As few of my co-authors have kindly pointed out, and probably many have thought, they are somehow a clear expression of an underlying ocd-like behavior... hence the name!

1 Implementation & Usage

1 `\package`

Required external packages:

```
2
3 \RequirePackage{etoolbox}
4
5 \RequirePackage{xargs}
6 \RequirePackage{xspace}
7 \RequirePackage{stringstrings}
8
```

Package options:

```
9
10 %% Auxiliary packages
11 \newif\ifaux@ \aux@false
12 \DeclareOption{aux}{\aux@true}
13 \DeclareOption{noaux}{\aux@false}
14
15 %% AMS defaults
16 \newif\ifamsdef@ \amsdef@true
17 \DeclareOption{noamsdef}{\amsdef@false}
18
19 %% AMS theorem tools
20 \newif\ifamsthm@ \amsthm@true
21 \DeclareOption{noamsthm}{\amsthm@false}
22
23 %% Extended theorem tools
24 \newif\ifthmtls@ \thmtls@true
25 \DeclareOption{nothmtls}{\thmtls@false}
26
27 %% Enumeration tools
28 \newif\ifenmtls@ \enmtls@true
29 \DeclareOption{noenmtls}{\enmtls@false}
30
31 %% Hyper reference
32 \newif\ifhympref@ \hympref@true
33 \DeclareOption{nohympref}{\hympref@false}
34
```

*This document describes version v0.32 of the fmocdmac package, last revised 2026/05/08.

```

35 %% Font tools
36 \newif\iffnttts@ \fnttts@true
37 \DeclareOption{nofnttts}{\fnttts@false}
38
39 %% Knowledge tools
40 \newif\ifkwtls@ \kwtls@true
41 \DeclareOption{nokwtls}{\kwtls@false}
42
43 %% Camera-ready version
44 \newif\ifcrv@ \crv@false
45 \DeclareOption{crv}{\crv@true}
46
47 %% Change bars
48 \newif\ifchgbar@ \chgbar@false
49 \DeclareOption{chgbar}{\chgbar@true}
50
51 %% Line numbers
52 \newif\iflinnum@ \linnum@false
53 \DeclareOption{linnum}{\linnum@true}
54
55
56 %% Text macro generation
57 \newif\iftxtgen@ \txtgen@false
58 \DeclareOption{txtgen}{\txtgen@true}
59 \DeclareOption{notxtgen}
60   {\txtgen@false\txt@false\com@false\gam@false\log@false\aut@false}
61
62 %% Math macro generation
63 \newif\ifmthgen@ \mthgen@false
64 \DeclareOption{mthgen}{\mthgen@true}
65 \DeclareOption{nomthgen}
66   {\mthgen@false\mth@false\gam@false\log@false\aut@false}
67
68
69 %% Elementary macros for text
70 \newif\iftxt@ \txt@false
71 \DeclareOption{txt}{\txt@true\txtgen@true}
72 \DeclareOption{notxt}{\txt@false}
73
74 %% Elementary macros for math
75 \newif\ifmth@ \mth@false
76 \DeclareOption{mth}{\mth@true\mthgen@true}
77 \DeclareOption{nomth}{\mth@false}
78
79
80 %% Macros for computational-complexity classes
81 \newif\ifcom@ \com@false
82 \DeclareOption{com}{\com@true\txtgen@true}
83 \DeclareOption{nocom}{\com@false}
84
85
86 %% Macros for graphs
87 \newif\ifgrp@ \grp@false
88 \DeclareOption{grp}{\grp@true\txtgen@true\mthgen@true}
89 \DeclareOption{nogrp}{\grp@false}
90
91 %% Macros for games
92 \newif\ifgam@ \gam@false
93 \DeclareOption{gam}{\gam@true\txtgen@true\mthgen@true}
94 \DeclareOption{nogam}{\gam@false}
95
96 %% Macros for logics
97 \newif\iflog@ \log@false

```

```

98 \DeclareOption{log}{\log@true\txtgen@true\mthgen@true}
99 \DeclareOption{nolog}{\log@false}
100
101 %% Macros for automata
102 \newif\ifaut@ \aut@false
103 \DeclareOption{aut}{\aut@true\txtgen@true\mthgen@true}
104 \DeclareOption{noaut}{\aut@false}
105
106
107 %% Format-related tricks
108 \newif\iffrm@ \frm@false
109 \DeclareOption{frm}{\frm@true}
110 \DeclareOption{nofrm}{\frm@false}
111
112
113 %% Figure-related tricks
114 \newif\iffig@ \fig@false
115 \DeclareOption{fig}{\fig@true}
116 \DeclareOption{nofig}{\fig@false}
117
118 %% Wrapfig package
119 \newif\ifwrpfig@ \wrpfig@true
120 \DeclareOption{nowrpfig}{\wrpfig@false}
121
122
123 %% Table-related tricks
124 \newif\iftab@ \tab@false
125 \DeclareOption{tab}{\tab@true}
126 \DeclareOption{notab}{\tab@false}
127
128
129 %% Algorithm-related tricks
130 \newif\ifalg@ \alg@false
131 \DeclareOption{alg}{\alg@true}
132 \DeclareOption{noalg}{\alg@false}
133

```

Option-processing code:

```

134
135 \DeclareOption*{\PackageWarning{fmocdmac}{Unknown~'\CurrentOption'}}%
136
137 \ExecuteOptions{aux,txtgen,mthgen,txt,mth,com,grp,gam,log,aut}%
138
139 \ProcessOptions\relax%
140
141 \ifcsdef{if@twocolumn}{\newif\if@twocolumn}
142

```

Package main body:

```

143
144 %%*****
145 %%** Auxiliary Alphabet Letters *****
146 %%*****

```

\omicron Auxiliary Greek lowercase letter: ... to do!

```

147 \csdef{omicron}{o}

```

\Alpha Auxiliary Greek uppercase letters: ... to do!

```

... 148 \csdef{Alpha}{A} \csdef{Beta}{B} \csdef{Epsilon}{E} \csdef{varEpsilon}{E}
149 \csdef{Zeta}{Z} \csdef{Eta}{H} \csdef{Iota}{I} \csdef{Kappa}{K}
150 \csdef{varKappa}{K} \csdef{Mu}{M} \csdef{Nu}{N} \csdef{Omicron}{O}
151 \csdef{Rho}{P} \csdef{varRho}{P} \csdef{Tau}{T} \csdef{Chi}{X}
152 %%*****

```

```

153 %*****%
154 %** Tools *****%
155 %*****%

```

\empchk Emptiness check: `\empchk{<A>}{<B>}` evaluates to the empty string, if Argument $\langle A \rangle$ is empty, and to Argument $\langle B \rangle$, otherwise.

- `\empchk{}{B}` = “”
- `\empchk{A}{B}` = “B”

```

156 \newrobustcmd{\empchk}[2]
157   {\if&#1&\else#2\fi}

```

\defval Default value: `\defval{<A>}{<B>}` evaluates to Argument $\langle B \rangle$, if Argument $\langle A \rangle$ is empty, and to Argument $\langle A \rangle$ itself, otherwise.

- `\defval{}{B}` = “B”
- `\defval{A}{B}` = “A”

```

158 \newrobustcmd{\defval}[2]
159   {\if&#1&#2\else#1\fi}

```

```

160 %*****%

```

\arglef Left extension: `\arglef{<A>}{<B>}` evaluates to the concatenation $\langle AB \rangle$ of the two arguments, if Argument $\langle B \rangle$ is non-empty, and to the empty string, otherwise.

- `\arglef{A}{}{}` = “”
- `\arglef{A}{B}` = “AB”

```

161 \newrobustcmd{\arglef}[2]
162   {\empchk{#2}{#1#2}}

```

\argrig Right extension: `\argrig{<A>}{<B>}` evaluates to the concatenation $\langle AB \rangle$ of the two arguments, if Argument $\langle A \rangle$ is non-empty, and to the empty string, otherwise.

- `\argrig{}{B}` = “”
- `\argrig{A}{B}` = “AB”

```

163 \newrobustcmd{\argrig}[2]
164   {\empchk{#1}{#1#2}}

```

\argmid Middle extension: `\argmid{<A>}{<B>}{<C>}` evaluates to the concatenation $\langle ABC \rangle$ of the three arguments, if Argument $\langle B \rangle$ is non-empty, and to the empty string, otherwise.

- `\argmid{A}{}{C}` = “”
- `\argmid{A}{B}{C}` = “ABC”

```

165 \newrobustcmd{\argmid}[3]
166   {\empchk{#2}{#1#2#3}}

```

\argsep Separators: `\argsep{<A>}{<B>}{<C>}` evaluates to Argument $\langle C \rangle$, if Argument $\langle A \rangle$ is empty, to Argument $\langle A \rangle$, if Argument $\langle C \rangle$ is empty, and to the concatenation $\langle ABC \rangle$, otherwise.

- `\argsep{}{B}{C}` = “C”
- `\argsep{A}{B}{}{}` = “A”
- `\argsep{A}{}{C}` = “AC”
- `\argsep{A}{B}{C}` = “ABC”

```

167 \newrobustcmd{\argsep}[3]
168   {\if&#1&#3\else#1\arglef{#2}{#3}\fi}

```

```

169 %*****%

```

\ifstarvar Command star variants: `\ifstarvar{<A>}{<B>}` ... to do!

```

170 \newrobustcmd{\ifstarvar}
171   {\@ifstar}

```

```

\ifexclavar Command exclamation variants: \ifexclavar{<A>}{<B>} ... to do!
172 \newrobustcmd{\ifexclavar}[1]
173   {\@ifnextchar!\{\@firstoftwo{#1}}}}

\ifquestvar Command questionmark variants: \ifquestvar{<A>}{<B>} ... to do!
174 \newrobustcmd{\ifquestvar}[1]
175   {\@ifnextchar?\{\@firstoftwo{#1}}}}

176 %%*****%

\varcmd Variadic commands: \varcmd{<A>}{<B>}{<C>}{<D>}{<E>}{<F>} ... to do!
177 \newrobustcmd{\varcmd}[6]
178   {\expandafter\newcommand\csname gobble#1arg\endcsname[2]
179     {\csname check#1arg\endcsname%
180       {\argsep{##1}{#4\allowbreak}{\empchk{##2}{##2}}}}}%
181   \expandafter\newcommand\csname check#1arg\endcsname[1]
182     {\csname @ifnextchar\endcsname%
183       \bgroup{\csname gobble#1arg\endcsname{##1}{#2{##1#5}#6}}}%
184   \expandafter\newcommand\csname#1\endcsname[1]
185     {\csname check#1arg\endcsname{#3#1}}}}

186 %%*****%

\seqoftag Sequence of tags: \seqoftag{<A>}{<B>}{<C>} ... to do!
187 \newrobustcmd{\seqoftag}[3]
188   {\@for\itr:={#1}\do%
189     {\expandafter\csedef{\itr#2}%
190       {\noexpand\csname #3\endcsname{\itr}}}%
191     \ifknwts@
192       \expandafter\csedef{in\itr#2}%
193         {\noexpand\csname #3\endcsname{\intro{\itr}}}%
194       \expandafter\csedef{kl\itr#2}%
195         {\noexpand\csname #3\endcsname{\kl{\itr}}}%
196     \fi}}

\seqofcmd Sequence of commands: \seqofcmd{<A>}{<B>}{<C>} ... to do!
197 \newrobustcmd{\seqofcmd}[3]
198   {\@for\itr:={#1}\do%
199     {\expandafter\csedef{\itr#2}%
200       {\noexpand\csname #3\endcsname{\csname \itr\endcsname}}}%
201     \ifknwts@
202       \expandafter\csedef{in\itr#2}%
203         {\noexpand\csname #3\endcsname{\intro{\csname \itr\endcsname}}}%
204       \expandafter\csedef{kl\itr#2}%
205         {\noexpand\csname #3\endcsname{\kl{\csname \itr\endcsname}}}%
206     \fi}}

207 %%*****%

\seqoflatlow Sequence of Latin lowercase letters: \seqoflatlow{<A>}{<B>} ... to do!
208 \newrobustcmd{\seqoflatlow}
209   {\seqoftag{a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z}}

\seqoflatupp Sequence of Latin uppercase letters: \seqoflatupp{<A>}{<B>} ... to do!
210 \newrobustcmd{\seqoflatupp}
211   {\seqoftag{A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z}}

\seqoflatlet Sequence of Latin letters: \seqoflatlet{<A>}{<B>} ... to do!
212 \newrobustcmd{\seqoflatlet}[2]
213   {\seqoflatlow{#1}{#2}\seqoflatupp{#1}{#2}}

214 %%*****%

```

```

\seqofgrklow Sequence of Greek lowercase letters: \seqofgrklow{⟨A⟩}{⟨B⟩} ... to do!
215 \newrobustcmd{\seqofgrklow}
216   {\seqofcmd{alpha,beta,gamma,delta,epsilon,varepsilon,zeta,eta,theta,vartheta,%
217    iota,kappa,varkappa,lambda,mu,nu,xi,omicron,pi,varpi,rho,varrho,sigma,%
218    varsigma,tau,upsilon,phi,varphi,chi,psi,omega}}

\seqofgrkupp Sequence of Greek uppercase letters: \seqofgrkupp{⟨A⟩}{⟨B⟩} ... to do!
219 \newrobustcmd{\seqofgrkupp}
220   {\seqofcmd{Alpha,Beta,Gamma,Delta,Epsilon,varEpsilon,Zeta,Eta,Theta,varTheta,%
221    Iota,Kappa,varKappa,Lambda,Mu,Nu,Xi,Omicron,Pi,varPi,Rho,varRho,Sigma,%
222    varSigma,Tau,Upsilon,Phi,varPhi,Chi,Psi,Omega}}

\seqofgrklet Sequence of Greek letters: \seqofgrklet{⟨A⟩}{⟨B⟩} ... to do!
223 \newrobustcmd{\seqofgrklet}[2]
224   {\seqofgrklow{#1}{#2}\seqofgrkupp{#1}{#2}}

225 %%*****

\seqoflow Sequence of lowercase letters: \seqoflow{⟨A⟩}{⟨B⟩} ... to do!
226 \newrobustcmd{\seqoflow}[2]
227   {\seqoflatlow{#1}{#2}\seqofgrklow{#1}{#2}}

\seqofupp Sequence of uppercase letters: \seqofupp{⟨A⟩}{⟨B⟩} ... to do!
228 \newrobustcmd{\seqofupp}[2]
229   {\seqoflatupp{#1}{#2}\seqofgrkupp{#1}{#2}}

\seqoflet Sequence of all letters: \seqoflet{⟨A⟩}{⟨B⟩} ... to do!
230 \newrobustcmd{\seqoflet}[2]
231   {\seqoflow{#1}{#2}\seqofupp{#1}{#2}}

232 %%*****
233 %%*****
234 %%** Auxiliary Packages *****
235 %%*****
236 \ifaux@
237
238 \ifamsdef@
239   % AMS Packages
240   \RequirePackage{mathtools}
241   \RequirePackage{amssymb}
242   \RequirePackage{stmaryrd}
243   \interdisplaylinepenalty=2500
244 \fi
245
246 \ifamsthm@
247   % AMS Theorem Tools
248   \RequirePackage{amsthm}
249 \fi
250
251 \ifthmtls@
252   % Extended Theorem Tools
253   \RequirePackage{thmtools}
254   \RequirePackage{thm-restate}
255 \fi
256
257 \ifenmtls@
258   % Enumeration Tools
259   \RequirePackage{paralist}
260 \fi
261
262 \ifhympref@
263   % Hyper Reference Tools
264   \RequirePackage{hyperref}

```

```

265 \hypersetup {
266   pdfsubject      = {},
267   pdfkeywords     = {},
268   pdfproducer     = {},
269   pdfcreator      = {},
270   pdfpagemode     = {UseNone},
271   pdfstartview    = {FitH},
272   urlcolor        = {blue},
273   colorlinks
274 }
275 \fi
276
277 \iffnttts@
278   % Font Tools
279   \RequirePackage[final]{microtype}
280 \fi
281
282 \ifknwtts@
283   % Knowledge Tools
284   \ifhyref@
285     \RequirePackage[xcolor,hyperref,notion]{knowledge}
286   \else
287     \RequirePackage[xcolor,notion]{knowledge}
288   \fi
289 \fi
290
291 \ifcrv@
292   % Camera-Ready Version
293
294   %%...
295
296 \else
297   % Draft Version
298
299   %%...
300
301   \ifchgbar@
302     % Change Bars
303     \RequirePackage{changebar}
304   \fi
305
306   \iflinnum@
307     % Line Numbers
308     \if@twocolumn
309       \RequirePackage[switch, columnwise, mathlines]{lineno}\linenumbers
310     \else
311       \RequirePackage[columnwise, mathlines]{lineno}\linenumbers
312     \fi
313   \fi
314
315   %%...
316
317 \fi
318
319 \fi
320 %%*****
321 %%*****
322 %** Auxiliary Font Declarations *****
323 %%*****

```

$\backslash\mathrm{mathbbo}$ **Bbo Math Font:** ... to do!

```

324 \ifdef{\mathbbo}{\}{\DeclareMathAlphabet{\mathbbo}{U}{bbold}{m}{n}}

```

```

\matheus Eus Math Font: ... to do!
325 \ifdef{\matheus}{\DeclareMathAlphabet{\matheus}{U}{eus}{m}{n}}

\mathpzc Pzc Math Font: ... to do!
326 \ifdef{\mathpzc}{\DeclareMathAlphabet{\mathpzc}{T1}{pzc}{m}{it}}

\mathscr Scr Math Font: ... to do!
327 \ifdef{\mathscr}{\DeclareMathAlphabet{\mathscr}{U}{rsfs}{m}{n}}

328 %*****%
329 %*****%
330 %** Text Meta Commands *****%
331 %*****%

\newtxt ... to do!
• \newtxt[\rmfamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxt[\sffamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxt[\ttfamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxt![\rmfamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxt![\sffamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxt![\ttfamily]{Name}[sub][sup][Ext] = "NamesubExt"
332 \DeclareRobustCommand{\newtxt}
333 {\ifexclavar{\@snewtxt}{\@newtxt}}
334 \DeclareRobustCommandx{\newtxt}[5][1=, 3=, 4=, 5=]
335 {\text{#1#2\textsubsup{#1}{#3}{#4}{#5}\xspace}}
336 \DeclareRobustCommandx{\@snewtxt}[5][1=, 3=, 4=, 5=]
337 {#1#2\textsubsup{#1}{#3}{#4}{#5}\normalfont\xspace}}

\newtxtsty ... to do!
• \newtxtsty[\rmfamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxtsty[\rmfamily][\sffamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxtsty[\rmfamily][\ttfamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxtsty!{\rmfamily}{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxtsty!{\rmfamily}[\sffamily]{Name}[sub][sup][Ext] = "NamesubExt"
• \newtxtsty!{\rmfamily}[\ttfamily]{Name}[sub][sup][Ext] = "NamesubExt"
338 \DeclareRobustCommand{\newtxtsty}
339 {\ifexclavar{\@snewtxtsty}{\@newtxtsty}}
340 \DeclareRobustCommandx{\newtxtsty}[2][2=]
341 {\newtxt[\defval{#2}{#1}]}
342 \DeclareRobustCommandx{\@snewtxtsty}[2][2=]
343 {\newtxt![\defval{#2}{#1}]}

\newtxtarg ... to do!
• \newtxtarg[\rmfamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
• \newtxtarg[\sffamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
• \newtxtarg[\ttfamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
• \newtxtarg![\rmfamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
• \newtxtarg![\sffamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
• \newtxtarg![\ttfamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesubExt1(Arg)Ext2"
344 \DeclareRobustCommand{\newtxtarg}
345 {\ifexclavar{\@snewtxtarg}{\@newtxtarg}}
346 \DeclareRobustCommandx{\newtxtarg}[7][1=, 3=, 4=, 5=, 7=]
347 {\newtxt{#1}{#2}{#3}{#4}[\argmid{#5}{#6}{#7}]}
348 \DeclareRobustCommandx{\@snewtxtarg}[7][1=, 3=, 4=, 5=, 7=]
349 {\newtxt!{#1}{#2}{#3}{#4}[\argmid{#5}{#6}{#7}]}

```


`\newtxtargsty ... to do!`

- `\newtxtargsty{\rmfamily}{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`
- `\newtxtargsty{\rmfamily}[\sffamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`
- `\newtxtargsty{\rmfamily}[\ttfamily]{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`
- `\newtxtargsty!\rmfamily}{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`
- `\newtxtargsty!\sffamily}{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`
- `\newtxtargsty!\ttfamily}{Name}[sub][sup][Ext1]{Arg}[Ext2] = "NamesupsubExt1(Arg)Ext2"`

```

350 \DeclareRobustCommand{\newtxtargsty}
351   {\ifexclavar{\@snewtxtargsty}{\@newtxtargsty}}
352 \DeclareRobustCommandx{\@newtxtargsty}[2][2=]
353   {\newtxtarg[\defval{#2}{#1}]}
354 \DeclareRobustCommandx{\@snewtxtargsty}[2][2=]
355   {\newtxtarg!\defval{#2}{#1}}

```

`\newtxtoarg ... to do!`

- `\newtxtoarg[\rmfamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoarg[\sffamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoarg[\ttfamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoarg!\rmfamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoarg!\sffamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoarg!\ttfamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`

```

356 \DeclareRobustCommand{\newtxtoarg}
357   {\ifexclavar{\@snewtxtoarg}{\@newtxtoarg}}
358 \DeclareRobustCommandx{\@newtxtoarg}[5][1=, 3=, 4=, 5=]
359   {\newtxtarg[#1]{#2}[#3][#4][#5]}
360 \DeclareRobustCommandx{\@snewtxtoarg}[5][1=, 3=, 4=, 5=]
361   {\newtxtarg!\#1{#2}[#3][#4][#5]}

```

`\newtxtoargsty ... to do!`

- `\newtxtoargsty{\rmfamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoargsty{\rmfamily}[\sffamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoargsty{\rmfamily}[\ttfamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoargsty!\rmfamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoargsty!\sffamily}{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`
- `\newtxtoargsty!\rmfamily}[\ttfamily]{Name}[sub][sup][Arg] = "Namesupsub(Arg)"`

```

362 \DeclareRobustCommand{\newtxtoargsty}
363   {\ifexclavar{\@snewtxtoargsty}{\@newtxtoargsty}}
364 \DeclareRobustCommandx{\@newtxtoargsty}[2][2=]
365   {\newtxtoarg[\defval{#2}{#1}]}
366 \DeclareRobustCommandx{\@snewtxtoargsty}[2][2=]
367   {\newtxtoarg!\defval{#2}{#1}}

```

`\newtxtpar ... to do!`

- `\newtxtpar[\rmfamily]{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`
- `\newtxtpar[\sffamily]{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`
- `\newtxtpar[\ttfamily]{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`
- `\newtxtpar!\rmfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`
- `\newtxtpar!\sffamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`
- `\newtxtpar!\ttfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = "NamesupsubExt1[Par]Ext2"`

```

368 \DeclareRobustCommand{\newtxtpar}
369   {\ifexclavar{\@snewtxtpar}{\@newtxtpar}}
370 \DeclareRobustCommandx{\@newtxtpar}[7][1=, 3=, 4=, 5=, 7=]
371   {\newtxt[#1]{#2}[#3][#4][\argmid{#5}{#6}{#7}]}
372 \DeclareRobustCommandx{\@snewtxtpar}[7][1=, 3=, 4=, 5=, 7=]
373   {\newtxt!\#1{#2}[#3][#4][\argmid{#5}{#6}{#7}]}

```

\newtxtparsty ... to do!

- \newtxtparsty{\rmfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”
- \newtxtparsty{\rmfamily}{\sffamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”
- \newtxtparsty{\rmfamily}{\ttfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”
- \newtxtparsty!\rmfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”
- \newtxtparsty!\sffamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”
- \newtxtparsty!\rmfamily}{\ttfamily}{Name}[sub][sup][Ext1]{Par}[Ext2] = “Name^{sup}_{sub}Ext1[Par]Ext2”

```

374 \DeclareRobustCommand{\newtxtparsty}
375   {\ifexclavar{\@snewtxtparsty}{\@newtxtparsty}}
376 \DeclareRobustCommandx{\@newtxtparsty}[2][2=]
377   {\newtxtpar[\defval{#2}{#1}]}
378 \DeclareRobustCommandx{\@snewtxtparsty}[2][2=]
379   {\newtxtpar!\defval{#2}{#1}}

```

\newtxtopar ... to do!

- \newtxtopar[\rmfamily]{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtopar[\sffamily]{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtopar[\ttfamily]{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtopar!\rmfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtopar!\sffamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtopar!\ttfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”

```

380 \DeclareRobustCommand{\newtxtopar}
381   {\ifexclavar{\@snewtxtopar}{\@newtxtopar}}
382 \DeclareRobustCommandx{\@newtxtopar}[5][1=, 3=, 4=, 5=]
383   {\newtxtpar[#1]{#2}{#3}[#4][#5]}
384 \DeclareRobustCommandx{\@snewtxtopar}[5][1=, 3=, 4=, 5=]
385   {\newtxtpar! [#1]{#2}{#3}[#4][#5]}

```

\newtxtoparsty ... to do!

- \newtxtoparsty{\rmfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtoparsty{\rmfamily}{\sffamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtoparsty{\rmfamily}{\ttfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtoparsty!\rmfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtoparsty!\rmfamily}{\sffamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”
- \newtxtoparsty!\rmfamily}{\ttfamily}{Name}[sub][sup][Par] = “Name^{sup}_{sub}[Par]”

```

386 \DeclareRobustCommand{\newtxtoparsty}
387   {\ifexclavar{\@snewtxtoparsty}{\@newtxtoparsty}}
388 \DeclareRobustCommandx{\@newtxtoparsty}[2][2=]
389   {\newtxtopar[\defval{#2}{#1}]}
390 \DeclareRobustCommandx{\@snewtxtoparsty}[2][2=]
391   {\newtxtopar!\defval{#2}{#1}}

```

\txtsubsup ... to do!

- \txtsubsup{sub}{ } = “_{sub}”; \txtsubsup{ }{sup} = “^{sup}”; \txtsubsup{sub}{sup} = “^{sup}_{sub}”
- \txtsubsup[\sffamily]{Aa}{Bb} = “_{Aa}^{Bb}”
- \txtsubsup[\ttfamily]{Aa}{Bb} = “_{Aa}^{Bb}”

```

392 \DeclareRobustCommand{\txtsubsup}[3][ ]
393   {\ensuremath{\empchk{#2}{\text{#1#2}}}\empchk{#3}{\text{#1#3}}}}

```

394 %*****

\txt ... to do!

- \txt{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \txt[\scshape]{Name}[sub][sup][Ext] = “NAME^{sup}_{SUB}EXT”
- \txt[\bfseries]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”

- $\text{\txt!}{Name}[sub][sup][Ext] = \text{“Name}_{sub}^{sup}Ext\text{”}$
- $\text{\txt!}[\scshape]{Name}[sub][sup][Ext] = \text{“NAME}_{SUB}^{sup}EXT\text{”}$
- $\text{\txt!}[\bfseries]{Name}[sub][sup][Ext] = \text{“Name}_{sub}^{sup}Ext\text{”}$

```

395 \DeclareRobustCommand{\txt}
396   {\ifexclavar{\newtxtsty!{\txtsty}}{\newtxtsty{\txtsty}}}

\txtgarg ... to do!


- $\text{\txtgarg}{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“Name}_{sub}^{sup}Ext1(Arg)Ext2\text{”}$
- $\text{\txtgarg}[\scshape]{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“NAME}_{SUB}^{sup}EXT1(ARG)EXT2\text{”}$
- $\text{\txtgarg}[\bfseries]{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“Name}_{sub}^{sup}Ext1(Arg)Ext2\text{”}$
- $\text{\txtgarg!}{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“Name}_{sub}^{sup}Ext1(Arg)Ext2\text{”}$
- $\text{\txtgarg!}[\scshape]{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“NAME}_{SUB}^{sup}EXT1(ARG)EXT2\text{”}$
- $\text{\txtgarg!}[\bfseries]{Name}[sub][sup][Ext1]{Arg}[Ext2] = \text{“Name}_{sub}^{sup}Ext1(Arg)Ext2\text{”}$



```

397 \DeclareRobustCommand{\txtgarg}
398 {\ifexclavar{\newtxtgargsty!{\txtsty}}{\newtxtgargsty{\txtsty}}}

\txtoarg ... to do!

- $\text{\txtoarg}{Name}[sub][sup][Arg] = \text{“Name}_{sub}^{sup}(Arg)\text{”}$
- $\text{\txtoarg}[\scshape]{Name}[sub][sup][Arg] = \text{“NAME}_{SUB}^{sup}(ARG)\text{”}$
- $\text{\txtoarg}[\bfseries]{Name}[sub][sup][Arg] = \text{“Name}_{sub}^{sup}(Arg)\text{”}$
- $\text{\txtoarg!}{Name}[sub][sup][Arg] = \text{“Name}_{sub}^{sup}(Arg)\text{”}$
- $\text{\txtoarg!}[\scshape]{Name}[sub][sup][Arg] = \text{“NAME}_{SUB}^{sup}(ARG)\text{”}$
- $\text{\txtoarg!}[\bfseries]{Name}[sub][sup][Arg] = \text{“Name}_{sub}^{sup}(Arg)\text{”}$


```

399 \DeclareRobustCommand{\txtoarg}
400   {\ifexclavar{\newtxtoargsty!{\txtsty}}{\newtxtoargsty{\txtsty}}}

\txtpar ... to do!


- $\text{\txtpar}{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“Name}_{sub}^{sup}Ext1[Par]Ext2\text{”}$
- $\text{\txtpar}[\scshape]{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“NAME}_{SUB}^{sup}EXT1[PAR]EXT2\text{”}$
- $\text{\txtpar}[\bfseries]{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“Name}_{sub}^{sup}Ext1[Par]Ext2\text{”}$
- $\text{\txtpar!}{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“Name}_{sub}^{sup}Ext1[Par]Ext2\text{”}$
- $\text{\txtpar!}[\scshape]{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“NAME}_{SUB}^{sup}EXT1[PAR]EXT2\text{”}$
- $\text{\txtpar!}[\bfseries]{Name}[sub][sup][Ext1]{Par}[Ext2] = \text{“Name}_{sub}^{sup}Ext1[Par]Ext2\text{”}$



```

401 \DeclareRobustCommand{\txtpar}
402 {\ifexclavar{\newtxtparsty!{\txtsty}}{\newtxtparsty{\txtsty}}}

\txtopar ... to do!

- $\text{\txtopar}{Name}[sub][sup][Par] = \text{“Name}_{sub}^{sup}[Par]\text{”}$
- $\text{\txtopar}[\scshape]{Name}[sub][sup][Par] = \text{“NAME}_{SUB}^{sup}[PAR]\text{”}$
- $\text{\txtopar}[\bfseries]{Name}[sub][sup][Par] = \text{“Name}_{sub}^{sup}[Par]\text{”}$
- $\text{\txtopar!}{Name}[sub][sup][Par] = \text{“Name}_{sub}^{sup}[Par]\text{”}$
- $\text{\txtopar!}[\scshape]{Name}[sub][sup][Par] = \text{“NAME}_{SUB}^{sup}[PAR]\text{”}$
- $\text{\txtopar!}[\bfseries]{Name}[sub][sup][Par] = \text{“Name}_{sub}^{sup}[Par]\text{”}$


```

403 \DeclareRobustCommand{\txtopar}
404   {\ifexclavar{\newtxtoparsty!{\txtsty}}{\newtxtoparsty{\txtsty}}}

\txtxtsty ... to do!


```

405 \DeclareRobustCommand{\txtxtsty}
406 {\mdseries\upshape\rmfamily}

407 %%%

\cmdtxt ... to do!
```


```


```


```


```


```

```

• \cmdtxt{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtNewCmd{Name}[sub][sup][Ext] = NAMESUBEXT
\txtNewCmd!{Name}[sub][sup][Ext] = NAMESUBEXT

408 \DeclareRobustCommand{\cmdtxt}[1]
409   {\csdef{txt#1}%
410     {\protect\ifexclavar%
411       {\newtxtsty!\csname txtsty#1\endcsname}}%
412     {\newtxtsty{\csname txtsty#1\endcsname}}}}

\cmdtxtarg ... to do!

• \cmdtxtarg{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtargetNewCmd{Name}[sub][sup][Ext1]{Arg}[Ext2] = NAMESUBEXT1(ARG)EXT2
\txtargetNewCmd!{Name}[sub][sup][Ext1]{Arg}[Ext2] = NAMESUBEXT1(ARG)EXT2

413 \DeclareRobustCommand{\cmdtxtarg}[1]
414   {\csdef{txtarget#1}%
415     {\protect\ifexclavar%
416       {\newtxtargetsty!\csname txtsty#1\endcsname}}%
417     {\newtxtargetsty{\csname txtsty#1\endcsname}}}}

\cmdtxttoarg ... to do!

• \cmdtxttoarg{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtoargNewCmd{Name}[sub][sup][Arg] = NAMESUB(ARG)
\txtoargNewCmd!{Name}[sub][sup][Arg] = NAMESUB(ARG)

418 \DeclareRobustCommand{\cmdtxttoarg}[1]
419   {\csdef{txtoarg#1}%
420     {\protect\ifexclavar%
421       {\newtxtoargsty!\csname txtsty#1\endcsname}}%
422     {\newtxtoargsty{\csname txtsty#1\endcsname}}}}

\cmdtxtpar ... to do!

• \cmdtxtpar{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtparNewCmd{Name}[sub][sup][Ext1]{Par}[Ext2] = NAMESUBEXT1[PAR]EXT2
\txtparNewCmd!{Name}[sub][sup][Ext1]{Par}[Ext2] = NAMESUBEXT1[PAR]EXT2

423 \DeclareRobustCommand{\cmdtxtpar}[1]
424   {\csdef{txtpar#1}%
425     {\protect\ifexclavar%
426       {\newtxtparsty!\csname txtsty#1\endcsname}}%
427     {\newtxtparsty{\csname txtsty#1\endcsname}}}}

\cmdtxtopar ... to do!

• \cmdtxtopar{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtoparNewCmd{Name}[sub][sup][Par] = NAMESUB[PAR]
\txtoparNewCmd!{Name}[sub][sup][Par] = NAMESUB[PAR]

428 \DeclareRobustCommand{\cmdtxtopar}[1]
429   {\csdef{txtopar#1}%
430     {\protect\ifexclavar%
431       {\newtxtoparsty!\csname txtsty#1\endcsname}}%
432     {\newtxtoparsty{\csname txtsty#1\endcsname}}}}

\cmdtxtall ... to do!

• \cmdtxtall{NewCmd}; \newcommand{txtstyNewCmd}{\scshape\ttfamily};
\txtNewCmd{Name}[sub][sup][Ext] = NAMESUBEXT
\txtargetNewCmd{Name}[sub][sup][Ext1]{Arg}[Ext2] = NAMESUBEXT1(ARG)EXT2
\txtoargNewCmd{Name}[sub][sup][Arg] = NAMESUB(ARG)
\txtparNewCmd{Name}[sub][sup][Ext1]{Par}[Ext2] = NAMESUBEXT1[PAR]EXT2
\txtoparNewCmd{Name}[sub][sup][Par] = NAMESUB[PAR]

433 \DeclareRobustCommand{\cmdtxtall}[1]
434   {\cmdtxt{#1}\cmdtxtarg{#1}\cmdtxttoarg{#1}\cmdtxtpar{#1}\cmdtxtopar{#1}}

435 %%*****%
```

\usrtxt ... to do!

- \usrtxt{cmdName}{Suf}{};
 $\backslash\text{cmdNameSuf} = \text{cmdName}$
 $\backslash\text{cmdNameSuf}^* = \text{cmdName}^*$
 $\backslash\text{usrtxt}\{ \text{cmdName} \} \{ \text{Suf} \} \{ \text{arg} \};$
 $\backslash\text{cmdNameSuf} \{ \text{Arg} \} = \text{cmdName}(\text{Arg})$
 $\backslash\text{cmdNameSuf} ! \{ \text{Arg} \} = \text{cmdName}(\text{Arg})$
 $\backslash\text{usrtxt}\{ \text{cmdName} \} \{ \text{Suf} \} \{ \text{par} \};$
 $\backslash\text{cmdNameSuf} \{ \text{Par} \} = \text{cmdName}[\text{Par}]$
 $\backslash\text{cmdNameSuf} ! \{ \text{Par} \} = \text{cmdName}[\text{Par}]$
- \usrtxt{cmdName}{Suf}{}[newName];
 $\backslash\text{cmdNameSuf} = \text{newName}$
 $\backslash\text{cmdNameSuf}^* = \text{newName}^*$
 $\backslash\text{usrtxt}\{ \text{cmdName} \} \{ \text{Suf} \} \{ \text{arg} \} [\text{newName}];$
 $\backslash\text{cmdNameSuf} \{ \text{Arg} \} = \text{newName}(\text{Arg})$
 $\backslash\text{cmdNameSuf} ! \{ \text{Arg} \} = \text{newName}(\text{Arg})$
 $\backslash\text{usrtxt}\{ \text{cmdName} \} \{ \text{Suf} \} \{ \text{par} \} [\text{newName}];$
 $\backslash\text{cmdNameSuf} \{ \text{Par} \} = \text{newName}[\text{Par}]$
 $\backslash\text{cmdNameSuf} ! \{ \text{Par} \} = \text{newName}[\text{Par}]$

```

436 \DeclareRobustCommandx{\usrtxt}[4][4=]
437   {\csdef{#1#2}%
438     {\protect\ifexclavar%
439       {\csname txt#3\endcsname!\defval{#4}{#1}}}%
440     {\csname txt#3\endcsname{\defval{#4}{#1}}}}}

441 %*****%
442 %*****%
443 %** Math Meta Commands *****%
444 %*****%
```

\newmth ... to do!

- \newmth[mathrm]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \newmth[mathsf]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \newmth[mathtt]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \newmth![mathrm]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \newmth![mathsf]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \newmth![mathtt]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \innewmth[mathrm]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \innewmth[mathsf]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \innewmth[mathtt]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \klnewmth[mathrm]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \klnewmth[mathsf]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”
- \klnewmth[mathtt]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”

```

445 \DeclareRobustCommand{\newmth}
446   {\ifexclavar{\@snewmth}{\@newmth}}
447 \DeclareRobustCommandx{\@newmth}[5][1=, 3=, 4=, 5=]
448   {\ensuremath{\csname#1\endcsname{#2}\mthsubsup{#3}{#4}{#5}}}
449 \DeclareRobustCommandx{\@snewmth}[5][1=, 3=, 4=, 5=]
450   {\ensuremath{\csname#1\endcsname #2\mthsubsup{#3}{#4}{#5}}}
451 \ifknwtls@
452   \DeclareRobustCommandx{\innewmth}[5][1=, 3=, 4=, 5=]
453     {\ensuremath{\intro{\csname#1\endcsname{#2}}\mthsubsup{#3}{#4}{#5}}}
454   \DeclareRobustCommandx{\klnewmth}[5][1=, 3=, 4=, 5=]
455     {\ensuremath{\kl{\csname#1\endcsname{#2}}\mthsubsup{#3}{#4}{#5}}}
456 \fi
```

\newmthsty ... to do!

- \newmthsty[mathrm]{Name}[sub][sup][Ext] = “Name^{sup}_{sub}Ext”

`\newmthargsty ... to do!`

- `\newmthargsty{\mathrm}{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\newmthargsty{\mathrm}[mathsf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\newmthargsty{\mathrm}[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\newmthargsty!\mathrm}{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\newmthargsty!\mathrm}[mathsf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\newmthargsty!\mathrm}[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\innewmthargsty{\mathrm}{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\innewmthargsty{\mathrm}[mathsf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\innewmthargsty{\mathrm}[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\klnewmthargsty{\mathrm}{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\klnewmthargsty{\mathrm}[mathsf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`
- `\klnewmthargsty{\mathrm}[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = “NamesupsubExt1(ArgExEx)Ext2”`

```

482 \DeclareRobustCommand{\newmthargsty}
483   {\ifexclavar{\@snewmthargsty}{\@newmthargsty}}
484 \DeclareRobustCommandx{\@newmthargsty}[2][2=]
485   {\newmtharg[\defval{#2}{#1}]}
486 \DeclareRobustCommandx{\@snewmthargsty}[2][2=]
487   {\newmtharg!\defval{#2}{#1}}
488 \ifknwtls@
489   \DeclareRobustCommandx{\innewmthargsty}[2][2=]
490     {\innewmtharg[\defval{#2}{#1}]}
491   \DeclareRobustCommandx{\klnewmthargsty}[2][2=]
492     {\klnewmtharg[\defval{#2}{#1}]}
493 \fi

```

`\newmthoarg ... to do!`

- `\newmthoarg{\mathrm}{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\newmthoarg[mathsf]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\newmthoarg[mathtt]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\newmthoarg!\mathrm}{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\newmthoarg!\mathsf]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\newmthoarg!\mathtt]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\innewmthoarg{\mathrm}{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\innewmthoarg[mathsf]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\innewmthoarg[mathtt]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\klnewmthoarg{\mathrm}{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\klnewmthoarg[mathsf]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`
- `\klnewmthoarg[mathtt]{Name}[sub][sup][Arg^{Ex^{Ex}}] = “Namesupsub(ArgExEx)”`

```

494 \DeclareRobustCommand{\newmthoarg}
495   {\ifexclavar{\@snewmthoarg}{\@newmthoarg}}
496 \DeclareRobustCommandx{\@newmthoarg}[5][1=, 3=, 4=, 5=]
497   {\newmtharg[#1]{#2}[#3][#4][#5][#6]}

```

\newmthoargsty ... to do!

- ```

506 \DeclareRobustCommand{\newmthoargsty}
507 {\ifxclavar{\@snewmthoargsty}{\@newmthoargsty}}
508 \DeclareRobustCommandx{\@newmthoargsty}[2][2=]
509 {\newmthoarg[\defval{#2}{#1}]}
510 \DeclareRobustCommandx{\@snewmthoargsty}[2][2=]
511 {\newmthoarg![\defval{#2}{#1}]}
512 \ifknwts@
513 \DeclareRobustCommandx{\innnewmthoargsty}[2][2=]
514 {\innnewmthoarg[\defval{#2}{#1}]}
515 \DeclareRobustCommandx{\klnnewmthoargsty}[2][2=]
516 {\klnnewmthoarg[\defval{#2}{#1}]}
517 \fi

```

\newmthpar ... to do!

- 16



- $\backslash\mathrm{klnewmthpar}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{klnewmthpar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$

```

518 \DeclareRobustCommand{\newmthpar}
519 {\ifexclavar{\@snewmthpar}{\@newmthpar}}
520 \DeclareRobustCommandx{\@newmthpar}[7][1=, 3=, 4=, 5=, 7=]
521 {\newmth[#1]{#2}{#3}{#4}[\argmid{#5}\!\left[\!]{#6}{\right]}\arglef{\!}{#7}}]
522 \DeclareRobustCommandx{\@snewmthpar}[7][1=, 3=, 4=, 5=, 7=]
523 {\newmth[#1]{#2}{#3}{#4}[\argmid{#5}{#6}{#7}]}
524 \ifknwts@
525 \DeclareRobustCommandx{\innewmthpar}[7][1=, 3=, 4=, 5=, 7=]
526 {\innewmth[#1]{#2}{#3}{#4}[\argmid{#5}\!\left[\!]{#6}{\right]}\arglef{\!}{#7}}]
527 \DeclareRobustCommandx{\klnewmthpar}[7][1=, 3=, 4=, 5=, 7=]
528 {\klnewmth[#1]{#2}{#3}{#4}[\argmid{#5}\!\left[\!]{#6}{\right]}\arglef{\!}{#7}}]
529 \fi

```

$\backslash\mathrm{newmthparsty}$  ... to do!

- $\backslash\mathrm{newmthparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{newmthparsty}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{newmthparsty}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{newmthparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{newmthparsty}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{newmthparsty}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{innewmthparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{innewmthparsty}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{innewmthparsty}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{klnewmthparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{klnewmthparsty}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$
- $\backslash\mathrm{klnewmthparsty}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Ext1}]{\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}[\mathrm{Ext2}] = \text{“Name}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2\text{”}$

```

530 \DeclareRobustCommand{\newmthparsty}
531 {\ifexclavar{\@snewmthparsty}{\@newmthparsty}}
532 \DeclareRobustCommandx{\@newmthparsty}[2][2=]
533 {\newmthpar[\defval{#2}{#1}]}
534 \DeclareRobustCommandx{\@snewmthparsty}[2][2=]
535 {\newmthpar!\defval{#2}{#1}}
536 \ifknwts@
537 \DeclareRobustCommandx{\innewmthparsty}[2][2=]
538 {\innewmthpar[\defval{#2}{#1}]}
539 \DeclareRobustCommandx{\klnewmthparsty}[2][2=]
540 {\klnewmthpar[\defval{#2}{#1}]}
541 \fi

```

$\backslash\mathrm{newmthpar}$  ... to do!

- $\backslash\mathrm{newmthpar}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$
- $\backslash\mathrm{newmthpar}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$
- $\backslash\mathrm{newmthpar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$
- $\backslash\mathrm{newmthpar}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$
- $\backslash\mathrm{newmthpar}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$
- $\backslash\mathrm{newmthpar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}}^{\mathrm{Ex}}{\mathrm{Ex}}] = \text{“Name}_{sub}^{sup}\left[Par^{Ex^{Ex}}\right]\text{”}$

- $\mathrm{innewmthopar}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{innewmthopar}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{innewmthopar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthopar}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthopar}[\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthopar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$

```

542 \DeclareRobustCommand{\newmthopar}
543 {\ifexclavar{\@snewmthopar}{\@newmthopar}}
544 \DeclareRobustCommand{\@newmthopar}[5][1=, 3=, 4=, 5=]
545 {\newmthpar[#1]{#2}[#3][#4][#5]}
546 \DeclareRobustCommand{\@snewmthopar}[5][1=, 3=, 4=, 5=]
547 {\newmthpar! [#1]{#2}[#3][#4][#5]}
548 \ifknwts@
549 \DeclareRobustCommand{\innewmthopar}[5][1=, 3=, 4=, 5=]
550 {\innewmthpar[#1]{#2}[#3][#4][#5]}
551 \DeclareRobustCommand{\klnewmthopar}[5][1=, 3=, 4=, 5=]
552 {\klnewmthpar[#1]{#2}[#3][#4][#5]}
553 \fi

```

$\mathrm{newmthoparsty}$  ... to do!

- $\mathrm{newmthoparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{newmthoparsty}[\mathrm{mathrm}][\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{newmthoparsty}[\mathrm{mathrm}][\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{newmthoparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{newmthoparsty}[\mathrm{mathrm}][\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{newmthoparsty}[\mathrm{mathrm}][\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{innewmthoparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{innewmthoparsty}[\mathrm{mathrm}][\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{innewmthoparsty}[\mathrm{mathrm}][\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthoparsty}[\mathrm{mathrm}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthoparsty}[\mathrm{mathrm}][\mathrm{mathsf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$
- $\mathrm{klnewmthoparsty}[\mathrm{mathrm}][\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}^{\mathrm{Ex}}}] = \text{“Name}_{sub}^{sup}[Par^{Ex^{Ex}}]”}$

```

554 \DeclareRobustCommand{\newmthoparsty}
555 {\ifexclavar{\@snewmthoparsty}{\@newmthoparsty}}
556 \DeclareRobustCommand{\@newmthoparsty}[2][2=]
557 {\newmthopar[\defval{#2}]{#1}}
558 \DeclareRobustCommand{\@snewmthoparsty}[2][2=]
559 {\newmthopar! [\defval{#2}]{#1}}
560 \ifknwts@
561 \DeclareRobustCommand{\innewmthoparsty}[2][2=]
562 {\innewmthopar[\defval{#2}]{#1}}
563 \DeclareRobustCommand{\klnewmthoparsty}[2][2=]
564 {\klnewmthopar[\defval{#2}]{#1}}
565 \fi

```

$\mathrm{mthsubsup}$  ... to do!

```

566 \DeclareRobustCommand{\mthsubsup}[2]
567 {\empchk{#1}{_}{#1}}\empchk{#2}{^}{#2}}

```

568 %\*\*\*\*\*%

\mth ... to do!

- \mth{Name}[sub][sup][Ext] = " $Name_{sub}^{sup}Ext$ "
- \mth[mathbf]{Name}[sub][sup][Ext] = " $\mathbf{Name}_{sub}^{sup}Ext$ "
- \mth[mathtt]{Name}[sub][sup][Ext] = " $Name_{sub}^{sup}Ext$ "
- \mth!{Name}[sub][sup][Ext] = " $Name_{sub}^{sup}Ext$ "
- \mth![mathbf]{Name}[sub][sup][Ext] = " $\mathbf{Name}_{sub}^{sup}Ext$ "
- \mth![mathtt]{Name}[sub][sup][Ext] = " $Name_{sub}^{sup}Ext$ "
- \inmth{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "
- \inmth[mathbf]{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "
- \inmth[mathtt]{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "
- \klmth{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "
- \klmth[mathbf]{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "
- \klmth[mathtt]{Name}[sub][sup][Ext] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext$ "

```
569 \DeclareRobustCommand{\mth}
570 {\ifexclavar{\newmthsty!\mthsty}}{\newmthsty{\mthsty}}
571 \ifknwtsl@
572 \DeclareRobustCommand{\inmth}
573 {\innewmthsty{\mthsty}}
574 \DeclareRobustCommand{\klmth}
575 {\klnewmthsty{\mthsty}}
576 \fi
```

\mtharg ... to do!

- \mtharg{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $Name_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \mtharg[mathbf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\mathbf{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \mtharg[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $Name_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \mtharg!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $Name_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \mtharg![mathbf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\mathbf{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \mtharg![mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $Name_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \inmtharg{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \inmtharg[mathbf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \inmtharg[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \klmtharg{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \klmtharg[mathbf]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "
- \klmtharg[mathtt]{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = " $\textcolor{brown}{Name}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$ "

```
577 \DeclareRobustCommand{\mtharg}
578 {\ifexclavar{\newmthargsty!\mthsty}}{\newmthargsty{\mthsty}}
579 \ifknwtsl@
580 \DeclareRobustCommand{\inmtharg}
581 {\innewmthargsty{\mthsty}}
582 \DeclareRobustCommand{\klmtharg}
583 {\klnewmthargsty{\mthsty}}
584 \fi
```

\mthoarg ... to do!

- \mthoarg{Name}[sub][sup][Arg^{Ex^{Ex}}] = " $Name_{sub}^{sup}\left(Arg^{Ex^{Ex}}\right)$ "



- $\backslash\mathrm{mthopar}[\mathrm{mathbf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{mthopar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{mthopar}!{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{mthopar}![\mathrm{mathbf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{mthopar}![\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{inmthopar}{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{inmthopar}[\mathrm{mathbf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{inmthopar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{klmthopar}{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{klmthopar}[\mathrm{mathbf}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$ ,
- $\backslash\mathrm{klmthopar}[\mathrm{mathtt}]{\mathrm{Name}}[\mathrm{sub}][\mathrm{sup}][\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}] = \mathrm{Name}_{\mathrm{sub}}^{\mathrm{sup}}[\mathrm{Par}^{\mathrm{Ex}\mathrm{Ex}}]$

```

601 \DeclareRobustCommand{\mthopar}
602 {\ifexclavar{\newmthoparsty!\mthsty}}{\newmthoparsty{\mthsty}}}
603 \ifknwts@
604 \DeclareRobustCommand{\inmthopar}
605 {\innewmthoparsty{\mthsty}}
606 \DeclareRobustCommand{\klmthopar}
607 {\klnewmthoparsty{\mthsty}}
608 \fi

```

$\backslash\mathrm{mthsty}$  ... to do!

```

609 \def\mthsty
610 {}
611 %*****%

```

$\backslash\mathrm{cmdmth}$  ... to do!

```

• \cmdmth{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};
 \mthNewCmd{Name}[sub][sup][Ext] = \Name_{sub}^{sup}Ext
 \mthNewCmd!{Name}[sub][sup][Ext] = \Name_{sub}^{sup}Ext
 \inmthNewCmd{Name}[sub][sup][Ext] = \Name_{sub}^{sup}Ext
 \klmthNewCmd{Name}[sub][sup][Ext] = \Name_{sub}^{sup}Ext
612 \DeclareRobustCommand{\cmdmth}[1]
613 {\csdef{mth#1}%
614 {\protect\ifexclavar{\newmthsty!\mthsty#1}}{\newmthsty{mthsty#1}}}%
615 \ifknwts@
616 \csdef{inmth#1}%
617 {\innewmthsty{mthsty#1}}%
618 \csdef{klmth#1}%
619 {\klnewmthsty{mthsty#1}}
620 \fi}

```

$\backslash\mathrm{cmdmtharg}$  ... to do!

```

• \cmdmtharg{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};
 \mthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \Name_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2
 \mthargNewCmd!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \Name_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2
 \inmthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \Name_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2
 \klmthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \Name_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2
621 \DeclareRobustCommand{\cmdmtharg}[1]
622 {\csdef{mtharg#1}%
623 {\protect\ifexclavar{\newmthargsty!\mthsty#1}}{\newmthargsty{mthsty#1}}}
624 \ifknwts@

```

```

625 \csdef{inmtharg#1}%
626 {\innewmthargsty{mthargsty#1}}%
627 \csdef{klmtharg#1}%
628 {\klnewmthargsty{mthargsty#1}}
629 \fi}

```

\cmdmthoarg ... to do!

```

• \cmdmthoarg{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};
\mthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Name_{sub}^{sup}(Arg^{Ex^{Ex}})
\mthoargNewCmd!{Name}[sub][sup][Arg^{Ex^{Ex}}] = Name_{sub}^{sup}(Arg^{Ex^{Ex}})
\inmthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Name_{sub}^{sup}(Arg^{Ex^{Ex}})
\klmthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Name_{sub}^{sup}(Arg^{Ex^{Ex}})

630 \DeclareRobustCommand{\cmdmthoarg}[1]
631 {\csdef{mthoarg#1}%
632 {\protect\ifexclavar{\newmthoargsty!{mthsty#1}}{\newmthoargsty{mthsty#1}}}
633 \ifknwts@
634 \csdef{inmthoarg#1}%
635 {\innewmthoargsty{mthargsty#1}}%
636 \csdef{klmthoarg#1}%
637 {\klnewmthoargsty{mthargsty#1}}
638 \fi}

```

\cmdmthpar ... to do!

```

• \cmdmthpar{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};
\mthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = Name_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2
\mthparNewCmd!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = Name_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2
\inmthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = Name_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2
\klmthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = Name_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2

639 \DeclareRobustCommand{\cmdmthpar}[1]
640 {\csdef{mthpar#1}%
641 {\protect\ifexclavar{\newmthparsty!{mthsty#1}}{\newmthparsty{mthsty#1}}}
642 \ifknwts@
643 \csdef{inmthpar#1}%
644 {\innewmthparsty{mthargsty#1}}%
645 \csdef{klmthpar#1}%
646 {\klnewmthparsty{mthargsty#1}}
647 \fi}

```

\cmdmthopar ... to do!

```

• \cmdmthopar{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};
\mthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Name_{sub}^{sup}[Par^{Ex^{Ex}}]
\mthoparNewCmd!{Name}[sub][sup][Par^{Ex^{Ex}}] = Name_{sub}^{sup}[Par^{Ex^{Ex}}]
\inmthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Name_{sub}^{sup}[Par^{Ex^{Ex}}]
\klmthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Name_{sub}^{sup}[Par^{Ex^{Ex}}]

648 \DeclareRobustCommand{\cmdmthopar}[1]
649 {\csdef{mthopar#1}%
650 {\protect\ifexclavar{\newmthoparsty!{mthsty#1}}{\newmthoparsty{mthsty#1}}}
651 \ifknwts@
652 \csdef{inmthopar#1}%
653 {\innewmthoparsty{mthargsty#1}}%
654 \csdef{klmthopar#1}%
655 {\klnewmthoparsty{mthargsty#1}}
656 \fi}

```

\cmdmthall ... to do!

- `\cmdmthall{NewCmd}; \newcommand{mthstyNewCmd}{\mathtt};`  
`\mthNewCmd{Name}[sub][sup][Ext] = NamesubExt`  
`\inmthNewCmd{Name}[sub][sup][Ext] = NamesubExt`  
`\klmthNewCmd{Name}[sub][sup][Ext] = NamesubExt`  
`\mthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesubExt1(Arg^{Ex^{Ex}})Ext2`  
`\mthargNewCmd!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesubExt1(Arg^{Ex^{Ex}})Ext2`  
`\inmthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesubExt1(Arg^{Ex^{Ex}})Ext2`  
`\klmthargNewCmd{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesubExt1(Arg^{Ex^{Ex}})Ext2`  
`\mthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Namesub(Arg^{Ex^{Ex}})`  
`\mthoargNewCmd!{Name}[sub][sup][Arg^{Ex^{Ex}}] = Namesub(Arg^{Ex^{Ex}})`  
`\inmthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Namesub(Arg^{Ex^{Ex}})`  
`\klmthoargNewCmd{Name}[sub][sup][Arg^{Ex^{Ex}}] = Namesub(Arg^{Ex^{Ex}})`  
`\mthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesubExt1[Par^{Ex^{Ex}}]Ext2`  
`\mthparNewCmd!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesubExt1[Par^{Ex^{Ex}}]Ext2`  
`\inmthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesubExt1[Par^{Ex^{Ex}}]Ext2`  
`\klmthparNewCmd{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesubExt1[Par^{Ex^{Ex}}]Ext2`  
`\mthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Namesub[Par^{Ex^{Ex}}]`  
`\mthoparNewCmd!{Name}[sub][sup][Par^{Ex^{Ex}}] = Namesub[Par^{Ex^{Ex}}]`  
`\inmthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Namesub[Par^{Ex^{Ex}}]`  
`\klmthoparNewCmd{Name}[sub][sup][Par^{Ex^{Ex}}] = Namesub[Par^{Ex^{Ex}}]`

657 `\DeclareRobustCommand{\cmdmthall}[1]`

658 `\{\cmdmth{#1}\cmdmtharg{#1}\cmdmthoarg{#1}\cmdmthpar{#1}\cmdmthopar{#1}\}`

659 `%%*****`

`\usrmth ... to do!`

- `\usrmth{cmdName}{Suf}{};`  
`\cmdNameSuf = cmdName`  
`\cmdNameSuf! = cmdName`  
`\incmdNameSuf = cmdName`  
`\klcmdNameSuf = cmdName`  
`\usrmth{cmdName}{Suf}{arg};`  
`\cmdNameSuf{Arg^{Ex^{Ex}}} = cmdName(Arg^{Ex^{Ex}})`  
`\cmdNameSuf!{Arg^{Ex^{Ex}}} = cmdName(Arg^{Ex^{Ex}})`  
`\incmdNameSuf{Arg^{Ex^{Ex}}} = cmdName(Arg^{Ex^{Ex}})`  
`\klcmdNameSuf{Arg^{Ex^{Ex}}} = cmdName(Arg^{Ex^{Ex}})`  
`\usrmth{cmdName}{Suf}{par};`  
`\cmdNameSuf{Par^{Ex^{Ex}}} = cmdName[Par^{Ex^{Ex}}]`  
`\cmdNameSuf!{Par^{Ex^{Ex}}} = cmdName[Par^{Ex^{Ex}}]`  
`\incmdNameSuf{Arg^{Ex^{Ex}}} = cmdName[Arg^{Ex^{Ex}}]`  
`\klcmdNameSuf{Arg^{Ex^{Ex}}} = cmdName[Arg^{Ex^{Ex}}]`
- `\usrmth{cmdName}{Suf}{}[newName];`  
`\cmdNameSuf = newName`  
`\cmdNameSuf! = newName`  
`\incmdNameSuf = newName`  
`\klcmdNameSuf = newName`  
`\usrmth{cmdName}{Suf}{arg}[newName];`  
`\cmdNameSuf{Arg^{Ex^{Ex}}} = newName(Arg^{Ex^{Ex}})`  
`\incmdNameSuf{Arg^{Ex^{Ex}}} = newName(Arg^{Ex^{Ex}})`  
`\klcmdNameSuf{Arg^{Ex^{Ex}}} = newName(Arg^{Ex^{Ex}})`  
`\usrmth{cmdName}{Suf}{par}[newName];`



```

\cmdNameSuf{Par^{Ex^{Ex}}} = newName[Par^{Ex^{Ex}}]
\cmdNameSuf{!{Par^{Ex^{Ex}}} = newName[Par^{Ex^{Ex}}]
\incmdNameSuf{Arg^{Ex^{Ex}}} = newName[Arg^{Ex^{Ex}}]
\klcmdNameSuf{Arg^{Ex^{Ex}}} = newName[Arg^{Ex^{Ex}}]

660 \DeclareRobustCommandx{\usrmth}[4][4=]
661 {\csdef{#1#2}%
662 {\protect\ifexclavar%
663 {\csname mth#3\endcsname!\defval{#4}{#1}}}%
664 {\csname mth#3\endcsname\defval{#4}{#1}}}%
665 \ifknwts@
666 \csdef{in#1#2}%
667 {\csname inmth#3\endcsname\defval{#4}{#1}}}%
668 \csdef{kl#1#2}%
669 {\csname klmth#3\endcsname\defval{#4}{#1}}}%
670 \fi}

671 %%%

\usrmthlatlow ... to do!

672 \DeclareRobustCommandx{\usrmthlatlow}[4][4=]
673 {\usrmth{#1}{#2}{#3}[#4]\seqoflatlow{#1#2}{mth#3}}

\usrmthlatupp ... to do!

674 \DeclareRobustCommandx{\usrmthlatupp}[4][4=]
675 {\usrmth{#1}{#2}{#3}[#4]\seqoflatupp{#1#2}{mth#3}}

\usrmthlatlet ... to do!

676 \DeclareRobustCommandx{\usrmthlatlet}[4][4=]
677 {\usrmth{#1}{#2}{#3}[#4]\seqoflatlet{#1#2}{mth#3}}

\usrmthgrklow ... to do!

678 \DeclareRobustCommandx{\usrmthgrklow}[4][4=]
679 {\usrmth{#1}{#2}{#3}[#4]\seqofgrklow{#1#2}{mth#3}}

\usrmthgrkupp ... to do!

680 \DeclareRobustCommandx{\usrmthgrkupp}[4][4=]
681 {\usrmth{#1}{#2}{#3}[#4]\seqofgrkupp{#1#2}{mth#3}}

\usrmthgrklet ... to do!

682 \DeclareRobustCommandx{\usrmthgrklet}[4][4=]
683 {\usrmth{#1}{#2}{#3}[#4]\seqofgrklet{#1#2}{mth#3}}

\usrmthlow ... to do!

684 \DeclareRobustCommandx{\usrmthlow}[4][4=]
685 {\usrmth{#1}{#2}{#3}[#4]\seqoflow{#1#2}{mth#3}}

\usrmthupp ... to do!

686 \DeclareRobustCommandx{\usrmthupp}[4][4=]
687 {\usrmth{#1}{#2}{#3}[#4]\seqofupp{#1#2}{mth#3}}

\usrmthlet ... to do!

688 \DeclareRobustCommandx{\usrmthlet}[4][4=]
689 {\usrmth{#1}{#2}{#3}[#4]\seqoflet{#1#2}{mth#3}}

690 %%%
691 %%%
692 %** Text Macro Generators %%%
693 %%%
694 \iftxtgen@

\txtdef ... to do!
\txtargdef
\txtpardef

```



- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext]} = Name_{sub}^{sup}Ext$
- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext1]\{Arg\}[Ext2]} = Name_{sub}^{sup}Ext1(Arg)Ext2$
- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext1]\{Par\}[Ext2]} = Name_{sub}^{sup}Ext1[Par]Ext2$

```

695 %% Style for Definitions
696 \cmdtxtall{def}
697 \DeclareRobustCommand{\txtstydef}{\normalfont\bfseries\em}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{cmdName\}};$   
 $\text{\texttt{cmdName}}[sub][sub][ext] = cmdName_{sub}^{sub}ext$
- $\text{\texttt{\texttt{def}}\{cmdName\}[newName]};$   
 $\text{\texttt{cmdName}}[sub][sub][ext] = newName_{sub}^{sub}ext$

```

698 \DeclareRobustCommandx{\cmdtxtdef}[2][2=]
699 {\usrtxt{\#1}\{\}\{def\}[{\#2}]}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{cmdName\}};$   
 $\text{\texttt{cmdName}}[sub][sub][ext1]\{arg\}[ext2] = cmdName_{sub}^{sub}ext1(arg)ext2$
- $\text{\texttt{\texttt{def}}\{cmdName\}[newName]};$   
 $\text{\texttt{cmdName}}[sub][sub][ext1]\{arg\}[ext2] = newName_{sub}^{sub}ext1(arg)ext2$

```

700 \DeclareRobustCommandx{\cmdtxtargdef}[2][2=]
701 {\usrtxt{\#1}\{\}\{argdef\}[{\#2}]}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{cmdName\}};$   
 $\text{\texttt{cmdName}}[sub][sub][arg] = cmdName_{sub}^{sub}(arg)$
- $\text{\texttt{\texttt{def}}\{cmdName\}[newName]};$   
 $\text{\texttt{cmdName}}[sub][sub][arg] = newName_{sub}^{sub}(arg)$

```

702 \DeclareRobustCommandx{\cmdtxtoargdef}[2][2=]
703 {\usrtxt{\#1}\{\}\{oargdef\}[{\#2}]}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{cmdName\}};$   
 $\text{\texttt{cmdName}}[sub][sub][ext1]\{par\}[ext2] = cmdName_{sub}^{sub}ext1[par]ext2$
- $\text{\texttt{\texttt{def}}\{cmdName\}[newName]};$   
 $\text{\texttt{cmdName}}[sub][sub][ext1]\{par\}[ext2] = newName_{sub}^{sub}ext1[par]ext2$

```

704 \DeclareRobustCommandx{\cmdtxtpardef}[2][2=]
705 {\usrtxt{\#1}\{\}\{pardef\}[{\#2}]}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{cmdName\}};$   
 $\text{\texttt{cmdName}}[sub][sub][par] = cmdName_{sub}^{sub}[par]$
- $\text{\texttt{\texttt{def}}\{cmdName\}[newName]};$   
 $\text{\texttt{cmdName}}[sub][sub][par] = newName_{sub}^{sub}[par]$

```

706 \DeclareRobustCommandx{\cmdtxtopardef}[2][2=]
707 {\usrtxt{\#1}\{\}\{opardef\}[{\#2}]}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext]} = Name_{sub}^{sup}Ext$
- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext1]\{Arg\}[Ext2]} = Name_{sub}^{sup}Ext1(Arg)Ext2$
- $\text{\texttt{\texttt{def}}\{Name\}[sub][sup][Ext1]\{Par\}[Ext2]} = Name_{sub}^{sup}Ext1[Par]Ext2$

```

708 %% Style for Abbreviations
709 \cmdtxtall{abr}
710 \DeclareRobustCommand{\txtstyabr}{\em}

```

$\text{\texttt{\texttt{def}}} \dots$  to do!

- $\backslash\text{cmdtxtabr}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext}$
- $\backslash\text{cmdtxtabr}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext}] = \text{newName}_{\text{sub}}^{\text{sub}}\text{ext}$

711  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtabr}\}[2][2=]$   
712  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{abr}\}\{\#2\}\}$

$\backslash\text{cmdtxtargabr}$  ... to do!

- $\backslash\text{cmdtxtargabr}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$
- $\backslash\text{cmdtxtargabr}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{newName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$

713  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtargabr}\}[2][2=]$   
714  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{argabr}\}\{\#2\}\}$

$\backslash\text{cmdtxtoargabr}$  ... to do!

- $\backslash\text{cmdtxtoargabr}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{arg}] = \text{cmdName}_{\text{sub}}^{\text{sub}}(\text{arg})$
- $\backslash\text{cmdtxtoargabr}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{arg}] = \text{newName}_{\text{sub}}^{\text{sub}}(\text{arg})$

715  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtoargabr}\}[2][2=]$   
716  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{oargabr}\}\{\#2\}\}$

$\backslash\text{cmdtxtparabr}$  ... to do!

- $\backslash\text{cmdtxtparabr}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$
- $\backslash\text{cmdtxtparabr}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{newName}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$

717  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtparabr}\}[2][2=]$   
718  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{parabr}\}\{\#2\}\}$

$\backslash\text{cmdtxtoparabr}$  ... to do!

- $\backslash\text{cmdtxtoparabr}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{par}] = \text{cmdName}_{\text{sub}}^{\text{sub}}[\text{par}]$
- $\backslash\text{cmdtxtoparabr}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{par}] = \text{newName}_{\text{sub}}^{\text{sub}}[\text{par}]$

719  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtoparabr}\}[2][2=]$   
720  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{oparabr}\}\{\#2\}\}$

721  $\%\text{*****\%}$

$\backslash\text{txtname}$  ... to do!

$\backslash\text{txtargname}$     •  $\backslash\text{txtname}\{\text{Name}\}[\text{sub}][\text{sup}][\text{Ext}] = \text{NAME}_{\text{SUB}}^{\text{SUP}}\text{EXT}$   
 $\backslash\text{txtparname}$     •  $\backslash\text{txtargname}\{\text{Name}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Arg}\}[\text{Ext2}] = \text{NAME}_{\text{SUB}}^{\text{SUP}}\text{EXT1}(\text{ARG})\text{EXT2}$   
                  •  $\backslash\text{txtparname}\{\text{Name}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Par}\}[\text{Ext2}] = \text{NAME}_{\text{SUB}}^{\text{SUP}}\text{EXT1}[\text{PAR}]\text{EXT2}$

722  $\%\%$  Style for Names  
723  $\backslash\text{cmdtxtall}\{\text{name}\}$   
724  $\backslash\text{DeclareRobustCommand}\{\backslash\text{txtstytname}\}\{\backslash\text{normalfont}\backslash\text{mdseries}\backslash\text{scshape}\backslash\text{sffamily}\}$

$\backslash\text{cmdtxtname}$  ... to do!

- $\backslash\text{cmdtxtname}\{\text{cmdName}\};$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext}] = \text{CMDNAME}_{\text{SUB}}^{\text{SUB}}\text{EXT}$
- $\backslash\text{cmdtxtname}\{\text{cmdName}\}[\text{newName}];$   
 $\backslash\text{cmdName}[\text{sub}][\text{sub}][\text{ext}] = \text{NEWNAME}_{\text{SUB}}^{\text{SUB}}\text{EXT}$

725  $\backslash\text{DeclareRobustCommand}\{\backslash\text{cmdtxtname}\}[2][2=]$   
726  $\{\backslash\text{usrtxt}\{\#1\}\{\}\{\text{name}\}\{\#2\}\}$

$\backslash\text{cmdtxtargname}$  ... to do!

- `\cmdtxtargname{cmdName};`  
`\cmdName[sub][sub][ext1]{arg}[ext2] = CMDNAMESUBSUBEXT1(ARG)EXT2`
- `\cmdtxtargname{cmdName}[newName];`  
`\cmdName[sub][sub][ext1]{arg}[ext2] = NEWNAMESUBSUBEXT1(ARG)EXT2`

727 `\DeclareRobustCommandx{\cmdtxtargname}[2][2=]`  
728 `{\usrtxt{#1}{-}{argname}[#2]}`

`\cmdtxttoargname ... to do!`

- `\cmdtxttoargname{cmdName};`  
`\cmdName[sub][sub][arg] = CMDNAMESUBSUB(ARG)`
- `\cmdtxttoargname{cmdName}[newName];`  
`\cmdName[sub][sub][arg] = NEWNAMESUBSUB(ARG)`

729 `\DeclareRobustCommandx{\cmdtxttoargname}[2][2=]`  
730 `{\usrtxt{#1}{-}{oargname}[#2]}`

`\cmdtxttparname ... to do!`

- `\cmdtxttparname{cmdName};`  
`\cmdName[sub][sub][ext1]{par}[ext2] = CMDNAMESUBSUBEXT1[PAR]EXT2`
- `\cmdtxttparname{cmdName}[newName];`  
`\cmdName[sub][sub][ext1]{par}[ext2] = NEWNAMESUBSUBEXT1[PAR]EXT2`

731 `\DeclareRobustCommandx{\cmdtxttparname}[2][2=]`  
732 `{\usrtxt{#1}{-}{parname}[#2]}`

`\cmdtxttoparname ... to do!`

- `\cmdtxttoparname{cmdName};`  
`\cmdName[sub][sub][par] = CMDNAMESUBSUB[PAR]`
- `\cmdtxttoparname{cmdName}[newName];`  
`\cmdName[sub][sub][par] = NEWNAMESUBSUB[PAR]`

733 `\DeclareRobustCommandx{\cmdtxttoparname}[2][2=]`  
734 `{\usrtxt{#1}{-}{oparname}[#2]}`

`\txtcom ... to do!`

`\txtargcom`    • `\txtcom{Name}[sub][sup][Ext] = NAMESUPSUBEXT`  
`\txtparcom`   • `\txtargcom{Name}[sub][sup][Ext1]{Arg}[Ext2] = NAMESUPSUBEXT1(ARG)EXT2`  
                • `\txtparcom{Name}[sub][sup][Ext1]{Par}[Ext2] = NAMESUPSUBEXT1[PAR]EXT2`

735 `%% Style for Complexities`  
736 `\cmdtxtall{com}`  
737 `\DeclareRobustCommand{\txtstycom}{\normalfont\mdseries\scshape\rmfamily}`

`\cmdtxtcom ... to do!`

- `\cmdtxtcom{cmdName};`  
`\cmdName[sub][sub][ext] = CMDNAMESUBSUBEXT`
- `\cmdtxtcom{cmdName}[newName];`  
`\cmdName[sub][sub][ext] = NEWNAMESUBSUBEXT`

738 `\DeclareRobustCommandx{\cmdtxtcom}[2][2=]`  
739 `{\usrtxt{#1}{-}{com}[#2]}`

`\cmdtxtargcom ... to do!`

- `\cmdtxtargcom{cmdName};`  
`\cmdName[sub][sub][ext1]{arg}[ext2] = CMDNAMESUBSUBEXT1(ARG)EXT2`
- `\cmdtxtargcom{cmdName}[newName];`  
`\cmdName[sub][sub][ext1]{arg}[ext2] = NEWNAMESUBSUBEXT1(ARG)EXT2`

740 `\DeclareRobustCommandx{\cmdtxtargcom}[2][2=]`  
741 `{\usrtxt{#1}{-}{argcom}[#2]}`

`\cmdtxttoargcom ... to do!`

```

 • \cmdtxtoargcom{cmdName};
 \cmdName[sub][sub][arg] = CMDNAMESUBSUB(ARG)
 • \cmdtxtoargcom{cmdName}[newName];
 \cmdName[sub][sub][arg] = NEWNAMESUBSUB(ARG)
742 \DeclareRobustCommandx{\cmdtxtoargcom}[2][2=]
743 {\usrtxt{#1}{-}{oargcom}[#2]}

\cmdtxtparcom ... to do!
 • \cmdtxtparcom{cmdName};
 \cmdName[sub][sub][ext1]{par}[ext2] = CMDNAMESUBSUBEXT1[PAR]EXT2
 • \cmdtxtparcom{cmdName}[newName];
 \cmdName[sub][sub][ext1]{par}[ext2] = NEWNAMESUBSUBEXT1[PAR]EXT2
744 \DeclareRobustCommandx{\cmdtxtparcom}[2][2=]
745 {\usrtxt{#1}{-}{parcom}[#2]}

\cmdtxtoparcom ... to do!
 • \cmdtxtoparcom{cmdName};
 \cmdName[sub][sub][par] = CMDNAMESUBSUB[PAR]
 • \cmdtxtoparcom{cmdName}[newName];
 \cmdName[sub][sub][par] = NEWNAMESUBSUB[PAR]
746 \DeclareRobustCommandx{\cmdtxtoparcom}[2][2=]
747 {\usrtxt{#1}{-}{oparcom}[#2]}

748 \fi
749 %*****%
750 %*****%
751 %** Math Macro Generators *****%
752 %*****%
753 \ifmthgen@

\mthname ... to do!
\mthargname • \mthname{NAME}[sub][sup][Ext] = $\mathcal{NAME}_{sub}^{sup}Ext$
\mthparname • \mthargname{NAME}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2$
 • \mthargname!{NAME}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2$
 • \mthparname{NAME}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1\left[Par^{Ex^{Ex}}\right]Ext2$
 • \mthparname!{NAME}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2$
754 %% Style for Names
755 \cmdmthall{name}
756 \DeclareRobustCommand{\mthstynname}{\mathcal}

\AName ... to do!
... A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
757 \seqoflatupp{Name}{mthname}

\cmdmthname ... to do!
 • \cmdmthname{CMDNAME};
 \CMDNAMEName[sub][sub][ext] = $CMDNAME_{sub}^{sub}ext$
 • \cmdmthname{cmdName}[NEWNAME];
 \cmdNameName[sub][sub][ext] = $\mathcal{NEWNAME}_{sub}^{sub}ext$
758 \DeclareRobustCommandx{\cmdmthname}[2][2=]
759 {\usrmth{#1}{Name}{name}[#2]}

\cmdmthargname ... to do!
 • \cmdmthargname{CMDNAME};
 \CMDNAMEName[sub][sub][ext1]{arg}[ext2] = $CMDNAME_{sub}^{sub}ext1(arg)ext2$

```

- $\backslash\text{cmdmthargname}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$

```
760 \DeclareRobustCommandx{\cmdmthargname}[2][2=]
761 {\usrmth\#1}\Name}\argname}\#2}
```

$\backslash\text{cmdmthoargname}$  ... to do!

- $\backslash\text{cmdmthoargname}\{\text{CMDNAME}\}$ ;  
 $\backslash\text{CMDNAMEName}[\text{sub}][\text{sub}][\text{arg}] = \mathcal{C}\mathcal{M}\mathcal{D}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}(\text{arg})$
- $\backslash\text{cmdmthoargname}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameName}[\text{sub}][\text{sub}][\text{arg}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}(\text{arg})$

```
762 \DeclareRobustCommandx{\cmdmthoargname}[2][2=]
763 {\usrmth\#1}\Name}\oargname}\#2}
```

$\backslash\text{cmdmthparname}$  ... to do!

- $\backslash\text{cmdmthparname}\{\text{CMDNAME}\}$ ;  
 $\backslash\text{CMDNAMEName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \mathcal{C}\mathcal{M}\mathcal{D}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$
- $\backslash\text{cmdmthparname}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameName}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$

```
764 \DeclareRobustCommandx{\cmdmthparname}[2][2=]
765 {\usrmth\#1}\Name}\parname}\#2}
```

$\backslash\text{cmdmthoparname}$  ... to do!

- $\backslash\text{cmdmthoparname}\{\text{CMDNAME}\}$ ;  
 $\backslash\text{CMDNAMEName}[\text{sub}][\text{sub}][\text{par}] = \mathcal{C}\mathcal{M}\mathcal{D}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}[\text{par}]$
- $\backslash\text{cmdmthoparname}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameName}[\text{sub}][\text{sub}][\text{par}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}[\text{par}]$

```
766 \DeclareRobustCommandx{\cmdmthoparname}[2][2=]
767 {\usrmth\#1}\Name}\oparname}\#2}
```

$\backslash\text{mthfam}$  ... to do!

```
\mthargfam
\mthparfam
```

- $\backslash\text{mthfam}\{\text{NAME}\}[\text{sub}][\text{sup}][\text{Ext}] = \mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sup}}\text{Ext}$
- $\backslash\text{mthargfam}\{\text{NAME}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Arg}^{\sim}\{\text{Ex}^{\sim}\{\text{Ex}\}\}\}[\text{Ext2}] = \mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sup}}\text{Ext1}\left(\text{Arg}^{\text{Ex}^{\text{Ex}}}\right)\text{Ext2}$
- $\backslash\text{mthargfam!}\{\text{NAME}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Arg}^{\sim}\{\text{Ex}^{\sim}\{\text{Ex}\}\}\}[\text{Ext2}] = \mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sup}}\text{Ext1}(\text{Arg}^{\text{Ex}^{\text{Ex}}})\text{Ext2}$
- $\backslash\text{mthparfam}\{\text{NAME}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Par}^{\sim}\{\text{Ex}^{\sim}\{\text{Ex}\}\}\}[\text{Ext2}] = \mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sup}}\text{Ext1}\left[\text{Par}^{\text{Ex}^{\text{Ex}}}\right]\text{Ext2}$
- $\backslash\text{mthparfam!}\{\text{NAME}\}[\text{sub}][\text{sup}][\text{Ext1}]\{\text{Par}^{\sim}\{\text{Ex}^{\sim}\{\text{Ex}\}\}\}[\text{Ext2}] = \mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sup}}\text{Ext1}[\text{Par}^{\text{Ex}^{\text{Ex}}}]\text{Ext2}$

```
768 %% Style for Families
769 \cmdmthall{fam}
770 \DeclareRobustCommand{\mthstyfam}\mathscr
```

$\backslash\text{AFam}$  ... to do!

...  $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z}$

```
771 \seqoflatupp{Fam}\mthfam}
```

$\backslash\text{cmdmthfam}$  ... to do!

- $\backslash\text{cmdmthfam}\{\text{CMDNAME}\}$ ;  
 $\backslash\text{CMDNAMEFam}[\text{sub}][\text{sub}][\text{ext}] = \mathcal{C}\mathcal{M}\mathcal{D}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext}$
- $\backslash\text{cmdmthfam}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameFam}[\text{sub}][\text{sub}][\text{ext}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext}$

```
772 \DeclareRobustCommandx{\cmdmthfam}[2][2=]
773 {\usrmth\#1}\Fam}\fam}\#2}
```

$\backslash\text{cmdmthargfam}$  ... to do!

- $\backslash\text{cmdmthargfam}\{\text{CMDNAME}\}$ ;  
 $\backslash\text{CMDNAMEFam}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \mathcal{C}\mathcal{M}\mathcal{D}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$
- $\backslash\text{cmdmthargfam}\{\text{cmdName}\}[\text{NEWNAME}]$ ;  
 $\backslash\text{cmdNameFam}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \mathcal{N}\mathcal{E}\mathcal{W}\mathcal{N}\mathcal{A}\mathcal{M}\mathcal{E}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$

```

774 \DeclareRobustCommandx{\cmdmthargfam}[2][2=]
775 {\usrmth{#1}{Fam}{argfam}{#2}}

\cmdmthoargfam ... to do!
 • \cmdmthoargfam{CMDNAME};
 \CMDNAMEFam[sub][sub][arg] = $\mathcal{CMDNAME}_{sub}^{sub}(arg)$
 • \cmdmthoargfam{cmdFam}{NEWNAME};
 \cmdFamFam[sub][sub][arg] = $\mathcal{NEWNAME}_{sub}^{sub}(arg)$
776 \DeclareRobustCommandx{\cmdmthoargfam}[2][2=]
777 {\usrmth{#1}{Fam}{oargfam}{#2}}

\cmdmthparfam ... to do!
 • \cmdmthparfam{CMDNAME};
 \CMDNAMEFam[sub][sub][ext1]{par}[ext2] = $\mathcal{CMDNAME}_{sub}^{sub}ext1[par]ext2$
 • \cmdmthparfam{cmdName}{NEWNAME};
 \cmdNameFam[sub][sub][ext1]{par}[ext2] = $\mathcal{NEWNAME}_{sub}^{sub}ext1[par]ext2$
778 \DeclareRobustCommandx{\cmdmthparfam}[2][2=]
779 {\usrmth{#1}{Fam}{parfam}{#2}}

\cmdmthoparfam ... to do!
 • \cmdmthoparfam{CMDNAME};
 \CMDNAMEFam[sub][sub][par] = $\mathcal{CMDNAME}_{sub}^{sub}[par]$
 • \cmdmthoparfam{cmdFam}{NEWNAME};
 \cmdFamFam[sub][sub][par] = $\mathcal{NEWNAME}_{sub}^{sub}[par]$
780 \DeclareRobustCommandx{\cmdmthoparfam}[2][2=]
781 {\usrmth{#1}{Fam}{oparfam}{#2}}

\mthcls ... to do!
\mthargcls
\mthparcls
 • \mthcls{NAME}[sub][sup][Ext] = $\mathcal{NAME}_{sub}^{sup}Ext$
 • \mthargcls{NAME}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2$
 • \mthargcls!{NAME}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1(Arg^{Ex^{Ex}})Ext2$
 • \mthparcls{NAME}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2$
 • \mthparcls!{NAME}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\mathcal{NAME}_{sub}^{sup}Ext1[Par^{Ex^{Ex}}]Ext2$
782 %% Style for Classes
783 \cmdmthall{cls}
784 \DeclareRobustCommand{\mthstyccls}{\matheus}

\ACls ... to do!
... A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
785 \seqoflatupp{Cls}{mthcls}

\cmdmthcls ... to do!
 • \cmdmthcls{CMDNAME};
 \CMDNAMECls[sub][sub][ext] = $\mathcal{CMDNAME}_{sub}^{sub}ext$
 • \cmdmthcls{cmdName}{NEWNAME};
 \cmdNameCls[sub][sub][ext] = $\mathcal{NEWNAME}_{sub}^{sub}ext$
786 \DeclareRobustCommandx{\cmdmthcls}[2][2=]
787 {\usrmth{#1}{Cls}{cls}{#2}}

\cmdmthargcls ... to do!
 • \cmdmthargcls{CMDNAME};
 \CMDNAMECls[sub][sub][ext1]{arg}[ext2] = $\mathcal{CMDNAME}_{sub}^{sub}ext1(arg)ext2$
 • \cmdmthargcls{cmdName}{NEWNAME};
 \cmdNameCls[sub][sub][ext1]{arg}[ext2] = $\mathcal{NEWNAME}_{sub}^{sub}ext1(arg)ext2$
788 \DeclareRobustCommandx{\cmdmthargcls}[2][2=]
789 {\usrmth{#1}{Cls}{argcls}{#2}}

```

```

\cmdmthoargcls ... to do!
 • \cmdmthoargcls{CMDNAME};
 \CMDNAMECcls[sub][sub][arg] = \CMDNAME\mathcal{E}_{sub}^{sub}(arg)
 • \cmdmthoargcls{cmdCcls}[NEWNAME];
 \cmdCclsCcls[sub][sub][arg] = \NEWNAME\mathcal{E}_{sub}^{sub}(arg)
790 \DeclareRobustCommandx{\cmdmthoargcls}[2][2=]
791 {\usrmth{#1}{Ccls}{oargcls}{#2}}

\cmdmthparcls ... to do!
 • \cmdmthparcls{CMDNAME};
 \CMDNAMECcls[sub][sub][ext1]{par}[ext2] = \CMDNAME\mathcal{E}_{sub}^{sub}ext1[par]ext2
 • \cmdmthparcls{cmdName}[NEWNAME];
 \cmdNameCcls[sub][sub][ext1]{par}[ext2] = \NEWNAME\mathcal{E}_{sub}^{sub}ext1[par]ext2
792 \DeclareRobustCommandx{\cmdmthparcls}[2][2=]
793 {\usrmth{#1}{Ccls}{parcls}{#2}}

\cmdmthoparcls ... to do!
 • \cmdmthoparcls{CMDNAME};
 \CMDNAMECcls[sub][sub][par] = \CMDNAME\mathcal{E}_{sub}^{sub}[par]
 • \cmdmthoparcls{cmdCcls}[NEWNAME];
 \cmdCclsCcls[sub][sub][par] = \NEWNAME\mathcal{E}_{sub}^{sub}[par]
794 \DeclareRobustCommandx{\cmdmthoparcls}[2][2=]
795 {\usrmth{#1}{Ccls}{oparcls}{#2}}

\mthsig ... to do!
\mthargsig
\mthparsig
 • \mthsig{Name}[sub][sup][Ext] = \mathcal{N}_{ame_{sub}^{sup}}Ext
 • \mthargsig{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \mathcal{N}_{ame_{sub}^{sup}}Ext1\left(Arg^{Ex^{Ex}}\right)Ext2
 • \mthargsig!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \mathcal{N}_{ame_{sub}^{sup}}Ext1(Arg^{Ex^{Ex}})Ext2
 • \mthparsig{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = \mathcal{N}_{ame_{sub}^{sup}}Ext1\left[Par^{Ex^{Ex}}\right]Ext2
 • \mthparsig!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = \mathcal{N}_{ame_{sub}^{sup}}Ext1[Par^{Ex^{Ex}}]Ext2
796 %% Style for Signatures
797 \cmdmthall{sig}
798 \DeclareRobustCommand{\mthstysig}{\mathpzc}

\asig ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
799 \seqoflatlet{Sig}{mthsig}\seqofgrklow{Sig}{mthsig}

\cmdmthsig ... to do!
 • \cmdmthsig{cmdName};
 \cmdNameSig[sub][sub][ext] = \cmd\mathcal{N}_{ame_{sub}^{sub}}ext
 • \cmdmthsig{cmdName}[NewName];
 \cmdNameSig[sub][sub][ext] = \new\mathcal{N}_{ame_{sub}^{sub}}ext
800 \DeclareRobustCommandx{\cmdmthsig}[2][2=]
801 {\usrmth{#1}{Sig}{sig}{#2}}

\cmdmthargsig ... to do!
 • \cmdmthargsig{cmdName};
 \cmdNameSig[sub][sub][ext1]{arg}[ext2] = \cmd\mathcal{N}_{ame_{sub}^{sub}}ext1(arg)ext2
 • \cmdmthargsig{cmdName}[NewName];
 \cmdNameSig[sub][sub][ext1]{arg}[ext2] = \new\mathcal{N}_{ame_{sub}^{sub}}ext1(arg)ext2
802 \DeclareRobustCommandx{\cmdmthargsig}[2][2=]
803 {\usrmth{#1}{Sig}{argsig}{#2}}

```

```

\cmdmthoargsig ... to do!
 • \cmdmthoargsig{cmdName};
 \cmdNameSig[sub][sub][arg] = cmd\namesub(arg)
 • \cmdmthoargsig{cmdSig}[NewName];
 \cmdSigSig[sub][sub][arg] = \ew\namesub(arg)
804 \DeclareRobustCommandx{\cmdmthoargsig}[2][2=]
805 {\usrmth{#1}{Sig}{oargsig}{#2}}

\cmdmthparsig ... to do!
 • \cmdmthparsig{cmdName};
 \cmdNameSig[sub][sub][ext1]{par}[ext2] = cmd\namesubext1[par]ext2
 • \cmdmthparsig{cmdName}[NewName];
 \cmdNameSig[sub][sub][ext1]{par}[ext2] = \ew\namesubext1[par]ext2
806 \DeclareRobustCommandx{\cmdmthparsig}[2][2=]
807 {\usrmth{#1}{Sig}{parsig}{#2}}

\cmdmthoparsig ... to do!
 • \cmdmthoparsig{cmdName};
 \cmdNameSig[sub][sub][par] = cmd\namesub[par]
 • \cmdmthoparsig{cmdSig}[NewName];
 \cmdSigSig[sub][sub][par] = \ew\namesub[par]
808 \DeclareRobustCommandx{\cmdmthoparsig}[2][2=]
809 {\usrmth{#1}{Sig}{oparsig}{#2}}

\mthstr ... to do!
\mthargstr
\mthparstr
 • \mthstr{Name}[sub][sup][Ext] = \namesupExt
 • \mthargstr{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \namesupExt1(ArgExEx)Ext2
 • \mthargstr!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = \namesupExt1(ArgExEx)Ext2
 • \mthparstr{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = \namesupExt1[ParExEx]Ext2
 • \mthparstr!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = \namesupExt1[ParExEx]Ext2
810 %% Style for Structures
811 \cmdmthall{str}
812 \DeclareRobustCommand{\mthstystyr}{\mathfrak}

\astr ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
813 \seqoflatlet{Str}{mthstr}\seqofgrklow{Str}{mthstr}

\cmdmthstr ... to do!
 • \cmdmthstr{cmdName};
 \cmdNameStr[sub][sub][ext] = cmd\namesubext
 • \cmdmthstr{cmdName}[NewName];
 \cmdNameStr[sub][sub][ext] = \ew\namesubext
814 \DeclareRobustCommandx{\cmdmthstr}[2][2=]
815 {\usrmth{#1}{Str}{str}{#2}}

\cmdmthargstr ... to do!
 • \cmdmthargstr{cmdName};
 \cmdNameStr[sub][sub][ext1]{arg}[ext2] = cmd\namesubext1(arg)ext2
 • \cmdmthargstr{cmdName}[NewName];
 \cmdNameStr[sub][sub][ext1]{arg}[ext2] = \ew\namesubext1(arg)ext2
816 \DeclareRobustCommandx{\cmdmthargstr}[2][2=]
817 {\usrmth{#1}{Str}{argstr}{#2}}

```



```

\cmdmthoargstr ... to do!
 • \cmdmthoargstr{cmdName};
 \cmdNameStr[sub][sub][arg] = cmdNamesubsub(arg)
 • \cmdmthoargstr{cmdStr}[NewName];
 \cmdStrStr[sub][sub][arg] = NewNamesubsub(arg)
818 \DeclareRobustCommandx{\cmdmthoargstr}[2][2=]
819 {\usrmth{#1}{Str}{oargstr}{#2}}

\cmdmthparstr ... to do!
 • \cmdmthparstr{cmdName};
 \cmdNameStr[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2
 • \cmdmthparstr{cmdName}[NewName];
 \cmdNameStr[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2
820 \DeclareRobustCommandx{\cmdmthparstr}[2][2=]
821 {\usrmth{#1}{Str}{parstr}{#2}}

\cmdmthoparstr ... to do!
 • \cmdmthoparstr{cmdName};
 \cmdNameStr[sub][sub][par] = cmdNamesubsub[par]
 • \cmdmthoparstr{cmdStr}[NewName];
 \cmdStrStr[sub][sub][par] = NewNamesubsub[par]
822 \DeclareRobustCommandx{\cmdmthoparstr}[2][2=]
823 {\usrmth{#1}{Str}{oparstr}{#2}}

\mthset ... to do!
\mthargset
\mthparset
 • \mthset{Name}[sub][sup][Ext] = NamesupsubExt
 • \mthargset{Name}[sub][sup][Ext1]{ArgEx{Ex}{Ex}}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthargset!{Name}[sub][sup][Ext1]{ArgEx{Ex}{Ex}}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthparset{Name}[sub][sup][Ext1]{ParEx{Ex}{Ex}}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
 • \mthparset!{Name}[sub][sup][Ext1]{ParEx{Ex}{Ex}}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
824 %% Style for Sets
825 \cmdmthall{set}
826 \DeclareRobustCommand{\mthstyset}{\mathrm}

\aset ... to do!
\mthargset a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
\mthparset A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
 α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϑ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
 Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Θ, Ι, Κ, Κ, Λ, Μ, Ν, Ξ, Ο, Π, ΙΙ, Ρ, Ρ, Σ, Σ, Τ, Υ, Φ, Φ, Χ, Ψ, Ω
827 \seqoflet{Set}{mthset}

\cmdmthset ... to do!
 • \cmdmthset{cmdName};
 \cmdNameSet[sub][sub][ext] = cmdNamesubsubext
 • \cmdmthset{cmdName}[NewName];
 \cmdNameSet[sub][sub][ext] = NewNamesubsubext
828 \DeclareRobustCommandx{\cmdmthset}[2][2=]
829 {\usrmth{#1}{Set}{set}{#2}}

\cmdmthargset ... to do!
 • \cmdmthargset{cmdName};
 \cmdNameSet[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2
 • \cmdmthargset{cmdName}[NewName];
 \cmdNameSet[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2
830 \DeclareRobustCommandx{\cmdmthargset}[2][2=]
831 {\usrmth{#1}{Set}{argset}{#2}}

```

\cmdmthoargset ... to do!

```

• \cmdmthoargset{cmdName};
 \cmdNameSet[sub][sub][arg] = cmdNamesub(arg)

• \cmdmthoargset{cmdSet}[NewName];
 \cmdSetSet[sub][sub][arg] = NewNamesub(arg)

832 \DeclareRobustCommandx{\cmdmthoargset}[2][2=]
833 {\usrmth{#1}{Set}{oargset}[#2]}

```

\cmdmthparset ... to do!

```

• \cmdmthparset{cmdName};
 \cmdNameSet[sub][sub][ext1]{par}[ext2] = cmdNamesubext1[par]ext2

• \cmdmthparset{cmdName}[NewName];
 \cmdNameSet[sub][sub][ext1]{par}[ext2] = NewNamesubext1[par]ext2

834 \DeclareRobustCommandx{\cmdmthparset}[2][2=]
835 {\usrmth{#1}{Set}{parset}[#2]}

```

\cmdmthoparset ... to do!

```

• \cmdmthoparset{cmdName};
 \cmdNameSet[sub][sub][par] = cmdNamesub[par]

• \cmdmthoparset{cmdSet}[NewName];
 \cmdSetSet[sub][sub][par] = NewNamesub[par]

836 \DeclareRobustCommandx{\cmdmthoparset}[2][2=]
837 {\usrmth{#1}{Set}{oparset}[#2]}

```

\cmdmthsetext ... to do!

```

838 \DeclareRobustCommandx{\cmdmthsetext}[3][2=, 3=]
839 {\cmdmthset{#1}[#2]\caselower[q]{#1}%
840 \usrmthlet{\thestring}{Sym}{sym}
841 [\defval{#3}{\defval{\empchk{#2}{\lowercase{#2}}}{\thestring}}}%
842 \usrmthlet{\thestring}{Elm}{elm}
843 [\defval{#3}{\defval{\empchk{#2}{\lowercase{#2}}}{\thestring}}]}

```

\mthrel ... to do!

```

\mthargrel
\mthparrel
• \mthrel{Name}[sub][sup][Ext] = NamesupsubExt

• \mthargrel{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2

• \mthargrel!{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2

• \mthparrel{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2

• \mthparrel!{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2

844 %% Style for Relations
845 \cmdmthall{rel}
846 \DeclareRobustCommand{\mthstyrel}{\mathit}

```

\aRel ... to do!

```

... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, L, T, U, V, W, X, Y, Z
α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϑ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
A, B, Γ, Δ, E, E, Z, H, Θ, Θ, I, K, K, A, M, N, Ξ, O, Π, Π, P, P, Σ, Σ, T, Υ, Φ, Φ, X, Ψ, Ω

847 \seqoflet{Rel}{mthrel}

```

\cmdmthrel ... to do!

```

• \cmdmthrel{cmdName};
 \cmdNameRel[sub][sub][ext] = cmdNamesubsubext

• \cmdmthrel{cmdName}[NewName];
 \cmdNameRel[sub][sub][ext] = NewNamesubsubext

848 \DeclareRobustCommandx{\cmdmthrel}[2][2=]
849 {\usrmth{#1}{Rel}{rel}[#2]}

```

```

\cmdmthargrel ... to do!
 • \cmdmthargrel{cmdName};
 \cmdNameRel[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2
 • \cmdmthargrel{cmdName}[NewName];
 \cmdNameRel[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2
850 \DeclareRobustCommandx{\cmdmthargrel}[2][2=]
851 {\usrmth{#1}{Rel}{argrel}{#2}}

\cmdmthoargrel ... to do!
 • \cmdmthoargrel{cmdName};
 \cmdNameRel[sub][sub][arg] = cmdNamesubsub(arg)
 • \cmdmthoargrel{cmdRel}[NewName];
 \cmdRelRel[sub][sub][arg] = NewNamesubsub(arg)
852 \DeclareRobustCommandx{\cmdmthoargrel}[2][2=]
853 {\usrmth{#1}{Rel}{oargrel}{#2}}

\cmdmthparrel ... to do!
 • \cmdmthparrel{cmdName};
 \cmdNameRel[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2
 • \cmdmthparrel{cmdName}[NewName];
 \cmdNameRel[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2
854 \DeclareRobustCommandx{\cmdmthparrel}[2][2=]
855 {\usrmth{#1}{Rel}{parrel}{#2}}

\cmdmthoparrel ... to do!
 • \cmdmthoparrel{cmdName};
 \cmdNameRel[sub][sub][par] = cmdNamesubsub[par]
 • \cmdmthoparrel{cmdRel}[NewName];
 \cmdRelRel[sub][sub][par] = NewNamesubsub[par]
856 \DeclareRobustCommandx{\cmdmthoparrel}[2][2=]
857 {\usrmth{#1}{Rel}{oparrel}{#2}}

\mthfun ... to do!
\mthargfun • \mthfun{Name}[sub][sup][Ext] = NamesupsubExt
\mthparfun • \mthargfun{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthargfun!{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthparfun{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
 • \mthparfun!{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
858 %% Style for Functions
859 \cmdmthall{fun}
860 \DeclareRobustCommand{\mthstyfun}{\mathsf{f}}

\mathFun ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
 $\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \varkappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega$
A, B, $\Gamma, \Delta, E, \Xi, Z, H, \Theta, \varnothing, I, K, \Lambda, M, N, \Xi, O, \Pi, \textit{II}, P, \Sigma, \varSigma, T, \Upsilon, \Phi, \varPhi, X, \Psi, \Omega$
861 \seqoflet{Fun}{mthfun}

\cmdmthfun ... to do!
 • \cmdmthfun{cmdName};
 \cmdNameFun[sub][sub][ext] = cmdNamesubsubext
 • \cmdmthfun{cmdName}[NewName];
 \cmdNameFun[sub][sub][ext] = NewNamesubsubext
862 \DeclareRobustCommandx{\cmdmthfun}[2][2=]
863 {\usrmth{#1}{Fun}{fun}{#2}}

```

```

\cmdmthargfun ... to do!
 • \cmdmthargfun{cmdName};
 \cmdNameFun[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2
 • \cmdmthargfun{cmdName}[NewName];
 \cmdNameFun[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2
864 \DeclareRobustCommandx{\cmdmthargfun}[2][2=]
865 {\usrmth{#1}{Fun}{argfun}[#2]}

\cmdmthoargfun ... to do!
 • \cmdmthoargfun{cmdName};
 \cmdNameFun[sub][sub][arg] = cmdNamesubsub(arg)
 • \cmdmthoargfun{cmdFun}[NewName];
 \cmdFunFun[sub][sub][arg] = NewNamesubsub(arg)
866 \DeclareRobustCommandx{\cmdmthoargfun}[2][2=]
867 {\usrmth{#1}{Fun}{oargfun}[#2]}

\cmdmthparfun ... to do!
 • \cmdmthparfun{cmdName};
 \cmdNameFun[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2
 • \cmdmthparfun{cmdName}[NewName];
 \cmdNameFun[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2
868 \DeclareRobustCommandx{\cmdmthparfun}[2][2=]
869 {\usrmth{#1}{Fun}{parfun}[#2]}

\cmdmthoparfun ... to do!
 • \cmdmthoparfun{cmdName};
 \cmdNameFun[sub][sub][par] = cmdNamesubsub[par]
 • \cmdmthoparfun{cmdFun}[NewName];
 \cmdFunFun[sub][sub][par] = NewNamesubsub[par]
870 \DeclareRobustCommandx{\cmdmthoparfun}[2][2=]
871 {\usrmth{#1}{Fun}{oparfun}[#2]}

\mthsym ... to do!
\mthargsym • \mthsym{Name}[sub][sup][Ext] = NamesupsubExt
\mthparsym • \mthargsym{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthargsym!{Name}[sub][sup][Ext1]{ArgEx{Ex}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthparsym{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
 • \mthparsym!{Name}[sub][sup][Ext1]{ParEx{Ex}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
872 %% Style for Symbols
873 \cmdmthall{sym}
874 \DeclareRobustCommand{\mthstysym}{\mathtt}

\asym ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \varkappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega$
A, B, $\Gamma, \Delta, E, \Xi, Z, H, \Theta, \ominus, I, K, \Lambda, M, N, \Xi, O, \Pi, \textit{II}, P, \Sigma, \textit{\textit{\Sigma}}, T, \Phi, \phi, X, \Psi, \Omega$
875 \seqoflet{Sym}{mthsym}

\cmdmthsym ... to do!
 • \cmdmthsym{cmdName};
 \cmdNameSym[sub][sub][ext] = cmdNamesubsubext
 • \cmdmthsym{cmdName}[NewName];
 \cmdNameSym[sub][sub][ext] = NewNamesubsubext
876 \DeclareRobustCommandx{\cmdmthsym}[2][2=]
877 {\usrmth{#1}{Sym}{sym}[#2]}

```

```

\cmdmthargsym ... to do!
 • \cmdmthargsym{cmdName};
 \cmdNameSym[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2
 • \cmdmthargsym{cmdName}[NewName];
 \cmdNameSym[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2
878 \DeclareRobustCommandx{\cmdmthargsym}[2][2=]
879 {\usrmth{#1}{Sym}{argsym}[#2]}

\cmdmthoargsym ... to do!
 • \cmdmthoargsym{cmdName};
 \cmdNameSym[sub][sub][arg] = cmdNamesubsub(arg)
 • \cmdmthoargsym{cmdSym}[NewName];
 \cmdSymSym[sub][sub][arg] = NewNamesubsub(arg)
880 \DeclareRobustCommandx{\cmdmthoargsym}[2][2=]
881 {\usrmth{#1}{Sym}{oargsym}[#2]}

\cmdmthparsym ... to do!
 • \cmdmthparsym{cmdName};
 \cmdNameSym[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2
 • \cmdmthparsym{cmdName}[NewName];
 \cmdNameSym[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2
882 \DeclareRobustCommandx{\cmdmthparsym}[2][2=]
883 {\usrmth{#1}{Sym}{parsym}[#2]}

\cmdmthoparsym ... to do!
 • \cmdmthoparsym{cmdName};
 \cmdNameSym[sub][sub][par] = cmdNamesubsub[par]
 • \cmdmthoparsym{cmdSym}[NewName];
 \cmdSymSym[sub][sub][par] = NewNamesubsub[par]
884 \DeclareRobustCommandx{\cmdmthoparsym}[2][2=]
885 {\usrmth{#1}{Sym}{oparsym}[#2]}

\mthelm ... to do!
\mthargelm • \mthelm{Name}[sub][sup][Ext] = NamesupsubExt
\mthparelm • \mthargelm{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthargelm!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = NamesupsubExt1(ArgExEx)Ext2
 • \mthparelm{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
 • \mthparelm!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = NamesupsubExt1[ParExEx]Ext2
886 %% Style for Elements
887 \cmdmthall{elm}
888 \DeclareRobustCommand{\mthstyelm}{\mathnormal}

\aeElm ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Θ, Ι, Κ, Κ, Λ, Μ, Ν, Ξ, Ο, Η, Η, Ρ, Ρ, Σ, Σ, Τ, Υ, Φ, Φ, Χ, Ψ, Ω
889 \seqoflet{Elm}{mthelm}

\cmdmthelm ... to do!
 • \cmdmthelm{cmdName};
 \cmdNameElm[sub][sub][ext] = cmdNamesubsubext
 • \cmdmthelm{cmdName}[NewName];
 \cmdNameElm[sub][sub][ext] = NewNamesubsubext
890 \DeclareRobustCommandx{\cmdmthelm}[2][2=]
891 {\usrmth{#1}{Elm}{elm}[#2]}

```

\cmdmthargelm ... to do!

- \cmdmthargelm{cmdName};  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$
- \cmdmthargelm{cmdName}[NewName];  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$

892 \DeclareRobustCommandx{\cmdmthargelm}[2][2=]  
893 {\usrmth{\#1}\Elm}\argelm{\#2}}

\cmdmthoargelm ... to do!

- \cmdmthoargelm{cmdName};  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{arg}] = \text{cmdName}_{\text{sub}}^{\text{sub}}(\text{arg})$
- \cmdmthoargelm{cmdElm}[NewName];  
 $\text{cmdElmElm}[\text{sub}][\text{sub}][\text{arg}] = \text{NewName}_{\text{sub}}^{\text{sub}}(\text{arg})$

894 \DeclareRobustCommandx{\cmdmthoargelm}[2][2=]  
895 {\usrmth{\#1}\Elm}\oargelm{\#2}}

\cmdmthparelm ... to do!

- \cmdmthparelm{cmdName};  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$
- \cmdmthparelm{cmdName}[NewName];  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext1}[\text{par}]\text{ext2}$

896 \DeclareRobustCommandx{\cmdmthparelm}[2][2=]  
897 {\usrmth{\#1}\Elm}\parelm{\#2}}

\cmdmthoparelm ... to do!

- \cmdmthoparelm{cmdName};  
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{par}] = \text{cmdName}_{\text{sub}}^{\text{sub}}[\text{par}]$
- \cmdmthoparelm{cmdElm}[NewName];  
 $\text{cmdElmElm}[\text{sub}][\text{sub}][\text{par}] = \text{NewName}_{\text{sub}}^{\text{sub}}[\text{par}]$

898 \DeclareRobustCommandx{\cmdmthoparelm}[2][2=]  
899 {\usrmth{\#1}\Elm}\oparelm{\#2}}

900 %%\*\*\*\*\*%

\cmdmthsymelm ... to do!

- \cmdmthsymelm{cmdName};  
 $\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext}$   
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext}$
- \cmdmthsymelm{cmdName}[NewName];  
 $\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext}$   
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext}$

901 \DeclareRobustCommandx{\cmdmthsymelm}[2][2=]  
902 {\cmdmthsym{\#1}\#2}%  
903 \cmdmthelm{\#1}\#2}}

\cmdmthargsymelm ... to do!

- \cmdmthargsymelm{cmdName};  
 $\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$   
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{cmdName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$
- \cmdmthargsymelm{cmdName}[NewName];  
 $\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$   
 $\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{arg}\}[\text{ext2}] = \text{NewName}_{\text{sub}}^{\text{sub}}\text{ext1}(\text{arg})\text{ext2}$

904 \DeclareRobustCommandx{\cmdmthargsymelm}[2][2=]  
905 {\cmdmthargsym{\#1}\#2}%  
906 \cmdmthargelm{\#1}\#2}}

\cmdmthoargsymelm ... to do!

- $\backslash\text{cmdmthoargsymelm}\{\text{cmdName}\};$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{arg}] = \text{cmdName}_{sub}^{sub}(\text{arg})$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{arg}] = \text{cmdName}_{sub}^{sub}(\text{arg})$
- $\backslash\text{cmdmthoargsymelm}\{\text{cmdName}\}[\text{NewName}];$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{arg}] = \text{NewName}_{sub}^{sub}(\text{arg})$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{arg}] = \text{NewName}_{sub}^{sub}(\text{arg})$

```

907 \DeclareRobustCommandx{\cmdmthoargsymelm}[2][2=]
908 {\cmdmthoargsym\#1}\#2}%
909 \cmdmthoargelm\#1}\#2}%

```

$\backslash\text{cmdmthparsymelm}$  ... to do!

- $\backslash\text{cmdmthparsymelm}\{\text{cmdName}\};$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{cmdName}_{sub}^{sub}\text{ext1}[\text{par}]\text{ext2}$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{cmdName}_{sub}^{sub}\text{ext1}[\text{par}]\text{ext2}$
- $\backslash\text{cmdmthparsymelm}\{\text{cmdName}\}[\text{NewName}];$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{NewName}_{sub}^{sub}\text{ext1}[\text{par}]\text{ext2}$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{ext1}]\{\text{par}\}[\text{ext2}] = \text{NewName}_{sub}^{sub}\text{ext1}[\text{par}]\text{ext2}$

```

910 \DeclareRobustCommandx{\cmdmthparsymelm}[2][2=]
911 {\cmdmthparsym\#1}\#2}%
912 \cmdmthparelm\#1}\#2}%

```

$\backslash\text{cmdmthoparsymelm}$  ... to do!

- $\backslash\text{cmdmthoparsymelm}\{\text{cmdName}\};$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{par}] = \text{cmdName}_{sub}^{sub}[\text{par}]$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{par}] = \text{cmdName}_{sub}^{sub}[\text{par}]$
- $\backslash\text{cmdmthoparsymelm}\{\text{cmdName}\}[\text{NewName}];$   
 $\backslash\text{cmdNameSym}[\text{sub}][\text{sub}][\text{par}] = \text{NewName}_{sub}^{sub}[\text{par}]$   
 $\backslash\text{cmdNameElm}[\text{sub}][\text{sub}][\text{par}] = \text{NewName}_{sub}^{sub}[\text{par}]$

```

913 \DeclareRobustCommandx{\cmdmthoparsymelm}[2][2=]
914 {\cmdmthoparsym\#1}\#2}%
915 \cmdmthoparelm\#1}\#2}%
916 %%*****%

```

$\backslash\text{mthluop}$  ... to do!

$\backslash\text{mthlbop}$

- $\backslash\text{mthluop}\{\backslash\text{oplus}\}[\text{sub}][\text{sup}][\text{Ext}] = \bigoplus_{sub}^{sup} \text{Ext}$
- $\backslash\text{mthlbop}\{\backslash\text{oplus}\}[\text{sub}][\text{sup}][\text{Ext}] = \bigoplus_{sub}^{sup} \text{Ext}$

```

917 %% Style for \LaTeX Operators
918 \cmdmth{luop}
919 \DeclareRobustCommand{\mthstyluop}[1]{\textstyle\mathop{\#1}}
920 \cmdmth{lbop}
921 \DeclareRobustCommand{\mthstylbop}[1]{\textstyle\mathbin{\#1}}

```

$\backslash\text{cmdmthluop}$  ... to do!

$\backslash\text{cmdmthlbop}$

- $\backslash\text{cmdmthluop}\{\text{cmdName}\};$   
 $\backslash\text{cmdNameUOp}[\text{sub}][\text{sub}][\text{ext}] = \text{cmdName}_{sub}^{sub}\text{ext}$
- $\backslash\text{cmdmthluop}\{\text{cmdName}\}[\backslash\text{oplus}];$   
 $\backslash\text{cmdNameUOp}[\text{sub}][\text{sub}][\text{ext}] = \bigoplus_{sub}^{sub} \text{ext}$
- $\backslash\text{cmdmthlbop}\{\text{cmdName}\};$   
 $\backslash\text{cmdNameBOp}[\text{sub}][\text{sub}][\text{ext}] = \text{cmdName}_{sub}^{sub}\text{ext}$
- $\backslash\text{cmdmthlbop}\{\text{cmdName}\}[\backslash\text{oplus}];$   
 $\backslash\text{cmdNameBOp}[\text{sub}][\text{sub}][\text{ext}] = \bigoplus_{sub}^{sub} \text{ext}$

```

922 \DeclareRobustCommandx{\cmdmthluop}[2][2=]
923 {\usrmth\#1}\{UOp}\{luop}\#2}%
924 \DeclareRobustCommandx{\cmdmthlbop}[2][2=]
925 {\usrmth\#1}\{BOp}\{lbop}\#2}%

```

$\backslash\text{mthlrel}$  ... to do!

- $\backslash\text{mthlrel}\{\backslash\text{preceq}\}[\text{sub}][\text{sup}][\text{Ext}] = \preceq_{sub}^{sup} \text{Ext}$

```

926 %% Style for \LaTeX Relations
927 \cmdmth{lrel}
928 \DeclareRobustCommand{\mthstylrel}{\mathrel}

\cmdmthlrel ... to do!

 • \cmdmthlrel{cmdName};
 \cmdNameRel[sub][sub][ext] = $\text{cmdName}_{sub}^{sub} \text{ext}$

 • \cmdmthlrel{cmdName}[\preceq];
 \cmdNameRel[sub][sub][ext] = $\preceq_{sub}^{sub} \text{ext}$

929 \DeclareRobustCommandx{\cmdmthlrel}[2][2=]
930 {\usrmth{#1}{Rel}{lrel}{#2}}

931 %%%

\mthsnt ... to do!
\mthargsnt
\mthparsnt
 • \mthsnt{Name}[sub][sup][Ext] = $\text{Name}_{sub}^{sup} \text{Ext}$
 • \mthargsnt{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}(\text{Arg}^{\text{Ex}^{\text{Ex}}}) \text{Ext2}$
 • \mthargsnt!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}(\text{Arg}^{\text{Ex}^{\text{Ex}}}) \text{Ext2}$
 • \mthparsnt{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}[\text{Par}^{\text{Ex}^{\text{Ex}}}] \text{Ext2}$
 • \mthparsnt!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}[\text{Par}^{\text{Ex}^{\text{Ex}}}] \text{Ext2}$

932 %% Style for Sentences
933 \cmdmthall{snt}
934 \DeclareRobustCommand{\mthstysnt}{\mathsf}

\aSnt ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega$
A, B, $\Gamma, \Delta, E, Z, H, \Theta, \vartheta, I, K, \Lambda, M, N, \Xi, O, \Pi, \text{II}, P, \Sigma, \varSigma, T, \Upsilon, \Phi, \varPhi, X, \Psi, \Omega$

935 \seqoflet{Snt}{mthsnt}

\cmdmthsnt ... to do!

 • \cmdmthsnt{cmdName};
 \cmdNameSnt[sub][sub][ext] = $\text{cmdName}_{sub}^{sub} \text{ext}$

 • \cmdmthsnt{cmdName}[NewName];
 \cmdNameSnt[sub][sub][ext] = $\text{NewName}_{sub}^{sub} \text{ext}$

936 \DeclareRobustCommandx{\cmdmthsnt}[2][2=]
937 {\usrmth{#1}{Snt}{snt}{#2}}

\cmdmthargsnt ... to do!

 • \cmdmthargsnt{cmdName};
 \cmdNameSnt[sub][sub][ext1]{arg}[ext2] = $\text{cmdName}_{sub}^{sub} \text{ext1}(\text{arg}) \text{ext2}$

 • \cmdmthargsnt{cmdName}[NewName];
 \cmdNameSnt[sub][sub][ext1]{arg}[ext2] = $\text{NewName}_{sub}^{sub} \text{ext1}(\text{arg}) \text{ext2}$

938 \DeclareRobustCommandx{\cmdmthargsnt}[2][2=]
939 {\usrmth{#1}{Snt}{argsnt}{#2}}

\cmdmthoargsnt ... to do!

 • \cmdmthoargsnt{cmdName};
 \cmdNameSnt[sub][sub][arg] = $\text{cmdName}_{sub}^{sub}(\text{arg})$

 • \cmdmthoargsnt{cmdName}[NewName];
 \cmdNameSnt[sub][sub][arg] = $\text{NewName}_{sub}^{sub}(\text{arg})$

940 \DeclareRobustCommandx{\cmdmthoargsnt}[2][2=]
941 {\usrmth{#1}{Snt}{oargsnt}{#2}}

\cmdmthparsnt ... to do!

```



- `\cmdmthparsnt{cmdName};`  
`\cmdNameSnt[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2`
- `\cmdmthparsnt{cmdName}[NewName];`  
`\cmdNameSnt[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2`

942 `\DeclareRobustCommandx{\cmdmthparsnt}[2][2=]`  
943 `{\usrmth{#1}{Snt}{parsnt}{#2}}`

`\cmdmthoparsnt ... to do!`

- `\cmdmthoparsnt{cmdName};`  
`\cmdNameSnt[sub][sub][par] = cmdNamesubsub[par]`
- `\cmdmthoparsnt{cmdName}[NewName];`  
`\cmdNameSnt[sub][sub][par] = NewNamesubsub[par]`

944 `\DeclareRobustCommandx{\cmdmthoparsnt}[2][2=]`  
945 `{\usrmth{#1}{Snt}{oparsnt}{#2}}`

`\mthfrm ... to do!`

`\mthargfrm`     • `\mthfrm{Name}[sub][sup][Ext] = NamesupsubExt`  
`\mthparfrm`     • `\mthargfrm{Name}[sub][sup][Ext1]{ArgExEx}[Ext2] = NamesupsubExt1(ArgExEx)Ext2`  
                  • `\mthargfrm!{Name}[sub][sup][Ext1]{ArgExEx}[Ext2] = NamesupsubExt1(ArgExEx)Ext2`  
                  • `\mthparfrm{Name}[sub][sup][Ext1]{ParExEx}[Ext2] = NamesupsubExt1[ParExEx]Ext2`  
                  • `\mthparfrm!{Name}[sub][sup][Ext1]{ParExEx}[Ext2] = NamesupsubExt1[ParExEx]Ext2`

946 `%% Style for Formulae`  
947 `\cmdmthall{frm}`  
948 `\DeclareRobustCommand{\mthstyfrm}{\mathit}`

`\aFrm ... to do!`

... *a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z*  
*A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z*  
*α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϑ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω*  
*A, B, Γ, Δ, E, Z, H, Θ, Θ, I, K, K, A, M, N, Ξ, O, Π, Π, P, P, Σ, Σ, T, Υ, Φ, Φ, X, Ψ, Ω*

949 `\seqoflet{Frm}{mthfrm}`

`\cmdmthfrm ... to do!`

- `\cmdmthfrm{cmdName};`  
`\cmdNameFrm[sub][sub][ext] = cmdNamesubsubext`
- `\cmdmthfrm{cmdName}[NewName];`  
`\cmdNameFrm[sub][sub][ext] = NewNamesubsubext`

950 `\DeclareRobustCommandx{\cmdmthfrm}[2][2=]`  
951 `{\usrmth{#1}{Frm}{frm}{#2}}`

`\cmdmthargfrm ... to do!`

- `\cmdmthargfrm{cmdName};`  
`\cmdNameFrm[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2`
- `\cmdmthargfrm{cmdName}[NewName];`  
`\cmdNameFrm[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2`

952 `\DeclareRobustCommandx{\cmdmthargfrm}[2][2=]`  
953 `{\usrmth{#1}{Frm}{argfrm}{#2}}`

`\cmdmthoargfrm ... to do!`

- `\cmdmthoargfrm{cmdName};`  
`\cmdNameFrm[sub][sub][arg] = cmdNamesubsub(arg)`
- `\cmdmthoargfrm{cmdName}[NewName];`  
`\cmdNameFrm[sub][sub][arg] = NewNamesubsub(arg)`

954 `\DeclareRobustCommandx{\cmdmthoargfrm}[2][2=]`  
955 `{\usrmth{#1}{Frm}{oargfrm}{#2}}`

```

\cmdmthparfrm ... to do!
 • \cmdmthparfrm{cmdName};
 \cmdNameFrm[sub][sub][ext1]{par}[ext2] = $\text{cmdName}_{sub}^{sub} \text{ext1}[par] \text{ext2}$

 • \cmdmthparfrm{cmdName}[NewName];
 \cmdNameFrm[sub][sub][ext1]{par}[ext2] = $\text{NewName}_{sub}^{sub} \text{ext1}[par] \text{ext2}$

956 \DeclareRobustCommandx{\cmdmthparfrm}[2][2=]
957 {\usrmth{#1}{Frm}{parfrm}[#2]}

\cmdmthoparfrm ... to do!
 • \cmdmthoparfrm{cmdName};
 \cmdNameFrm[sub][sub][par] = $\text{cmdName}_{sub}^{sub}[par]$

 • \cmdmthoparfrm{cmdName}[NewName];
 \cmdNameFrm[sub][sub][par] = $\text{NewName}_{sub}^{sub}[par]$

958 \DeclareRobustCommandx{\cmdmthoparfrm}[2][2=]
959 {\usrmth{#1}{Frm}{oparfrm}[#2]}

960 %%*****%

\mthmat ... to do!
\mthargmat
\mthparmat
 • \mthmat{Name}[sub][sup][Ext] = $\text{Name}_{sub}^{sup} \text{Ext}$

 • \mthargmat{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}(\text{Arg}^{Ex^{Ex}}) \text{Ext2}$

 • \mthargmat!{Name}[sub][sup][Ext1]{Arg^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}(\text{Arg}^{Ex^{Ex}}) \text{Ext2}$

 • \mthparmat{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}[\text{Par}^{Ex^{Ex}}] \text{Ext2}$

 • \mthparmat!{Name}[sub][sup][Ext1]{Par^{Ex^{Ex}}}[Ext2] = $\text{Name}_{sub}^{sup} \text{Ext1}[\text{Par}^{Ex^{Ex}}] \text{Ext2}$

961 %% Style for Matrices
962 \cmdmthall{mat}
963 \DeclareRobustCommand{\mthstymat}[1]{\boldsymbol{\mathsf{#1}}}

\Mat ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
 $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega$
A, B, $\Gamma, \Delta, E, Z, H, \Theta, \Theta, I, K, K, \Lambda, M, N, \Xi, O, \Pi, \Pi, P, P, \Sigma, \Sigma, T, T, \Phi, \Phi, X, \Psi, \Omega$

964 \seqoflet{Mat}{mthmat}

\cmdmthmat ... to do!
 • \cmdmthmat{cmdName};
 \cmdNameMat[sub][sub][ext] = $\text{cmdName}_{sub}^{sub} \text{ext}$

 • \cmdmthmat{cmdName}[NewName];
 \cmdNameMat[sub][sub][ext] = $\text{NewName}_{sub}^{sub} \text{ext}$

965 \DeclareRobustCommandx{\cmdmthmat}[2][2=]
966 {\usrmth{#1}{Mat}{mat}[#2]}

\cmdmthargmat ... to do!
 • \cmdmthargmat{cmdName};
 \cmdNameMat[sub][sub][ext1]{arg}[ext2] = $\text{cmdName}_{sub}^{sub} \text{ext1}(arg) \text{ext2}$

 • \cmdmthargmat{cmdName}[NewName];
 \cmdNameMat[sub][sub][ext1]{arg}[ext2] = $\text{NewName}_{sub}^{sub} \text{ext1}(arg) \text{ext2}$

967 \DeclareRobustCommandx{\cmdmthargmat}[2][2=]
968 {\usrmth{#1}{Mat}{argmat}[#2]}

\cmdmthoargmat ... to do!
 • \cmdmthoargmat{cmdName};
 \cmdNameMat[sub][sub][arg] = $\text{cmdName}_{sub}^{sub}(arg)$

 • \cmdmthoargmat{cmdName}[NewName];
 \cmdNameMat[sub][sub][arg] = $\text{NewName}_{sub}^{sub}(arg)$

```

```

969 \DeclareRobustCommandx{\cmdmthoargmat}[2][2=]
970 {\usrmth{#1}\{Mat\}\{oargmat\}[#2]}

\cmdmthparmat ... to do!
 • \cmdmthparmat{cmdName};
 \cmdNameMat[sub][sub][ext1]{par}[ext2] = cmdNamesubsubext1[par]ext2
 • \cmdmthparmat{cmdName}[NewName];
 \cmdNameMat[sub][sub][ext1]{par}[ext2] = NewNamesubsubext1[par]ext2
971 \DeclareRobustCommandx{\cmdmthparmat}[2][2=]
972 {\usrmth{#1}\{Mat\}\{parmat\}[#2]}

\cmdmthoparmat ... to do!
 • \cmdmthoparmat{cmdName};
 \cmdNameMat[sub][sub][par] = cmdNamesubsub[par]
 • \cmdmthoparmat{cmdName}[NewName];
 \cmdNameMat[sub][sub][par] = NewNamesubsub[par]
973 \DeclareRobustCommandx{\cmdmthoparmat}[2][2=]
974 {\usrmth{#1}\{Mat\}\{oparmat\}[#2]}

\mthvec ... to do!
\mthargvec
\mthparvec
 • \mthvec{Name}[sub][sup][Ext] = NamesubsupExt
 • \mthargvec{Name}[sub][sup][Ext1]{ArgEx{Ex}}}[Ext2] = NamesubsupExt1(ArgExExt)Ext2
 • \mthargvec!{Name}[sub][sup][Ext1]{ArgEx{Ex}}}[Ext2] = NamesubsupExt1(ArgExExt)Ext2
 • \mthparvec{Name}[sub][sup][Ext1]{ParEx{Ex}}}[Ext2] = NamesubsupExt1[ParExExt]Ext2
 • \mthparvec!{Name}[sub][sup][Ext1]{ParEx{Ex}}}[Ext2] = NamesubsupExt1[ParExExt]Ext2
975 %% Style for Vectors
976 \cmdmthall{vec}
977 \DeclareRobustCommand{\mthstyvec}[1]{\boldsymbol{\mathit{#1}}}

\Vec ... to do!
... a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
α, β, γ, δ, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, ϕ, χ, ψ, ω
A, B, Γ, Δ, E, E, Z, H, Θ, Θ, I, K, K, Λ, M, N, Ξ, O, Π, Π, P, P, Σ, Σ, T, Υ, Φ, Φ, X, Ψ, Ω
978 \seqoflet{Vec}{mthvec}

\cmdmthvec ... to do!
 • \cmdmthvec{cmdName};
 \cmdNameVec[sub][sub][ext] = cmdNamesubsubext
 • \cmdmthvec{cmdName}[NewName];
 \cmdNameVec[sub][sub][ext] = NewNamesubsubext
979 \DeclareRobustCommandx{\cmdmthvec}[2][2=]
980 {\usrmth{#1}\{Vec\}\{vec\}[#2]}

\cmdmthargvec ... to do!
 • \cmdmthargvec{cmdName};
 \cmdNameVec[sub][sub][ext1]{arg}[ext2] = cmdNamesubsubext1(arg)ext2
 • \cmdmthargvec{cmdName}[NewName];
 \cmdNameVec[sub][sub][ext1]{arg}[ext2] = NewNamesubsubext1(arg)ext2
981 \DeclareRobustCommandx{\cmdmthargvec}[2][2=]
982 {\usrmth{#1}\{Vec\}\{argvec\}[#2]}

\cmdmthoargvec ... to do!
 • \cmdmthoargvec{cmdName};
 \cmdNameVec[sub][sub][arg] = cmdNamesubsub(arg)

```

```

 • \cmdmthoargvec{cmdName}[NewName];
 \cmdNameVec[sub][sub][arg] = NewNamesub(arg)
983 \DeclareRobustCommandx{\cmdmthoargvec}[2][2=]
984 {\usrmth{#1}{Vec}{oargvec}[#2]}

\cmdmthparvec ... to do!
 • \cmdmthparvec{cmdName};
 \cmdNameVec[sub][sub][ext1]{par}[ext2] = cmdNamesubext1[par]ext2
 • \cmdmthparvec{cmdName}[NewName];
 \cmdNameVec[sub][sub][ext1]{par}[ext2] = NewNamesubext1[par]ext2
985 \DeclareRobustCommandx{\cmdmthparvec}[2][2=]
986 {\usrmth{#1}{Vec}{parvec}[#2]}

\cmdmthoparvec ... to do!
 • \cmdmthoparvec{cmdName};
 \cmdNameVec[sub][sub][par] = cmdNamesub[par]
 • \cmdmthoparvec{cmdName}[NewName];
 \cmdNameVec[sub][sub][par] = NewNamesub[par]
987 \DeclareRobustCommandx{\cmdmthoparvec}[2][2=]
988 {\usrmth{#1}{Vec}{oparvec}[#2]}

989 \fi
990 %*****%
991 %*****%
992 %** Elementary Macros for Text *****%
993 %*****%
994 \iftxt@

\dotcheck • A\dotcheck A\dotcheck.A = A.A. A
995 \newrobustcmd{\dotcheck}
996 {\@ifnextchar.{\.\@}}

997 %** Latin Abbreviations *****%

\adhoc • \adhoc = ad hoc
998 \cmdtxtabr{adhoc}[ad hoc]

\afortiori • \afortiori = a fortiori
999 \cmdtxtabr{afortiori}[a fortiori]

\apriori • \apriori = a priori
1000 \cmdtxtabr{apriori}[a priori]

\aposteriori • \aposteriori = a posteriori
1001 \cmdtxtabr{aposteriori}[a posteriori]

\cf • \cf = cf.
1002 \cmdtxtabr{cf}[cf.\@]

\dedicto • \dedicto = de dicto
1003 \cmdtxtabr{dedicto}[de dicto]

\defacto • \defacto = de facto
1004 \cmdtxtabr{defacto}[de facto]

\dere • \dere = de re
1005 \cmdtxtabr{dere}[de re]

\divideetimpera • \divideetimpera = divide et impera
1006 \cmdtxtabr{divideetimpera}[divide et impera]

```

`\eg`      • `\eg` = *e.g.*

1007 `\cmdtxtabr{eg}[e.g.\@]`

`\ergo`      • `\ergo` = *ergo*

1008 `\cmdtxtabr{ergo}`

`\errata`      • `\errata` = *errata*

1009 `\cmdtxtabr{errata}`

`\erratum`      • `\erratum` = *erratum*

1010 `\cmdtxtabr{erratum}`

`\etal`      • `\etal` = *et al.*

1011 `\cmdtxtabr{etal}[et al.\@]`

`\etc`      • `\etc` = *etc.*

1012 `\cmdtxtabr{etc}[etc.\@]`

`\ie`      • `\ie` = *i.e.*

1013 `\cmdtxtabr{ie}[i.e.\@]`

`\mutatismutandis`      • `\mutatismutandis` = *mutatis mutandis*

1014 `\cmdtxtabr{mutatismutandis}[mutatis mutandis]`

`\percontra`      • `\percontra` = *per contra*

1015 `\cmdtxtabr{percontra}[per contra]`

`\primafacie`      • `\primafacie` = *prima facie*

1016 `\cmdtxtabr{primafacie}[prima facie]`

`\viceversa`      • `\viceversa` = *vice versa*

1017 `\cmdtxtabr{viceversa}[vice versa]`

`\vs`      • `\vs` = *vs.*

1018 `\cmdtxtabr{vs}[vs.\@]`

`\viz`      • `\viz` = *viz.*

1019 `\cmdtxtabr{viz}[viz.\@]`

1020 `%%*****%`

`\Afortiori`      • `\Afortiori` = *A fortiori*

1021 `\cmdtxtabr{Afortiori}[A fortiori]`

`\Apriori`      • `\Apriori` = *A priori*

1022 `\cmdtxtabr{Apriori}[A priori]`

`\Aposteriori`      • `\Aposteriori` = *A posteriori*

1023 `\cmdtxtabr{Aposteriori}[A posteriori]`

`\Dedicto`      • `\Dedicto` = *De dicto*

1024 `\cmdtxtabr{Dedicto}[De dicto]`

`\Defacto`      • `\Defacto` = *De facto*

1025 `\cmdtxtabr{Defacto}[De facto]`

`\Dere`      • `\Dere` = *De re*

1026 `\cmdtxtabr{Dere}[De re]`

`\Divideetimpera`     • `\Divideetimpera = Divide et impera`  
1027 `\cmdtxtabr{Divideetimpera}[Divide et impera]`

`\Eg`     • `\Eg = E.g.`  
1028 `\cmdtxtabr{Eg}[E.g.\@]`

`\Errata`     • `\Errata = Errata`  
1029 `\cmdtxtabr{Errata}`

`\Erratum`     • `\Erratum = Erratum`  
1030 `\cmdtxtabr{Erratum}`

`\Mutatismutandis`     • `\Mutatismutandis = Mutatis mutandis`  
1031 `\cmdtxtabr{Mutatismutandis}[Mutatis mutandis]`

`\Percontra`     • `\Percontra = Per contra`  
1032 `\cmdtxtabr{Percontra}[Per contra]`

`\Primafacie`     • `\Primafacie = Prima facie`  
1033 `\cmdtxtabr{Primafacie}[Prima facie]`

`\Viceversa`     • `\Viceversa = Vice versa`  
1034 `\cmdtxtabr{Viceversa}[Vice versa]`

1035 `%%** Italian Abbreviations *****%`  
...  
1036 `%%*****%`  
...  
1037 `%%** French Abbreviations *****%`

`\ala`     • `\ala = à la`  
1038 `\cmdtxtabr{ala}[\`a la]`

`\naif`     • `\naif = naïf`  
1039 `\cmdtxtabr{naif}[na\`{i}f]`

`\naive`     • `\naive = naïve`  
1040 `\cmdtxtabr{naive}[na\`{i}ve]`

`\role`     • `\role = rôle`  
1041 `\cmdtxtabr{role}[r\`{o}le]`  
1042 `%%*****%`

`\Role`     • `\Role = Rôle`  
1043 `\cmdtxtabr{Role}[R\`{o}le]`  
1044 `%%** English Abbreviations *****%`

`\aka`     • `\aka = a.k.a.`  
1045 `\cmdtxtabr{aka}[a.k.a.\@]`

`\contd`     • `\contd = contd.`  
1046 `\cmdtxtabr{contd}[contd.\@]`

`\iff`     • `\iff = iff`  
1047 `\cmdtxtabr{iff}`

```

\iht • \iht = i.h.t.
1048 \cmdtxtabr{iht}[i.h.t.\@]

\stx • \stx = s.t.
1049 \cmdtxtabr{stx}[s.t.\@]

\resp • \resp = resp.
1050 \cmdtxtabr{resp}[resp.\@]

\wrt • \wrt = w.r.t.
1051 \cmdtxtabr{wrt}[w.r.t.\@]

\wlogx • \wlogx = w.l.o.g.
1052 \cmdtxtabr{wlogx}[w.l.o.g.\@]

1053 %%*****

\Contd • \Contd = Contd.
1054 \cmdtxtabr{Contd}[Contd.\@]

\Wlogx • \Wlogx = W.l.o.g.
1055 \cmdtxtabr{Wlogx}[W.l.o.g.\@]

1056 \fi
1057 %%*****

1058 %%*****
1059 %** Elementary Macros for Math *****
1060 %%*****
1061 \ifmth@

1062 %** General Notation *****

\defeq ...
\seteq 1063 \DeclareRobustCommand{\defeq}
1064 {\ifstarvar%
1065 {\mthlbop{\stackrel{\text{\textup{def}}}{=}}{}}%
1066 {\mthlbop{\triangleq}}}
1067 \DeclareRobustCommand{\seteq}
1068 {\ifstarvar{\mthlbop{\Coloneqq}}{\mthlbop{\coloneqq}}}

1069 %%*****

\limp ...
\lcoimp 1070 \DeclareRobustCommand{\limp}
1071 {\mthlbop{\rightarrow}}
1072 \DeclareRobustCommand{\lcoimp}
1073 {\mthlbop{\leftrightarrow}}

1074 %%*****

\implies ...
\notimplies 1075 \DeclareRobustCommand{\implies}
1076 {\mthlrel{\rightarrow}}
1077 \DeclareRobustCommand{\notimplies}
1078 {\mthlrel{\not\rightarrow}}

\implied ...
\notimplied 1079 \DeclareRobustCommand{\implied}
1080 {\mthlrel{\Leftarrow}}
1081 \DeclareRobustCommand{\notimplied}
1082 {\mthlrel{\not\Leftarrow}}

```

```

\coimplies ...
\notcoimplies 1083 \DeclareRobustCommand{\coimplies}
 1084 {\mthlrel{\Leftrightarrow}}
 1085 \DeclareRobustCommand{\notcoimplies}
 1086 {\mthlrel{\not\!\Leftrightarrow}}
 1087 %%*****%%

\cmodels ...
\notcmodels 1088 \DeclareRobustCommand{\cmodels}
 1089 {\mthlrel{\models}}
 1090 \DeclareRobustCommand{\notcmodels}
 1091 {\mthlrel{\not\models}}

\cequiv ...
\notcequiv 1092 \DeclareRobustCommand{\cequiv}
 1093 {\mthlrel{\equiv}}
 1094 \DeclareRobustCommand{\notcequiv}
 1095 {\mthlrel{\not\equiv}}
 1096 %%*****%%

\denot ...
 1097 \DeclareRobustCommand{\denot}
 1098 {\ifexclavar{\@edenot}{\@denot}}
 1099 \DeclareRobustCommand{\@denot}[1]
 1100 {\mth{\argmid{\left\llbracket}{\#1}{\right\rrbracket}}}
 1101 \DeclareRobustCommand{\@edenot}[1]
 1102 {\mth!\{\argmid{\llbracket}{\#1}{\rrbracket}}}
 1103 %%*****%%

\comp ...
\dual 1104 \DeclareRobustCommand{\comp}
\adj 1105 {\ifstarvar{\@scompdual}{\@compdual}}
\der 1106 \DeclareRobustCommand{\dual}
\trn 1107 {\ifstarvar{\@scompdual}{\@compdual}}
 1108 \DeclareRobustCommand{\@compdual}[1]
 1109 {\mth{\overline{\#1}}}
 1110 \DeclareRobustCommand{\@scompdual}[1]
 1111 {\mth{\widetilde{\#1}}}
 1112 \DeclareRobustCommand{\adj}[1]
 1113 {\mth{\mathring{\#1}}}
 1114 \DeclareRobustCommand{\der}[1]
 1115 {\mth{\widehat{\#1}}}
 1116 \DeclareRobustCommand{\trn}[1]
 1117 {\mth{\widetilde{\#1}}}

\vec ...
 1118 \DeclareRobustCommand{\vec}
 1119 {\ifstarvar{\@svec}{\@vec}}
 1120 \DeclareRobustCommand{\@vec}[1]
 1121 {\mth{\mathaccent"017E{\#1}}}
 1122 \DeclareRobustCommand{\@svec}[1]
 1123 {\mth{\overline{\#1}}}
 1124 %%*****%%

\enumeration ...
 1125 \DeclareRobustCommand{\enumeration}
 1126 {\ifstarvar{\@senumeration}{\@enumeration}}
 1127 \varcmd{\@enumeration}{\mth!}{\{,\}{\{}}
 1128 \varcmd{\@senumeration}{\mth!}{\{;\}{\{}}

```



```

\sequence ...
\sequence1 1129 \DeclareRobustCommand{\sequence}
\sequencer 1130 {\ifstarvar{\@ssequence}{\@sequence}}
1131 \DeclareRobustCommand{\@sequence}
1132 {\ifexclavar{\@e@sequence}{\@@sequence}}
1133 \DeclareRobustCommand{\@ssequence}
1134 {\ifexclavar{\@e@ssequence}{\@@ssequence}}
1135 \varcmd{\@sequence}{\mth}{\left[}{,}{\right]}{}
1136 \varcmd{\@e@sequence}{\mth!}{}{,}{}
1137 \varcmd{\@@sequence}{\mth}{\left[}{;}{\right]}{}
1138 \varcmd{\@e@ssequence}{\mth!}{}{;}{}
1139 \DeclareRobustCommand{\sequence1}
1140 {\ifstarvar{\@ssequence1}{\@sequence1}}
1141 \DeclareRobustCommand{\@sequence1}
1142 {\ifexclavar{\@e@sequence1}{\@@sequence1}}
1143 \DeclareRobustCommand{\@ssequence1}
1144 {\ifexclavar{\@e@ssequence1}{\@@ssequence1}}
1145 \varcmd{\@sequence1}{\mth}{\left[}{,}{\right.}{}
1146 \varcmd{\@e@sequence1}{\mth!}{}{,}{}
1147 \varcmd{\@@sequence1}{\mth}{\left[}{;}{\right.}{}
1148 \varcmd{\@e@ssequence1}{\mth!}{}{;}{}
1149 \DeclareRobustCommand{\sequencer}
1150 {\ifstarvar{\@ssequencer}{\@sequencer}}
1151 \DeclareRobustCommand{\@sequencer}
1152 {\ifexclavar{\@e@sequencer}{\@@sequencer}}
1153 \DeclareRobustCommand{\@ssequencer}
1154 {\ifexclavar{\@e@ssequencer}{\@@ssequencer}}
1155 \varcmd{\@sequencer}{\mth}{\left.}{,}{\right]}{}
1156 \varcmd{\@e@sequencer}{\mth!}{}{,}{}
1157 \varcmd{\@@sequencer}{\mth}{\left.}{;}{\right]}{}
1158 \varcmd{\@e@ssequencer}{\mth!}{}{;}{}

\tuple ...
\tuple1 1159 \DeclareRobustCommand{\tuple}
\tupler 1160 {\ifstarvar{\@stuple}{\@tuple}}
1161 \DeclareRobustCommand{\@tuple}
1162 {\ifexclavar{\@e@tuple}{\@@tuple}}
1163 \DeclareRobustCommand{\@stuple}
1164 {\ifexclavar{\@e@stuple}{\@@stuple}}
1165 \varcmd{\@tuple}{\mth}{\left\langle}{,}{\right\rangle}{}
1166 \varcmd{\@e@tuple}{\mth!}{\langle}{,}{\rangle}{}
1167 \varcmd{\@@tuple}{\mth}{\left\langle}{;}{\right\rangle}{}
1168 \varcmd{\@e@stuple}{\mth!}{\langle}{;}{\rangle}{}
1169 \DeclareRobustCommand{\tuple1}
1170 {\ifstarvar{\@stuple1}{\@tuple1}}
1171 \DeclareRobustCommand{\@tuple1}
1172 {\ifexclavar{\@e@tuple1}{\@@tuple1}}
1173 \DeclareRobustCommand{\@stuple1}
1174 {\ifexclavar{\@e@stuple1}{\@@stuple1}}
1175 \varcmd{\@tuple1}{\mth}{\left\langle}{,}{\right.}{}
1176 \varcmd{\@e@tuple1}{\mth!}{\langle}{,}{}
1177 \varcmd{\@@tuple1}{\mth}{\left\langle}{;}{\right.}{}
1178 \varcmd{\@e@stuple1}{\mth!}{\langle}{;}{}
1179 \DeclareRobustCommand{\tupler}
1180 {\ifstarvar{\@stupler}{\@tupler}}
1181 \DeclareRobustCommand{\@tupler}
1182 {\ifexclavar{\@e@tupler}{\@@tupler}}
1183 \DeclareRobustCommand{\@stupler}
1184 {\ifexclavar{\@e@stupler}{\@@stupler}}
1185 \varcmd{\@tupler}{\mth}{\left.}{,}{\right\rangle}{}
1186 \varcmd{\@e@tupler}{\mth!}{}{,}{\rangle}{}
1187 \varcmd{\@@stupler}{\mth}{\left.}{;}{\right\rangle}{}
1188 \varcmd{\@e@stupler}{\mth!}{}{;}{\rangle}{}

```

```

1189 %** Sets *****%

\set ...{ $1^2 \mid 1^2$ }{ $1^2 : 1^2$ }{ $1^2 \mid 1^2$ }{ $1^2 : 1^2$ }
\setl ...{ $1^2 \mid 1^2$ }
\setr ...{ $1^2 : 1^2$ }
...{ $1^2 \mid 1^2$ }
...{ $1^2 : 1^2$ }

1190 \DeclareRobustCommand{\set}
1191 {\ifstarvar{\@sset}{\@set}}
1192 \DeclareRobustCommand{\@set}
1193 {\ifexclavar{\@e@xset{\vert}}{\@xset{\vert}}}
1194 \DeclareRobustCommand{\@sset}
1195 {\ifexclavar{\@e@xset{\!:\!}}{\@xset{.\!:\!}}}
1196 \DeclareRobustCommand{\@xset}[3]
1197 {\mth{\argmid{\left\lbrace}{\argsep{#2}{\,\middle#1\,}{#3}}{\right\rbrace}}}
1198 \DeclareRobustCommand{\@e@xset}[3]
1199 {\mth!\{\argmid{\lbrace}{\argsep{#2}{\,\,#1\,}{#3}}{\rbrace}}}
1200 \DeclareRobustCommand{\setl}
1201 {\ifstarvar{\@ssetl}{\@setl}}
1202 \DeclareRobustCommand{\@setl}
1203 {\ifexclavar{\@e@xsetl{\,\,\vert}}{\@xsetl{\vert}}}
1204 \DeclareRobustCommand{\@ssetl}
1205 {\ifexclavar{\@e@xsetl{\!:\!}}{\@xsetl{.\!:\!}}}
1206 \DeclareRobustCommand{\@xsetl}[2]
1207 {\mth{\argmid{\left\lbrace}{#2}{\,\,\right#1}}}
1208 \DeclareRobustCommand{\@e@xsetl}[2]
1209 {\mth!\{\argmid{\lbrace}{#2}{\,\,\,#1}}}
1210 \DeclareRobustCommand{\setr}
1211 {\ifstarvar{\@setr}{\@setr}}
1212 \DeclareRobustCommand{\@setr}
1213 {\ifexclavar{\@e@setr}{\@xsetr}}
1214 \DeclareRobustCommand{\@xsetr}[1]
1215 {\mth{\argmid{\left.}{#1}{\right\rbrace}}}
1216 \DeclareRobustCommand{\@e@setr}[1]
1217 {\mth!\{\argmid{\lbrace}{#1}{\rbrace}}}

\card ...

1218 \DeclareRobustCommand{\card}
1219 {\ifexclavar{\@ecard}{\@card}}
1220 \DeclareRobustCommand{\@card}[1]
1221 {\mth{\argmid{\left\lvert}{#1}{\right\rvert}}}
1222 \DeclareRobustCommand{\@ecard}[1]
1223 {\mth!\{\argmid{\lvert}{#1}{\rvert}}}

\pow ...

1224 \DeclareRobustCommand{\pow}
1225 {\ifstarvar{\@spow}{\@pow}}
1226 \DeclareRobustCommand{\@pow}[1]
1227 {\mth{2^{\defval{#1}{\cdot}}}}
1228 \DeclareRobustCommand{\@spow}
1229 {\ifexclavar{\@e@spow}{\@xspow}}
1230 \DeclareRobustCommand{\@e@spow}[1]
1231 {\mthargfun![mathscr]{P}{\defval{#1}{\cdot}}}
1232 \DeclareRobustCommand{\@xspow}[1]
1233 {\mthargfun[mathscr]{P}{\defval{#1}{\cdot}}}

1234 %** Relations *****%

\emptyrel ...

1235 \DeclareRobustCommand{\emptyrel}
1236 {\mth{\varnothing}}

1237 %*****%

```

```

\dom ...
\cod 1238 \usrmth{dom}{-}{argfun}
\rng 1239 \usrmth{cod}{-}{argfun}
\img 1240 \usrmth{rng}{-}{argfun}
 1241 \usrmth{img}{-}{argfun}

\deg ...
 1242 \usrmth{deg}{-}{argfun}

 1243 %%*****

\prj ...
 1244 \DeclareRobustCommand{\prj}
 1245 {\mthlbop{\downarrow}}

\rst ...
 1246 \DeclareRobustCommand{\rst}
 1247 {\mthlbop{\upharpoonright}}

\cmp ...
 1248 \DeclareRobustCommand{\cmp}
 1249 {\mthlbop{\circ}}

 1250 %%** Functions *****

\emptyfun ...
 1251 \DeclareRobustCommand{\emptyfun}
 1252 {\mth{\varnothing}}

 1253 %%*****

\pto ...
\pmapsto 1254 \DeclareMathOperator{\pto}
 1255 {\ensuremath{\rightharpoonup}}
 1256 \DeclareMathOperator{\pmapsto}
 1257 {\ensuremath{\mathrel{\raisebox{0.5ex}{\footnotesize$\llcorner$}}}%
 1258 \kern-1.5ex\rightharpoonup}}
 1259 %%*****

\fix ...
\ifp 1260 \usrmth{fix}{-}{fun}
\lfp 1261 \usrmth{lfp}{-}{fun}
\gfp 1262 \usrmth{lfp}{-}{fun}
 1263 \usrmth{gfp}{-}{fun}

 1264 %%*****

\Aomega ...
\A0omega 1265 \usrmth{Aomega}{-}{argset}[\omega]
 1266 \usrmth{A0omega}{-}{argset}[\Omega]

\Atheta ...
\ATheta 1267 \usrmth{Atheta}{-}{argset}[\theta]
 1268 \usrmth{ATheta}{-}{argset}[\Theta]

\Aomicron ...
\A0micron 1269 \usrmth{Aomicron}{-}{argset}[\omicron]
 1270 \usrmth{A0micron}{-}{argset}[\Omicron]

 1271 %%** Numbers *****

\SetB ...
 1272 \DeclareRobustCommand{\SetB}
 1273 {\mthset[\mathbb]{B}}

```

```

\SetF ...
1274 \DeclareRobustCommand{\SetF}
1275 {\mthset [mathbb]{F}}

\SetN ...
\SetNI 1276 \DeclareRobustCommand{\SetN}
1277 {\mthset [mathbb]{N}}
1278 \DeclareRobustCommand{\SetNI}[1] []
1279 {\SetN[\infty #1]}

\SetZ ...
\SetZI 1280 \DeclareRobustCommand{\SetZ}
\SetZPI 1281 {\mthset [mathbb]{Z}}
\SetZNI 1282 \DeclareRobustCommand{\SetZI}[1] []
1283 {\SetZ[\pm\infty #1]}
1284 \DeclareRobustCommand{\SetZPI}[1] []
1285 {\SetZ[+\infty #1]}
1286 \DeclareRobustCommand{\SetZNI}[1] []
1287 {\SetZ[-\infty #1]}

\SetQ ...
\SetQI 1288 \DeclareRobustCommand{\SetQ}
\SetQPI 1289 {\mthset [mathbb]{Q}}
\SetQNI 1290 \DeclareRobustCommand{\SetQI}[1] []
1291 {\SetQ[\pm\infty #1]}
1292 \DeclareRobustCommand{\SetQPI}[1] []
1293 {\SetQ[+\infty #1]}
1294 \DeclareRobustCommand{\SetQNI}[1] []
1295 {\SetQ[-\infty #1]}

\SetR ...
\SetRI 1296 \DeclareRobustCommand{\SetR}
\SetRPI 1297 {\mthset [mathbb]{R}}
\SetRNI 1298 \DeclareRobustCommand{\SetRI}[1] []
1299 {\SetR[\pm\infty #1]}
1300 \DeclareRobustCommand{\SetRPI}[1] []
1301 {\SetR[+\infty #1]}
1302 \DeclareRobustCommand{\SetRNI}[1] []
1303 {\SetR[-\infty #1]}

\SetC ...
\SetCI 1304 \DeclareRobustCommand{\SetC}
1305 {\mthset [mathbb]{C}}
1306 \DeclareRobustCommand{\SetCI}[1] []
1307 {\SetC[\infty #1]}
1308 %%*****%

\num ...
\numcc 1309 \DeclareRobustCommand{\num}[1]
\numco 1310 {\mth{[#1]}}
\numoc 1311 \DeclareRobustCommand{\numcc}[2]
\numoo 1312 {\mth{[\argsep{#1}{,}{#2}]}}
1313 \DeclareRobustCommand{\numco}[2]
1314 {\mth{[\argsep{#1}{,}{#2}]}}
1315 \DeclareRobustCommand{\numoc}[2]
1316 {\mth{(\argsep{#1}{,}{#2})}}
1317 \DeclareRobustCommand{\numoo}[2]
1318 {\mth{(\argsep{#1}{,}{#2})}}
1319 %%*****%

```

```

\abs ...
\norm 1320 \DeclareRobustCommand{\abs}
1321 {\ifexclavar{\@eabs}{\@abs}}
1322 \DeclareRobustCommand{\@abs}[1]
1323 {\mth{\argmid{\left\lvert}{#1}{\right\rvert}}}
1324 \DeclareRobustCommand{\@eabs}[1]
1325 {\mth!\{\argmid{\lvert}{#1}{\rvert}}}
1326 \DeclareRobustCommand{\norm}
1327 {\ifexclavar{\@enorm}{\@norm}}
1328 \DeclareRobustCommand{\@norm}[1]
1329 {\mth{\argmid{\left\lVert}{#1}{\right\rVert}}}
1330 \DeclareRobustCommand{\@enorm}[1]
1331 {\mth!\{\argmid{\lVert}{#1}{\rVert}}}

\floor ...
\ceil 1332 \DeclareRobustCommand{\floor}
1333 {\ifexclavar{\@efloor}{\@floor}}
1334 \DeclareRobustCommand{\@efloor}[1]
1335 {\mth{\argmid{\left\lfloor}{#1}{\right\rfloor}}}
1336 \DeclareRobustCommand{\@efloor}[1]
1337 {\mth!\{\argmid{\lfloor}{#1}{\rfloor}}}
1338 \DeclareRobustCommand{\ceil}
1339 {\ifexclavar{\@eceil}{\@ceil}}
1340 \DeclareRobustCommand{\@eceil}[1]
1341 {\mth{\argmid{\left\lceil}{#1}{\right\rceil}}}
1342 \DeclareRobustCommand{\@eceil}[1]
1343 {\mth!\{\argmid{\lceil}{#1}{\rceil}}}
1344 %%*****

\arg ...
1345 \usrmth{arg}{\}{fun}

\evn ...
\odd 1346 \usrmth{evn}{\}{fun}
1347 \usrmth{odd}{\}{fun}

\bst ...
\argbst 1348 \usrmth{bst}{\}{fun}
1349 \usrmth{argbst}{\}{fun}[arg,bst]

\min ...
\max 1350 \usrmth{min}{\}{fun}
\argmin 1351 \usrmth{max}{\}{fun}
\argmax 1352 \usrmth{argmin}{\}{fun}[arg,min]
1353 \usrmth{argmax}{\}{fun}[arg,max]

\inf ...
\sup 1354 \usrmth{inf}{\}{fun}
1355 \usrmth{sup}{\}{fun}

\gcd ...
\lcm 1356 \usrmth{gcd}{\}{fun}
1357 \usrmth{lcm}{\}{fun}
1358 %%** Sequences *****

\emptyseq ...
1359 \DeclareRobustCommand{\emptyseq}
1360 {\mth{\varepsilon}}

```

```

\len ...
1361 \DeclareRobustCommand{\len}
1362 {\ifexclavar{\@elen}{\@len}}
1363 \DeclareRobustCommand{\@len}[1]
1364 {\mth{\argmid{\left\lvert\right\lvert}{\#1}{\right\rvert}}}}
1365 \DeclareRobustCommand{\@elen}[1]
1366 {\mth!\{\argmid{\lvert\right\lvert}{\#1}{\rvert}}}}

\fst ...
\lst 1367 \usrmth{fst}{\}{argfun}
1368 \usrmth{lst}{\}{argfun}

1369 \fi
1370 %%*****%
1371 %%*****%
1372 %%** Macros for Computational-Complexity Classes *****%
1373 %%*****%
1374 \ifcom@

```

\defcomcls ... to do!

- \defcomcls{CompClass};  

$$\text{CompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{COMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$$
- \defcomcls{CompClass}[NewClass];  

$$\text{CompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{NEWCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$$

```

1375 \DeclareRobustCommandx{\defcomcls}[2][2=]
1376 {\csdef{\#1}{\txttoargcom{\defval{\#2}{\#1}}}}

```

\defcomclsgrp ... to do!

- \defcomclsgrp{CompClass};  

$$\begin{aligned}
&\text{CompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{COMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{COCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{COMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoCompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{COCOMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{COMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoCompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{COCOMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{COMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoCompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{COCOMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
\\
&\text{DCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{DCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoDCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{CODCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{DCompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{DCOMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoDCompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{CODCOMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{DCompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{DCOMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoDCompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{CODCOMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{DCompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{DCOMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoDCompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{CODCOMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
\\
&\text{NCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{NCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoNCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{CONCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{NCompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{NCOMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoNCompClassE}[\text{sub}][\text{sup}][\text{arg}] = \text{CONCOMPCLASS-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{NCompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{NCOMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoNCompClassH}[\text{sub}][\text{sup}][\text{arg}] = \text{CONCOMPCLASS-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{NCompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{NCOMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoNCompClassC}[\text{sub}][\text{sup}][\text{arg}] = \text{CONCOMPCLASS-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
\\
&\text{UCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{UCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG}) \\
&\text{CoUCompClass}[\text{sub}][\text{sup}][\text{arg}] = \text{COUCOMPCLASS}_{\text{SUB}}^{\text{SUP}}(\text{ARG})
\end{aligned}$$

```

\UCompClassE[sub][sup][arg] = UCOMPCLASS-EASYSUB(ARG)
\CoUCompClassE[sub][sup][arg] = CoUCOMPCLASS-EASYSUB(ARG)
\UCompClassH[sub][sup][arg] = UCOMPCLASS-HARDSUB(ARG)
\CoUCompClassH[sub][sup][arg] = CoUCOMPCLASS-HARDSUB(ARG)
\UCompClassC[sub][sup][arg] = UCOMPCLASS-COMPLETESUB(ARG)
\CoUCompClassC[sub][sup][arg] = CoUCOMPCLASS-COMPLETESUB(ARG)

\ACompClass[sub][sup][arg] = ACOMPCLASSSUB(ARG)
\CoACompClass[sub][sup][arg] = CoACOMPCLASSSUB(ARG)
\ACompClassE[sub][sup][arg] = ACOMPCLASS-EASYSUB(ARG)
\CoACompClassE[sub][sup][arg] = CoACOMPCLASS-EASYSUB(ARG)
\ACompClassH[sub][sup][arg] = ACOMPCLASS-HARDSUB(ARG)
\CoACompClassH[sub][sup][arg] = CoACOMPCLASS-HARDSUB(ARG)
\ACompClassC[sub][sup][arg] = ACOMPCLASS-COMPLETESUB(ARG)
\CoACompClassC[sub][sup][arg] = CoACOMPCLASS-COMPLETESUB(ARG)

• \defcomclsgrp{CompClass}[NewClass];

\CompClass[sub][sup][arg] = NEWCLASSSUB(ARG)
\CoCompClass[sub][sup][arg] = CoNEWCLASSSUB(ARG)
\CompClassE[sub][sup][arg] = NEWCLASS-EASYSUB(ARG)
\CoCompClassE[sub][sup][arg] = CoNEWCLASS-EASYSUB(ARG)
\CompClassH[sub][sup][arg] = NEWCLASS-HARDSUB(ARG)
\CoCompClassH[sub][sup][arg] = CoNEWCLASS-HARDSUB(ARG)
\CompClassC[sub][sup][arg] = NEWCLASS-COMPLETESUB(ARG)
\CoCompClassC[sub][sup][arg] = CoNEWCLASS-COMPLETESUB(ARG)

\DCompClass[sub][sup][arg] = DNEWCLASSSUB(ARG)
\CoDCompClass[sub][sup][arg] = CoDNEWCLASSSUB(ARG)
\DCompClassE[sub][sup][arg] = DNEWCLASS-EASYSUB(ARG)
\CoDCompClassE[sub][sup][arg] = CoDNEWCLASS-EASYSUB(ARG)
\DCompClassH[sub][sup][arg] = DNEWCLASS-HARDSUB(ARG)
\CoDCompClassH[sub][sup][arg] = CoDNEWCLASS-HARDSUB(ARG)
\DCompClassC[sub][sup][arg] = DNEWCLASS-COMPLETESUB(ARG)
\CoDCompClassC[sub][sup][arg] = CoDNEWCLASS-COMPLETESUB(ARG)

\NCompClass[sub][sup][arg] = NNEWCLASSSUB(ARG)
\CoNCompClass[sub][sup][arg] = CoNNEWCLASSSUB(ARG)
\NCompClassE[sub][sup][arg] = NNEWCLASS-EASYSUB(ARG)
\CoNCompClassE[sub][sup][arg] = CoNNEWCLASS-EASYSUB(ARG)
\NCompClassH[sub][sup][arg] = NNEWCLASS-HARDSUB(ARG)
\CoNCompClassH[sub][sup][arg] = CoNNEWCLASS-HARDSUB(ARG)
\NCompClassC[sub][sup][arg] = NNEWCLASS-COMPLETESUB(ARG)
\CoNCompClassC[sub][sup][arg] = CoNNEWCLASS-COMPLETESUB(ARG)

\UCompClass[sub][sup][arg] = UNEWCLASSSUB(ARG)
\CoUCompClass[sub][sup][arg] = CoUNEWCLASSSUB(ARG)
\UCompClassE[sub][sup][arg] = UNEWCLASS-EASYSUB(ARG)
\CoUCompClassE[sub][sup][arg] = CoUNEWCLASS-EASYSUB(ARG)
\UCompClassH[sub][sup][arg] = UNEWCLASS-HARDSUB(ARG)
\CoUCompClassH[sub][sup][arg] = CoUNEWCLASS-HARDSUB(ARG)
\UCompClassC[sub][sup][arg] = UNEWCLASS-COMPLETESUB(ARG)
\CoUCompClassC[sub][sup][arg] = CoUNEWCLASS-COMPLETESUB(ARG)

\ACompClass[sub][sup][arg] = ANEWCLASSSUB(ARG)
\CoACompClass[sub][sup][arg] = CoANEWCLASSSUB(ARG)
\ACompClassE[sub][sup][arg] = ANEWCLASS-EASYSUB(ARG)
\CoACompClassE[sub][sup][arg] = CoANEWCLASS-EASYSUB(ARG)
\ACompClassH[sub][sup][arg] = ANEWCLASS-HARDSUB(ARG)
\CoACompClassH[sub][sup][arg] = CoANEWCLASS-HARDSUB(ARG)
\ACompClassC[sub][sup][arg] = ANEWCLASS-COMPLETESUB(ARG)
\CoACompClassC[sub][sup][arg] = CoANEWCLASS-COMPLETESUB(ARG)

1377 \DeclareRobustCommandx{\defcomclsgrp}[2][2=]
1378 {\defcomclsgrpsem{#1}{\defval{#2}{#1}}%
1379 \defcomclsgrpsem{#1}{\defval{#2}{#1}}[Co]}

```

```

1380 \DeclareRobustCommandx{\defcomclsggrpsem}[3][3=]
1381 {\defcomclsgrpred{#3#1}{#2}[#3]%
1382 \defcomclsgrpred{#3D#1}{#2}[#3D]%
1383 \defcomclsgrpred{#3N#1}{#2}[#3N]%
1384 \defcomclsgrpred{#3U#1}{#2}[#3U]%
1385 \defcomclsgrpred{#3A#1}{#2}[#3A]}
1386 \DeclareRobustCommandx{\defcomclsgrpred}[3][3=]
1387 {\defcomclsgrpcomd{#1}{#2}[#3]%
1388 \defcomclsgrpcomd{#1E}{#2}[#3][-easy]%
1389 \defcomclsgrpcomd{#1H}{#2}[#3][-hard]%
1390 \defcomclsgrpcomd{#1C}{#2}[#3][-complete]}%
1391 \DeclareRobustCommandx{\defcomclsgrpcomd}[4][3=, 4=]
1392 {\csdef{#1}{\txttoargcom{#3#2#4}}}

```

\defcomhrc ... to do!

- \defcomhrc{CompHierarchy};
- $\text{CompHierarchy}_{\text{sub}}^{\text{sup}}[\text{par}] = \text{COMPHIERARCHY}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$
- \defcomhrc{CompHierarchy}[NewHierarchy];
- $\text{CompHierarchy}_{\text{sub}}^{\text{sup}}[\text{par}] = \text{NEWHIERARCHY}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

```

1393 \DeclareRobustCommandx{\defcomhrc}[2][2=]
1394 {\csdef{#1}{\txttoparcom{\defval{#2}{#1}}}}
1395 %%*****%

```

\Easy

\Hard

\Complete

```

1396 \cmdtxtcom{Easy}
1397 \cmdtxtcom{Hard}
1398 \cmdtxtcom{Complete}
1399 %%*****%

```

\FPT

\FPLin

\FPQdr

\FPCub

- \FPT[sub][sup][arg] =  $\text{FPT}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \FPLin[sub][sup][arg] =  $\text{FPL}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \FPQdr[sub][sup][arg] =  $\text{FPQ}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \FPCub[sub][sup][arg] =  $\text{FPC}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$

```

1400 \defcomcls{FPT}
1401 \defcomcls{FPLin}[FPL]
1402 \defcomcls{FPQdr}[FPQ]
1403 \defcomcls{FPCub}[FPC]
1404 %%*****%

```

\Time(E/H/C)

\DTime(E/H/C)

\NTime(E/H/C)

\UTime(E/H/C)

\ATime(E/H/C)

- \Time[sub][sup][arg] =  $\text{TIME}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \TimeE[sub][sup][arg] =  $\text{TIME-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \TimeH[sub][sup][arg] =  $\text{TIME-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \TimeC[sub][sup][arg] =  $\text{TIME-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \DTime[sub][sup][arg] =  $\text{DTIME}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \DTimeE[sub][sup][arg] =  $\text{DTIME-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \DTimeH[sub][sup][arg] =  $\text{DTIME-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \DTimeC[sub][sup][arg] =  $\text{DTIME-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \NTime[sub][sup][arg] =  $\text{NTIME}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \NTimeE[sub][sup][arg] =  $\text{NTIME-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \NTimeH[sub][sup][arg] =  $\text{NTIME-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \NTimeC[sub][sup][arg] =  $\text{NTIME-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \UTime[sub][sup][arg] =  $\text{UTIME}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \UTimeE[sub][sup][arg] =  $\text{UTIME-EASY}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \UTimeH[sub][sup][arg] =  $\text{UTIME-HARD}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- \UTimeC[sub][sup][arg] =  $\text{UTIME-COMPLETE}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$







|                            |                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------|
| $\backslash$ QTime(E/H/C)  | • $\backslash$ QTime[sub][sup][arg] = QPTIME <sub>SUB</sub> <sup>SUP</sup> (ARG)           |
| $\backslash$ DQTime(E/H/C) | $\backslash$ QTimeE[sub][sup][arg] = QPTIME-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)       |
| $\backslash$ NQTime(E/H/C) | $\backslash$ QTimeH[sub][sup][arg] = QPTIME-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)       |
| $\backslash$ UQTime(E/H/C) | $\backslash$ QTimeC[sub][sup][arg] = QPTIME-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG)   |
| $\backslash$ AQTime(E/H/C) | • $\backslash$ DQTime[sub][sup][arg] = DQPTIME <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                            | $\backslash$ DQTimeE[sub][sup][arg] = DQPTIME-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ DQTimeH[sub][sup][arg] = DQPTIME-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ DQTimeC[sub][sup][arg] = DQPTIME-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                            | • $\backslash$ NQTime[sub][sup][arg] = NQPTIME <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                            | $\backslash$ NQTimeE[sub][sup][arg] = NQPTIME-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ NQTimeH[sub][sup][arg] = NQPTIME-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ NQTimeC[sub][sup][arg] = NQPTIME-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                            | • $\backslash$ UQTime[sub][sup][arg] = UQPTIME <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                            | $\backslash$ UQTimeE[sub][sup][arg] = UQPTIME-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ UQTimeH[sub][sup][arg] = UQPTIME-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ UQTimeC[sub][sup][arg] = UQPTIME-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                            | • $\backslash$ AQTime[sub][sup][arg] = AQPTIME <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                            | $\backslash$ AQTimeE[sub][sup][arg] = AQPTIME-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ AQTimeH[sub][sup][arg] = AQPTIME-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                            | $\backslash$ AQTimeC[sub][sup][arg] = AQPTIME-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |

1411  $\backslash$ defcomclsgrp{QTime}

|                              |                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------|
| $\backslash$ QPSpace(E/H/C)  | • $\backslash$ QPSpace[sub][sup][arg] = QPSPACE <sub>SUB</sub> <sup>SUP</sup> (ARG)           |
| $\backslash$ DQPSpace(E/H/C) | $\backslash$ QPSpaceE[sub][sup][arg] = QPSPACE-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)       |
| $\backslash$ NQPSpace(E/H/C) | $\backslash$ QPSpaceH[sub][sup][arg] = QPSPACE-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)       |
| $\backslash$ UQPSpace(E/H/C) | $\backslash$ QPSpaceC[sub][sup][arg] = QPSPACE-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG)   |
| $\backslash$ AQPSpace(E/H/C) | • $\backslash$ DQPSpace[sub][sup][arg] = DQPSPACE <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                              | $\backslash$ DQPSpaceE[sub][sup][arg] = DQPSPACE-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ DQPSpaceH[sub][sup][arg] = DQPSPACE-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ DQPSpaceC[sub][sup][arg] = DQPSPACE-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                              | • $\backslash$ NQPSpace[sub][sup][arg] = NQPSPACE <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                              | $\backslash$ NQPSpaceE[sub][sup][arg] = NQPSPACE-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ NQPSpaceH[sub][sup][arg] = NQPSPACE-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ NQPSpaceC[sub][sup][arg] = NQPSPACE-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                              | • $\backslash$ UQPSpace[sub][sup][arg] = UQPSPACE <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                              | $\backslash$ UQPSpaceE[sub][sup][arg] = UQPSPACE-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ UQPSpaceH[sub][sup][arg] = UQPSPACE-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ UQPSpaceC[sub][sup][arg] = UQPSPACE-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |
|                              | • $\backslash$ AQPSpace[sub][sup][arg] = AQPSPACE <sub>SUB</sub> <sup>SUP</sup> (ARG)         |
|                              | $\backslash$ AQPSpaceE[sub][sup][arg] = AQPSPACE-EASY <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ AQPSpaceH[sub][sup][arg] = AQPSPACE-HARD <sub>SUB</sub> <sup>SUP</sup> (ARG)     |
|                              | $\backslash$ AQPSpaceC[sub][sup][arg] = AQPSPACE-COMPLETE <sub>SUB</sub> <sup>SUP</sup> (ARG) |

1412  $\backslash$ defcomclsgrp{QPSpace}

|                              |                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------|
| $\backslash$ ExpTime(E/H/C)  | • $\backslash$ ExpTime[sub][sup][arg] = EXP <sub>SUB</sub> <sup>SUP</sup> TIME(ARG)           |
| $\backslash$ DExpTime(E/H/C) | $\backslash$ ExpTimeE[sub][sup][arg] = EXP <sub>SUB</sub> <sup>SUP</sup> TIME-EASY(ARG)       |
| $\backslash$ NExpTime(E/H/C) | $\backslash$ ExpTimeH[sub][sup][arg] = EXP <sub>SUB</sub> <sup>SUP</sup> TIME-HARD(ARG)       |
| $\backslash$ UExpTime(E/H/C) | $\backslash$ ExpTimeC[sub][sup][arg] = EXP <sub>SUB</sub> <sup>SUP</sup> TIME-COMPLETE(ARG)   |
| $\backslash$ AExpTime(E/H/C) | • $\backslash$ DExpTime[sub][sup][arg] = DEXP <sub>SUB</sub> <sup>SUP</sup> TIME(ARG)         |
|                              | $\backslash$ DExpTimeE[sub][sup][arg] = DEXP <sub>SUB</sub> <sup>SUP</sup> TIME-EASY(ARG)     |
|                              | $\backslash$ DExpTimeH[sub][sup][arg] = DEXP <sub>SUB</sub> <sup>SUP</sup> TIME-HARD(ARG)     |
|                              | $\backslash$ DExpTimeC[sub][sup][arg] = DEXP <sub>SUB</sub> <sup>SUP</sup> TIME-COMPLETE(ARG) |
|                              | • $\backslash$ NExpTime[sub][sup][arg] = NEXP <sub>SUB</sub> <sup>SUP</sup> TIME(ARG)         |
|                              | $\backslash$ NExpTimeE[sub][sup][arg] = NEXP <sub>SUB</sub> <sup>SUP</sup> TIME-EASY(ARG)     |
|                              | $\backslash$ NExpTimeH[sub][sup][arg] = NEXP <sub>SUB</sub> <sup>SUP</sup> TIME-HARD(ARG)     |
|                              | $\backslash$ NExpTimeC[sub][sup][arg] = NEXP <sub>SUB</sub> <sup>SUP</sup> TIME-COMPLETE(ARG) |
|                              | • $\backslash$ UExpTime[sub][sup][arg] = UEXP <sub>SUB</sub> <sup>SUP</sup> TIME(ARG)         |
|                              | $\backslash$ UExpTimeE[sub][sup][arg] = UEXP <sub>SUB</sub> <sup>SUP</sup> TIME-EASY(ARG)     |
|                              | $\backslash$ UExpTimeH[sub][sup][arg] = UEXP <sub>SUB</sub> <sup>SUP</sup> TIME-HARD(ARG)     |
|                              | $\backslash$ UExpTimeC[sub][sup][arg] = UEXP <sub>SUB</sub> <sup>SUP</sup> TIME-COMPLETE(ARG) |

- $\backslash\text{ExpTime}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpTime}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{ExpTimeE}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpTime-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{ExpTimeH}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpTime-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{ExpTimeC}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpTime-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$

1413  $\backslash\text{defcomclsgroup}\{\text{ExpTime}\}$

- $\backslash\text{ExpSpace}(\text{E/H/C})$  •  $\backslash\text{ExpSpace}[\text{sub}][\text{sup}][\text{arg}] = \text{ExpSpace}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{DExpSpace}(\text{E/H/C})$  •  $\backslash\text{ExpSpaceE}[\text{sub}][\text{sup}][\text{arg}] = \text{ExpSpace-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{NExpSpace}(\text{E/H/C})$  •  $\backslash\text{ExpSpaceH}[\text{sub}][\text{sup}][\text{arg}] = \text{ExpSpace-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{UExpSpace}(\text{E/H/C})$  •  $\backslash\text{ExpSpaceC}[\text{sub}][\text{sup}][\text{arg}] = \text{ExpSpace-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{AExpSpace}(\text{E/H/C})$  •  $\backslash\text{DExpSpace}[\text{sub}][\text{sup}][\text{arg}] = \text{DExpSpace}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{DExpSpaceE}[\text{sub}][\text{sup}][\text{arg}] = \text{DExpSpace-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{DExpSpaceH}[\text{sub}][\text{sup}][\text{arg}] = \text{DExpSpace-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{DExpSpaceC}[\text{sub}][\text{sup}][\text{arg}] = \text{DExpSpace-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{NExpSpace}[\text{sub}][\text{sup}][\text{arg}] = \text{NExpSpace}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{NExpSpaceE}[\text{sub}][\text{sup}][\text{arg}] = \text{NExpSpace-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{NExpSpaceH}[\text{sub}][\text{sup}][\text{arg}] = \text{NExpSpace-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{NExpSpaceC}[\text{sub}][\text{sup}][\text{arg}] = \text{NExpSpace-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{UExpSpace}[\text{sub}][\text{sup}][\text{arg}] = \text{UExpSpace}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{UExpSpaceE}[\text{sub}][\text{sup}][\text{arg}] = \text{UExpSpace-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{UExpSpaceH}[\text{sub}][\text{sup}][\text{arg}] = \text{UExpSpace-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{UExpSpaceC}[\text{sub}][\text{sup}][\text{arg}] = \text{UExpSpace-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{AExpSpace}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpSpace}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{AExpSpaceE}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpSpace-Easy}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{AExpSpaceH}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpSpace-Hard}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$
- $\backslash\text{AExpSpaceC}[\text{sub}][\text{sup}][\text{arg}] = \text{AExpSpace-Complete}_{\text{SUB}}^{\text{SUP}}(\text{ARG})$

1414  $\backslash\text{defcomclsgroup}\{\text{ExpSpace}\}$

1415  $\%*****\%$

$\backslash\text{PH}$  •  $\backslash\text{PH}[\text{sub}][\text{sup}][\text{par}] = \text{PH}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1416  $\backslash\text{defcomhrc}\{\text{PH}\}$

$\backslash\text{WH}$  •  $\backslash\text{WH}[\text{sub}][\text{sup}][\text{par}] = \text{W}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1417  $\backslash\text{defcomhrc}\{\text{WH}\}[\text{W}]$

$\backslash\text{AH}$  •  $\backslash\text{AH}[\text{sub}][\text{sup}][\text{par}] = \text{A}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1418  $\backslash\text{defcomhrc}\{\text{AH}\}[\text{A}]$

$\backslash\text{DLH}$  •  $\backslash\text{DLH}[\text{sub}][\text{sup}][\text{par}] = \Delta_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

$\backslash\text{DBH}$  •  $\backslash\text{DBH}[\text{sub}][\text{sup}][\text{par}] = \mathbf{\Delta}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1419  $\backslash\text{defcomhrc}\{\text{DLH}\}[\{\backslash\text{mth}\{\Delta\}\}]$

1420  $\backslash\text{defcomhrc}\{\text{DBH}\}[\{\backslash\text{mth}[\text{mathbf}\{\Delta\}]\}]$

$\backslash\text{ELH}$  •  $\backslash\text{ELH}[\text{sub}][\text{sup}][\text{par}] = \Sigma_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

$\backslash\text{EBH}$  •  $\backslash\text{EBH}[\text{sub}][\text{sup}][\text{par}] = \mathbf{\Sigma}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1421  $\backslash\text{defcomhrc}\{\text{ELH}\}[\{\backslash\text{mth}\{\Sigma\}\}]$

1422  $\backslash\text{defcomhrc}\{\text{EBH}\}[\{\backslash\text{mth}[\text{mathbf}\{\Sigma\}]\}]$

$\backslash\text{ULH}$  •  $\backslash\text{ULH}[\text{sub}][\text{sup}][\text{par}] = \Pi_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

$\backslash\text{UBH}$  •  $\backslash\text{UBH}[\text{sub}][\text{sup}][\text{par}] = \mathbf{\Pi}_{\text{SUB}}^{\text{SUP}}[\text{PAR}]$

1423  $\backslash\text{defcomhrc}\{\text{ULH}\}[\{\backslash\text{mth}\{\Pi\}\}]$

1424  $\backslash\text{defcomhrc}\{\text{UBH}\}[\{\backslash\text{mth}[\text{mathbf}\{\Pi\}]\}]$

1425  $\backslash\text{fi}$

1426  $\%*****\%$

1427  $\%*****\%$

1428  $\%** \text{Macros for Graphs} *****\%$

1429  $\%*****\%$

1430  $\backslash\text{ifgrp@}$

```

1431 %%** Syntax *****%%

\GrpName ...
... 1432 \newcommand{\grpname}{G}
1433 \usrmthlatupp{Grp}{Name}{name}[\grpname]

\VerSet ...
... 1434 \newcommand{\versym}{v}
1435 \newcommand{\verset}{V}
1436 \cmdmthsetext{Ver}[\verset][\versym]
1437 \cmdmthsymelm{iver}[\versym_{I}]
1438 \cmdmthsymelm{fver}[\versym_{F}]

\EdgRel ...
1439 \newcommand{\edgrel}{E}
1440 \cmdmthrel{Edg}[\edgrel]

1441 %%** Semantics *****%%

\PthSet ...
\pthFun 1442 \newcommand{\pthsym}{\pi}
1443 \newcommand{\pthset}{Pth}
1444 \cmdmthsetext{Pth}[\pthset][\pthsym]
1445 \usrmth{path}{-}{argfun}

\pre ...
\suc 1446 \usrmth{pre}{-}{oargfun}
1447 \usrmth{suc}{-}{oargfun}

1448 \fi
1449 %*****%%

1450 %*****%%
1451 %%** Macros for Games *****%%
1452 %*****%%
1453 \ifgam@

1454 %%** Logic Games *****%%

\SATG ...
... 1455 %% Satisfiability Games
1456 \cmdtxtoparname{SATG}[Sat]
1457
1458 %% Validity Games
1459 \cmdtxtoparname{VALG}[Val]
1460
1461 %% Evaluation Games
1462 \cmdtxtoparname{EVLG}[Evl]
1463
1464 %% Synthesis Games
1465 \cmdtxtoparname{SYNG}[Syn]
1466
1467 %% Model-Checking Games
1468 \cmdtxtoparname{MCG}[MC]
1469
1470 %% Ehrenfeucht-Fraisse Games
1471 \cmdtxtoparname{EFG}[EF]
1472 %%** Syntax *****%%

\PlrSym ...
\OppSym 1473 \newcommand{\plrsym}{E}
1474 \cmdmthsym{Plr}[\plrsym]
1475 \newcommand{\oppsym}{A}
1476 \cmdmthsym{Opp}[\oppsym]

```

```

\ArenaName ...
... 1477 \newcommand{\arenaname}{A}
 1478 \usrmthlatupp{Arena}{Name}{name}[\arenaname]

\PosSet ...
... 1479 \newcommand{\possym}{v}
 1480 \newcommand{\posset}{Ps}
 1481 \cmdmthsetext{Pos}[\posset][\possym]
 1482 \cmdmthsymelm{ipos}[\possym_{I}]
 1483 \cmdmthsymelm{fpos}[\possym_{F}]
 1484 \cmdmthset{PPos}[\posset_{\PlrSym}]
 1485 \cmdmthsymelm{ppos}[\possym_{\PlrSym}]
 1486 \cmdmthset{OPos}[\posset_{\OppSym}]
 1487 \cmdmthsymelm{opos}[\possym_{\OppSym}]

\PlrFun ...
 1488 \newcommand{\plrfun}{pl}
 1489 \cmdmthfun{plr}[\plrfun]

\MovRel ...
 1490 \newcommand{\movrel}{Mv}
 1491 \cmdmthrel{Mov}[\movrel]

\GameName ...
... 1492 \newcommand{\gamename}{\Game}
 1493 \usrmthlatupp{Game}{Name}{name}[\gamename]

\WinSet ...
 1494 \newcommand{\winset}{Wn}
 1495 \cmdmthset{Win}[\winset]

\ObsSet ...
\obsFun 1496 \newcommand{\obsset}{Ob}
 1497 \cmdmthset{Obs}[\obsset]
 1498 \cmdmthfun{obs}

 1499 %** Semantics *****%

\HstSet ...
... 1500 \newcommand{\hstsym}{\varpi}
 1501 \newcommand{\hstset}{Hst}
 1502 \cmdmthsetext{Hst}[\hstset][\hstsym]
 1503 \cmdmthset{PHst}[\hstset_{\PlrSym}]
 1504 \cmdmthsymelm{phst}[\hstsym_{\PlrSym}]
 1505 \cmdmthset{OHst}[\hstset_{\OppSym}]
 1506 \cmdmthsymelm{ohst}[\hstsym_{\OppSym}]
 1507 \usrmth{play}{\}{argfun}

\PlaySet ...
\playFun 1508 \newcommand{\playsym}{\pi}
 1509 \newcommand{\playset}{Play}
 1510 \cmdmthsetext{Play}[\playset][\playsym]
 1511 \usrmth{hst}{\}{argfun}

\StrSet ...
... 1512 \newcommand{\strsym}{\sigma}
 1513 \newcommand{\strset}{Str}
 1514 \cmdmthsetext{Str}[\strset][\strsym]
 1515 \cmdmthset{PStr}[\strset_{\PlrSym}]
 1516 \cmdmthsymelm{pstr}[\strsym_{\PlrSym}]
 1517 \cmdmthset{OStr}[\strset_{\OppSym}]
 1518 \cmdmthsymelm{ostr}[\strsym_{\OppSym}]

```

```

\PrfSet ...
\prfFun 1519 \newcommand{\prfsym}{\xi}
 1520 \newcommand{\prfset}{Prf}
 1521 \cmdmthsetext{Prf}[\prfset][\prfsym]

\ent ...
\esc 1522 \usrmth{ent}{\}{oargfun}
 1523 \usrmth{esc}{\}{oargfun}

\int ...
\out 1524 \usrmth{int}{\}{oargfun}
 1525 \usrmth{out}{\}{oargfun}

\atr ...
\rch 1526 \usrmth{atr}{\}{oargfun}
 1527 \usrmth{rch}{\}{oargfun}

\lift ...
 1528 \usrmth{lift}{\}{oargfun}

\sol ...
 1529 \usrmth{sol}{\}{oargfun}

 1530 %** Qualitative Games on Graph *****%%

\BG ...
... 1531 %% Buchi Games
 1532 \cmdtxttoparname{BG}
 1533
 1534 %% Co-Buchi Games
 1535 \cmdtxttoparname{CG}
 1536
 1537 %% Parity Games
 1538 \cmdtxttoparname{PG}
 1539
 1540 %% Rabin Games
 1541 \cmdtxttoparname{RG}
 1542
 1543 %% Streett Games
 1544 \cmdtxttoparname{SG}
 1545
 1546 %% Muller Games
 1547 \cmdtxttoparname{MG}
 1548 %** Syntax *****%%

\EvnSym ...
\OddSym 1549 \newcommand{\evnsym}{0}
 1550 \cmdmthsym{Evnsym}[\evnsym]
 1551 \newcommand{\oddsym}{1}
 1552 \cmdmthsym{Oddsym}[\oddsym]

\PrtSet ...
\prtFun 1553 \newcommand{\prtsym}{p}
 1554 \newcommand{\prtset}{Pr}
 1555 \cmdmthsetext{Prt}[\prtset][\prtsym]
 1556 \cmdmthfun{prt}[pr]

 1557 %** Semantics *****%%

...

 1558 %** Quantitative Games on Graph *****%%

```

```

\EG ...
... 1559 %% Energy Games
1560 \cmdtxttoparname{EG}
1561
1562 %% Mean-Payoff Games
1563 \cmdtxttoparname{MPG}
1564
1565 %% Discounted-Payoff Games
1566 \cmdtxttoparname{DPG}
1567 %** Syntax *****%%

\MaxSym ...
\MinSym 1568 \newcommand{\maxsym}{\oplus}
1569 \cmdmthsym{Max}[\maxsym]
1570 \newcommand{\minsym}{\boxminus}
1571 \cmdmthsym{Min}[\minsym]

\WghSet ...
\wghFun 1572 \newcommand{\wghsym}{w}
1573 \newcommand{\wghset}{Wg}
1574 \cmdmthsetext{Wgh}[\wghset][\wghsym]
1575 \cmdmthfun{wgh}[wg]
1576 %** Semantics *****%%
...
1577 \fi
1578 %*****%%
1579 %*****%%
1580 %** Macros for Logics *****%%
1581 %*****%%
1582 \iflog@
1583 %** Propositional Logics *****%%

\BF ...
\QBF 1584 % Boolean Formulae
... 1585 \cmdtxttoparname{BF}
1586
1587 % Quantified Boolean Formulae
1588 \DeclareRobustCommand{\QBF}
1589 {\{\txname{Q}\}\BF}
1590 \DeclareRobustCommand{\EBF}
1591 {\ensuremath{\exists}\BF}
1592 \DeclareRobustCommand{\UBF}
1593 {\ensuremath{\forall}\BF}
1594 %** Syntax *****%%

\LogSig ...
... 1595 \newcommand{\logsig}{L}
1596 \usrmthlatupp{Log}{Sig}{sig}[\logsig]

\Tt ...
\Ff 1597 \newcommand{\ttsym}{\top}
1598 \usrmth{Tt}{\}{sym}[\ttsym]
1599 \newcommand{\ffsym}{\bot}
1600 \usrmth{Ff}{\}{sym}[\ffsym]

\LNeg ...
\LNot 1601 \newcommand{\lnegsym}{\neg}
1602 \usrmth{LNeg}{\}{luop}[\lnegsym]
1603 \newcommand{\lnotsym}{\sim}
1604 \usrmth{LNot}{\}{luop}[\lnotsym]

```



```

\LCon ...
\LDis 1605 \newcommand{\lconsym}{\land}
 1606 \usrmth{LCon}{-}{\lbop}{\lconsym}
 1607 \newcommand{\ldissym}{\lor}
 1608 \usrmth{LDis}{-}{\lbop}{\ldissym}

\LImp ...
\LCoi 1609 \newcommand{\limpsym}{\rightarrow}
 1610 \usrmth{LImp}{-}{\lbop}{\limpsym}
 1611 \newcommand{\lcoisym}{\leftrightharpoonup}
 1612 \usrmth{LCoi}{-}{\lbop}{\lcoisym}

\LExs ...
\LAll 1613 \newcommand{\lexssym}{\exists}
 1614 \usrmth{LExs}{-}{\luop}{\lexssym}
 1615 \newcommand{\lallsym}{\forall}
 1616 \usrmth{LAll}{-}{\luop}{\lallsym}

\APSet ...
... 1617 \newcommand{\apsym}{\wp}
 1618 \newcommand{\apset}{\mathcal{A}}
 1619 \cmdmthsetext{AP}{\apset} [\apsym]
 1620 \usrmth{ap}{-}{argfun}

\sub ...
 1621 \usrmth{sub}{-}{argfun}

\Cnt ...
\Qnt 1622 \usrmth{Cnt}{-}{sym}[C]
\Sym 1623 \usrmth{Qnt}{-}{sym}[Q]
 1624 \usrmth{Sym}{-}{sym}[\odot]

\QAE ...
\QEA 1625 \usrmth{QAE}{-}{sym}[\forall\exists]
 1626 \usrmth{QEA}{-}{sym}[\exists\forall]

\QntSet ...
... 1627 \newcommand{\qntsym}{\wp}
 1628 \newcommand{\qntset}{\mathcal{Q}}
 1629 \cmdmthsetext{Qnt}{\qntset} [\qntsym]

\free ...
\bound 1630 \usrmth{free}{-}{argfun}
 1631 \usrmth{bound}{-}{argfun}

\dep ...
\alt 1632 \usrmth{dep}{-}{argfun}
 1633 \usrmth{alt}{-}{argfun}

\cnf ...
\dnf 1634 \cmdtxtabr{cnf}
... 1635 \cmdtxtabr{dnf}
 1636 \cmdtxtabr{pnf}
 1637 \cmdtxtabr{nnf}
 1638 %%** Semantics *****%%

\LogStr ...
... 1639 \newcommand{\logstr}{\mathcal{L}}
 1640 \usrmthlatupp{Log}{Str}{str}[\logstr]

```

```

\ValSet ...
... 1641 \newcommand{\valsym}{\xi}
 1642 \newcommand{\valset}{\Val}
 1643 \cmdmthsetext{Val}[\valset][\valsym]

\AsgSet ...
... 1644 \newcommand{\asgsym}{\chi}
 1645 \newcommand{\asgset}{\Asg}
 1646 \cmdmthsetext{Asg}[\asgset][\asgsym]
 1647 %** First-Order Logics I *****%%

\FOL ...
... 1648 % First-Order Logic
 1649 \cmdtxtoparname{FOL}[Fol]
 1650 \cmdtxtoparname{FO}[FO]
 1651
 1652 % Monadic First-Order Logic
 1653 \DeclareRobustCommand{\MFOL}
 1654 {\txtname{M}\FOL}
 1655 \DeclareRobustCommand{\MFO}
 1656 {\txtname{M}\FO}
 1657 %** Syntax *****%%

\VarSig ...
... 1658 \newcommand{\varsig}{V}
 1659 \usrmthlatupp{Var}{Sig}{sig}[\varsig]
 1660 \newcommand{\varsym}{x}
 1661 \newcommand{\varset}{Vr}
 1662 \cmdmthsetext{Var}[\varset][\varsym]
 1663 \usrmth{var}{}{argfun}[vr]
 1664 \usrmth{dim}{}{argfun}[dm]

\ConSig ...
... 1665 \newcommand{\consig}{C}
 1666 \usrmthlatupp{Con}{Sig}{sig}[\consig]
 1667 \newcommand{\consym}{c}
 1668 \newcommand{\conset}{Cn}
 1669 \cmdmthsetext{Con}[\conset][\consym]
 1670 \usrmth{con}{}{argfun}[cn]

\FunSig ...
... 1671 \newcommand{\funsig}{F}
 1672 \usrmthlatupp{Fun}{Sig}{sig}[\funsig]
 1673 \newcommand{\funsym}{f}
 1674 \newcommand{\funset}{Fn}
 1675 \cmdmthsetext{Fun}[\funset][\funsym]
 1676 \usrmth{fun}{}{argfun}[fn]
 1677 \usrmth{art}{}{argfun}[ar]

\TerSig ...
... 1678 \newcommand{\tersig}{T}
 1679 \usrmthlatupp{Ter}{Sig}{sig}[\tersig]
 1680 \newcommand{\tersym}{t}
 1681 \newcommand{\terset}{Tr}
 1682 \cmdmthsetext{Ter}[\terset][\tersym]
 1683 \usrmth{ter}{}{argfun}

\RelSig ...
... 1684 \newcommand{\relsig}{R}
 1685 \usrmthlatupp{Rel}{Sig}{sig}[\relsig]
 1686 \newcommand{\relsym}{r}
 1687 \newcommand{\relset}{Rl}

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1688 \cmdnthsetext{Rel}[\relset][\relsym]
1689 \usrmth{rel}{\}{argfun}[rl]

\skm ...
1690 \usrmth{skm}{\}{argfun}

1691 %** Semantics *****%

\ConStr ...
... 1692 \newcommand{\constr}{C}
1693 \usrmthlatupp{Con}{Str}{str}[\constr]

\FunStr ...
... 1694 \newcommand{\funstr}{F}
1695 \usrmthlatupp{Fun}{Str}{str}[\funstr]

\TerStr ...
... 1696 \newcommand{\terstr}{T}
1697 \usrmthlatupp{Ter}{Str}{str}[\terstr]

\RelStr ...
... 1698 \newcommand{\relstr}{R}
1699 \usrmthlatupp{Rel}{Str}{str}[\relstr]

1700 %** First-Order Logics II *****%

\DF ...
\IF 1701 % Dependence-Friendly Logic
... 1702 \cmdtxtoparname{DF}
1703
1704 % Independence-Friendly Logic
1705 \cmdtxtoparname{IF}
1706
1707 % Dependence/Independence-Friendly Logic
1708 \cmdtxtoparname{DIF}
1709
1710 % Dependence Logic
1711 \cmdtxtoparname{DL}
1712
1713 % Team Logic
1714 \cmdtxtoparname{TL}
1715
1716 % Alternating Dependence-Friendly Logic
1717 \cmdtxtoparname{ADF}
1718
1719 % Alternating Independence-Friendly Logic
1720 \cmdtxtoparname{AIF}
1721
1722 % Alternating Dependence/Independence-Friendly Logic
1723 \cmdtxtoparname{ADIF}
...
1724 %** Syntax *****%

\LEExs ...
\LAAll 1725 \newcommand{\leexssym}{\Sigma}
1726 \usrmth{LEExs}{\}{luop}[\leexssym]
1727 \newcommand{\laallsym}{\Pi}
1728 \usrmth{LAAll}{\}{luop}[\laallsym]

1729 %** Semantics *****%

...
1730 %** Second-Order Logics I *****%

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\SOL ...
... 1731 % Second-Order Logic
1732 \cmdtxttoparname{SOL}[Sol]
1733 \cmdtxttoparname{SO}
1734
1735 % Weak Second-Order Logic
1736 \DeclareRobustCommand{\WSOL}
1737 {\{\txtrname{W}\}\SOL}
1738 \DeclareRobustCommand{\WSO}
1739 {\{\txtrname{W}\}\SO}
1740
1741 % coWeak Second-Order Logic
1742 \DeclareRobustCommand{\coWSOL}
1743 {\{\txtrname{coW}\}\SOL}
1744 \DeclareRobustCommand{\coWSO}
1745 {\{\txtrname{coW}\}\SO}
1746
1747 % Monadic Second-Order Logic
1748 \DeclareRobustCommand{\MSOL}
1749 {\{\txtrname{M}\}\SOL}
1750 \DeclareRobustCommand{\MSO}
1751 {\{\txtrname{M}\}\SO}
1752
1753 % Weak Monadic Second-Order Logic
1754 \DeclareRobustCommand{\WMSOL}
1755 {\{\txtrname{W}\}\MSOL}
1756 \DeclareRobustCommand{\WMSO}
1757 {\{\txtrname{W}\}\MSO}
1758
1759 % coWeak Monadic Second-Order Logic
1760 \DeclareRobustCommand{\coWMSOL}
1761 {\{\txtrname{coW}\}\MSOL}
1762 \DeclareRobustCommand{\coWMSO}
1763 {\{\txtrname{coW}\}\MSO}
1764 %** Syntax *****%

\FVarSet ...
... 1765 \newcommand{\fvarsym}{x}
1766 \newcommand{\fvarset}{FVr}
1767 \cmdmthsetext{FVar}[\fvarset][\fvarsym]

\SVarSet ...
... 1768 \newcommand{\svarsym}{X}
1769 \newcommand{\svarset}{SVr}
1770 \cmdmthsetext{SVar}[\svarset][\svarsym]
1771 %** Semantics *****%

...

1772 %** Second-Order Logics II *****%

\TL ...
\CL 1773 % Tree Logic
\PL 1774 \cmdtxttoparname{TL}
... 1775
1776 % Weak Tree Logic
1777 \DeclareRobustCommand{\WTL}
1778 {\{\txtrname{W}\}\TL}
1779
1780 % coWeak Tree Logic
1781 \DeclareRobustCommand{\coWTL}
1782 {\{\txtrname{coW}\}\TL}
1783

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1784 % Monadic Tree Logic
1785 \DeclareRobustCommand{\MTL}
1786 {\txtname{M}}\TL}
1787
1788 % Weak Monadic Tree Logic
1789 \DeclareRobustCommand{\WMTL}
1790 {\txtname{W}}\MTL}
1791
1792 % coWeak Monadic Tree Logic
1793 \DeclareRobustCommand{\coWMTL}
1794 {\txtname{coW}}\MTL}
1795
1796 % Chain Logic
1797 \cmdtxttoparname{CL}
1798
1799 % Weak Chain Logic
1800 \DeclareRobustCommand{\WCL}
1801 {\txtname{W}}\CL}
1802
1803 % coWeak Chain Logic
1804 \DeclareRobustCommand{\coWCL}
1805 {\txtname{coW}}\CL}
1806
1807 % Monadic Chain Logic
1808 \DeclareRobustCommand{\MCL}
1809 {\txtname{M}}\CL}
1810
1811 % Weak Monadic Chain Logic
1812 \DeclareRobustCommand{\WMCL}
1813 {\txtname{W}}\MCL}
1814
1815 % coWeak Monadic Chain Logic
1816 \DeclareRobustCommand{\coWMCL}
1817 {\txtname{coW}}\MCL}
1818
1819 % Path Logic
1820 \cmdtxttoparname{PL}
1821
1822 % Weak Path Logic
1823 \DeclareRobustCommand{\WPL}
1824 {\txtname{W}}\PL}
1825
1826 % coWeak Path Logic
1827 \DeclareRobustCommand{\coWPL}
1828 {\txtname{coW}}\PL}
1829
1830 % Monadic Path Logic
1831 \DeclareRobustCommand{\MPL}
1832 {\txtname{M}}\PL}
1833
1834 % Weak Monadic Path Logic
1835 \DeclareRobustCommand{\WMPL}
1836 {\txtname{W}}\MPL}
1837
1838 % coWeak Monadic Path Logic
1839 \DeclareRobustCommand{\coWMPL}
1840 {\txtname{coW}}\MPL}
1841 %** Syntax *****%%
...
1842 %** Semantics *****%%
...
1843 %** Modal Logics I *****%%

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\ML ...
\GML 1844 % Modal Logic
... 1845 \cmdtxttoparname{ML}
1846
1847 % Graded Modal Logic
1848 \DeclareRobustCommand{\GML}
1849 {\txtname{G}\ML}
1850
1851 % Quantified Modal Logic
1852 \DeclareRobustCommand{\QML}
1853 {\txtname{Q}\ML}
1854 \DeclareRobustCommand{\EML}
1855 {\ensuremath{\exists}\ML}
1856 \DeclareRobustCommand{\UML}
1857 {\ensuremath{\forall}\ML}
1858 %** Syntax *****%

\Opr ...
1859 \usrmth{Opr}{\sym}[Op]

\DMod ...
\BMod 1860 \usrmth{DMod}{\sym}[\Diamond]
1861 \usrmth{BMod}{\sym}[\Box]

\Exs ...
\All 1862 \DeclareRobustCommand{\Exs}
1863 {\ifstarvar{\@sexs}{\@exs}}
1864 \DeclareRobustCommand{\@sexs}[1]
1865 {\mth{DMod}[#1]}
1866 \DeclareRobustCommand{\@exs}[1]
1867 {\mth{defval{\argmid{\langle}{#1}{\rangle}}{\DMod}}}
1868 \DeclareRobustCommand{\All}
1869 {\ifstarvar{\@sall}{\@all}}
1870 \DeclareRobustCommand{\@sall}[1]
1871 {\mth{BMod}[#1]}
1872 \DeclareRobustCommand{\@all}[1]
1873 {\mth{defval{\argmid{\left[]{\#1}{\right]}}{\BMod}}}
1874 %** Semantics *****%

\KrpStr ...
... 1875 \newcommand{\krpstr}{K}
1876 \usrmthlatupp{Krp}{Str}{str}[\krpstr]

\WrlSet ...
... 1877 \newcommand{\wrlsym}{w}
1878 \newcommand{\wrlset}{W}
1879 \cmdmthsetext{Wrl}[\wrlset][\wrlsym]
1880 \cmdmthsymelm{iwrl}[\wrlsym_{I}]

\AccRel ...
\TrnRel 1881 \newcommand{\accsym}{R}
1882 \cmdmthrel{Acc}[\accsym]
1883 \cmdmthrel{Trn}[\accsym]

\labFun ...
1884 \newcommand{\labsym}{\lambda}
1885 \cmdmthfun{lab}[\labsym]

\PthSet ...
... 1886 \providecommand{\pthsym}{\pi}
1887 \providecommand{\pthset}{Pth}
1888 \cmdmthsetext{Pth}[\pthset][\pthsym]
1889 \usrmth{path}{\argfun}

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1890 %%** Modal Logics II *****%%
\MC ...
\GMC 1891 % Mu Calculus
... 1892 \cmdtxtopname{MC}[\ensuremath{\mu}-Calculus]
1893
1894 % Graded Mu Calculus
1895 \DeclareRobustCommand{\GMC}
1896 {\txtname{G}\MC}
1897
1898 % Quantified Mu Calculus
1899 \DeclareRobustCommand{\QMC}
1900 {\txtname{Q}\MC}
1901 \DeclareRobustCommand{\EMC}
1902 {\ensuremath{\exists}\MC}
1903 \DeclareRobustCommand{\UMC}
1904 {\ensuremath{\forall}\MC}
1905
1906 % Alternation-Free Mu Calculus
1907 \DeclareRobustCommand{\AFMC}
1908 {\txtname{AF}\MC}
1909
1910 % Alternation-Free Graded Mu Calculus
1911 \DeclareRobustCommand{\AFGMC}
1912 {\txtname{AF}\GMC}
1913
1914 % Quantified Alternation-Free Mu Calculus
1915 \DeclareRobustCommand{\QAFMC}
1916 {\txtname{Q}\AFMC}
1917 \DeclareRobustCommand{\EAFMC}
1918 {\ensuremath{\exists}\AFMC}
1919 \DeclareRobustCommand{\UAFMC}
1920 {\ensuremath{\forall}\AFMC}
1921
1922 %%** Syntax *****%%
...
1923 %%** Semantics *****%%
...
1924 %%** Temporal Logics I *****%%
\PTL ...
\LTL 1925 % Propositional Temporal Logic
... 1926 \cmdtxtopname{PTL}
1927
1928 % Quantified Propositional Temporal Logic
1929 \DeclareRobustCommand{\QPTL}
1930 {\txtname{Q}\PTL}
1931 \DeclareRobustCommand{\EPTL}
1932 {\ensuremath{\exists}\PTL}
1933 \DeclareRobustCommand{\UPTL}
1934 {\ensuremath{\forall}\PTL}
1935
1936 % Linear Temporal Logic
1937 \cmdtxtopname{LTL}
1938
1939 % Quantified Linear Temporal Logic
1940 \DeclareRobustCommand{\QLTL}
1941 {\txtname{Q}\LTL}
1942 \DeclareRobustCommand{\ELTL}
1943 {\ensuremath{\exists}\LTL}
1944 \DeclareRobustCommand{\ULTL}
1945 {\ensuremath{\forall}\LTL}

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1946 %%** Syntax *****%%

\X ...
... 1947 \usrmth{X}{-}{sym}[X\,]
 1948 \usrmth{F}{-}{sym}[F\,]
 1949 \usrmth{G}{-}{sym}[G\,]
 1950 \usrmth{U}{-}{sym}[\,U\,]
 1951 \usrmth{R}{-}{sym}[\,R\,]

\Y ...
... 1952 \usrmth{Y}{-}{sym}[Y\,]
 1953 \usrmth{P}{-}{sym}[P\,]\let\SavePilcrow\P
 1954 \usrmth{H}{-}{sym}[H\,]\let\SaveDoubleAcute\H
 1955 \usrmth{S}{-}{sym}[\,S\,]\let\SaveSectionSymbol\S
 1956 \usrmth{B}{-}{sym}[\,B\,]

1957 %%** Semantics *****%%
...

1958 %%** Temporal Logics II *****%%

\PDL ...
\CTL 1959 % Propositional Dynamic Logic
... 1960 \cmdtxtopname{PDL}
 1961
 1962 % Computation Tree Logic
 1963 \cmdtxtopname{CTL}
 1964
 1965 % Weak Computation Tree Logic
 1966 \DeclareRobustCommand{\WCTL}
 1967 {\txtrname{W}}\CTL}
 1968
 1969 % Quantified Computation Tree Logic
 1970 \DeclareRobustCommand{\QCTL}
 1971 {\txtrname{Q}}\CTL}
 1972 \DeclareRobustCommand{\ECTL}
 1973 {\ensuremath{\exists}\CTL}
 1974 \DeclareRobustCommand{\UCTL}
 1975 {\ensuremath{\forall}\CTL}
 1976
 1977 % Improved Computation Tree Logic
 1978 \cmdtxtopname{CTLP}[CTL^{+}]
 1979
 1980 % Weak Improved Computation Tree Logic
 1981 \DeclareRobustCommand{\WCTLP}
 1982 {\txtrname{W}}\CTLP}
 1983
 1984 % Quantified Improved Computation Tree Logic
 1985 \DeclareRobustCommand{\QCTLP}
 1986 {\txtrname{Q}}\CTLP}
 1987 \DeclareRobustCommand{\ECTLP}
 1988 {\ensuremath{\exists}\CTLP}
 1989 \DeclareRobustCommand{\UCTLP}
 1990 {\ensuremath{\forall}\CTLP}
 1991
 1992 % Full Computation Tree Logic
 1993 \cmdtxtopname{CTLS}[CTL*]
 1994
 1995 % Weak Full Computation Tree Logic
 1996 \DeclareRobustCommand{\WCTLS}
 1997 {\txtrname{W}}\CTLS}
 1998
 1999 % Quantified Full Computation Tree Logic
 2000 \DeclareRobustCommand{\QCTLS}

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2001 {\txtname{Q}}\CTLs}
2002 \DeclareRobustCommand{\ECTLS}
2003 {\ensuremath{\exists}\CTLs}
2004 \DeclareRobustCommand{\UCTLS}
2005 {\ensuremath{\forall}\CTLs}
2006 %%** Syntax *****%%

\E ...
\A 2007 \usrmth{E}{\sym}
2008 \usrmth{A}{\sym}
2009 %%** Semantics *****%%
...
2010 %%** Game Logics I *****%%

\ATL ...
... 2011 % Alternating Temporal Logic
2012 \cmdtxttoparname{ATL}
2013
2014 % Weak Alternating Tree Logic
2015 \DeclareRobustCommand{\WATL}
2016 {\txtname{W}}\ATL}
2017
2018 % Quantified Alternating Temporal Logic
2019 \DeclareRobustCommand{\QATL}
2020 {\txtname{Q}}\ATL}
2021 \DeclareRobustCommand{\EATL}
2022 {\ensuremath{\exists}\ATL}
2023 \DeclareRobustCommand{\UATL}
2024 {\ensuremath{\forall}\ATL}
2025
2026 % Improved Alternating Temporal Logic
2027 \cmdtxttoparname{ATLP}[ATL^{+}]
2028
2029 % Weak Improved Alternating Tree Logic
2030 \DeclareRobustCommand{\WATLP}
2031 {\txtname{W}}\ATLP}
2032
2033 % Quantified Improved Alternating Temporal Logic
2034 \DeclareRobustCommand{\QATLP}
2035 {\txtname{Q}}\ATLP}
2036 \DeclareRobustCommand{\EATLP}
2037 {\ensuremath{\exists}\ATLP}
2038 \DeclareRobustCommand{\UATLP}
2039 {\ensuremath{\forall}\ATLP}
2040
2041 % Full Alternating Temporal Logic
2042 \cmdtxttoparname{ATLS}[ATL*]
2043
2044 % Weak Full Alternating Tree Logic
2045 \DeclareRobustCommand{\WATLS}
2046 {\txtname{W}}\ATLS}
2047
2048 % Quantified Full Alternating Temporal Logic
2049 \DeclareRobustCommand{\QATLS}
2050 {\txtname{Q}}\ATLS}
2051 \DeclareRobustCommand{\EATLS}
2052 {\ensuremath{\exists}\ATLS}
2053 \DeclareRobustCommand{\UATLS}
2054 {\ensuremath{\forall}\ATLS}
2055 %%** Syntax *****%%

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\EEs ...
\AA11 2056 \DeclareRobustCommand{\EEs}[1]
 2057 {\mth{\argmid{\langle\!\langle}{\defval{#1}{\emptyset}}{\rangle\!\rangle}}
 2058 \DeclareRobustCommand{\AA11}[1]
 2059 {\mth{\argmid{\left[\left[}{\defval{#1}{\emptyset}}{\right]\right]}}}

 2060 %%** Semantics *****%%

\CGS ...
 2061 \cmdtxtname{CGS}

\CGSStr ...
 ... 2062 \newcommand{\cgsstr}{G}
 2063 \usrmthlatupp{CGS}{Str}{str}[\cgsstr]

\AgnSet ...
 ... 2064 \newcommand{\agnsym}{a}
 2065 \newcommand{\agnset}{Ag}
 2066 \cmdmthsetext{Agn}[\agnset][\agnsym]

\ActSet ...
 ... 2067 \newcommand{\actsym}{c}
 2068 \newcommand{\actset}{Ac}
 2069 \cmdmthsetext{Act}[\actset][\actsym]

\PosSet ...
 ... 2070 \providecommand{\possym}{v}
 2071 \providecommand{\posset}{Ps}
 2072 \cmdmthsetext{Pos}[\posset][\possym]
 2073 \cmdmthsymelm{ipos}[\possym_{I}]
 2074 \cmdmthsymelm{fpos}[\possym_{F}]
 2075 \cmdmthset{PPos}[\posset_{\PlrSym}]
 2076 \cmdmthsymelm{ppos}[\possym_{\PlrSym}]
 2077 \cmdmthset{OPos}[\posset_{\OppSym}]
 2078 \cmdmthsymelm{opos}[\possym_{\OppSym}]

\SttSet ...
 ... 2079 \newcommand{\sttsym}{s}
 2080 \newcommand{\sttset}{St}
 2081 \cmdmthsetext{Stt}[\sttset][\sttsym]
 2082 \cmdmthset{IStt}[\sttset_{I}]
 2083 \cmdmthsymelm{istt}[\sttsym_{I}]
 2084 \cmdmthset{FStt}[\sttset_{F}]
 2085 \cmdmthsymelm{fstt}[\sttsym_{F}]

\DecSet ...
 ... 2086 \newcommand{\decsym}{d}
 2087 \newcommand{\decset}{Dc}
 2088 \cmdmthsetext{Dec}[\decset][\decsym]

\movFun ...
\movRel 2089 \newcommand{\movsym}{\tau}
 2090 \cmdmthfun{mov}[\movsym]
 2091 \cmdmthrel{mov}[\movsym]

\trnFun ...
\trnRel 2092 \newcommand{\trnsym}{\delta}
 2093 \cmdmthfun{trn}[\trnsym]
 2094 \cmdmthrel{trn}[\trnsym]

\PrfSet ...
 2095 \providecommand{\prfsym}{\xi}
 2096 \providecommand{\prfset}{Prf}
 2097 \cmdmthsetext{Prf}[\prfset][\prfsym]

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\HstSet ...
... 2098 \providecommand{\hstsym}{\varpi}
2099 \providecommand{\hstset}{Hst}
2100 \cmdmthsettext{Hst}[\hstset][\hstsym]
2101 \cmdmthset{PHst}[\hstset_{\PlrSym}]
2102 \cmdmthsymelm{phst}[\hstsym_{\PlrSym}]
2103 \cmdmthset{OHst}[\hstset_{\OppSym}]
2104 \cmdmthsymelm{ohst}[\hstsym_{\OppSym}]
2105 \usrmth{hst}{-}{argfun}

\PlaySet ...
... 2106 \providecommand{\playsym}{\pi}
2107 \providecommand{\playset}{Play}
2108 \cmdmthsettext{Play}[\playset][\playsym]
2109 \usrmth{play}{-}{argfun}

\PlnSet ...
... 2110 \providecommand{\plnsym}{\rho}
2111 \providecommand{\plnset}{Pln}
2112 \cmdmthsettext{Pln}[\plnset][\plnsym]
2113 \cmdmthset{PPln}[\plnset_{\PlrSym}]
2114 \cmdmthsymelm{pPln}[\plnsym_{\PlrSym}]
2115 \cmdmthset{OPln}[\plnset_{\OppSym}]
2116 \cmdmthsymelm{oPln}[\plnsym_{\OppSym}]

\StrSet ...
... 2117 \providecommand{\strsym}{\sigma}
2118 \providecommand{\strset}{Str}
2119 \cmdmthsettext{Str}[\strset][\strsym]
2120 \cmdmthset{PStr}[\strset_{\PlrSym}]
2121 \cmdmthsymelm{pstr}[\strsym_{\PlrSym}]
2122 \cmdmthset{OStr}[\strset_{\OppSym}]
2123 \cmdmthsymelm{ostr}[\strsym_{\OppSym}]

2124 %** Game Logics II *****%

\PL ...
... 2125 % Plan Logic
2126 \cmdtxttoparname{PL}
2127
2128 \DeclareRobustCommand{\EPL}
2129 {\ensuremath{\exists}\PL}
2130 \DeclareRobustCommand{\UPL}
2131 {\ensuremath{\forall}\PL}
2132
2133 \DeclareRobustCommand{\FPL}
2134 {\{\textname{F}\}\PL}
2135
2136 \DeclareRobustCommand{\EFPL}
2137 {\ensuremath{\exists}\FPL}
2138 \DeclareRobustCommand{\UFPL}
2139 {\ensuremath{\forall}\FPL}
2140
2141 % One-Goal Plan Logic
2142 \DeclareRobustCommandx{\OGPL}[3][1=, 2=, 3=]
2143 {\PL[#1][#2][1g\arglef{,}{#3}]}
2144
2145 \DeclareRobustCommand{\EOGPL}
2146 {\ensuremath{\exists}\OGPL}
2147 \DeclareRobustCommand{\UOGPL}
2148 {\ensuremath{\forall}\OGPL}
2149
2150 \DeclareRobustCommand{\FOGPL}

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2151 {{\txtname{F}}\OGPL}
2152
2153 \DeclareRobustCommand{\EFOGPL}
2154 {\ensuremath{\exists}\FOGPL}
2155 \DeclareRobustCommand{\UFOGPL}
2156 {\ensuremath{\forall}\FOGPL}
2157
2158 % Conjunctive-Goal Plan Logic
2159 \DeclareRobustCommandx{\CGPL}[3][1=, 2=, 3=]
2160 {\PL[#1][#2][cg\argleft{,}{#3}]}
2161
2162 \DeclareRobustCommand{\ECGPL}
2163 {\ensuremath{\exists}\CGPL}
2164 \DeclareRobustCommand{\UCGPL}
2165 {\ensuremath{\forall}\CGPL}
2166
2167 \DeclareRobustCommand{\FCGPL}
2168 {{\txtname{F}}\CGPL}
2169
2170 \DeclareRobustCommand{\EFCGPL}
2171 {\ensuremath{\exists}\FCGPL}
2172 \DeclareRobustCommand{\UFCGPL}
2173 {\ensuremath{\forall}\FCGPL}
2174
2175 % Disjunctive-Goal Plan Logic
2176 \DeclareRobustCommandx{\DGPL}[3][1=, 2=, 3=]
2177 {\PL[#1][#2][dg\argleft{,}{#3}]}
2178
2179 \DeclareRobustCommand{\EDGPL}
2180 {\ensuremath{\exists}\DGPL}
2181 \DeclareRobustCommand{\UDGPL}
2182 {\ensuremath{\forall}\DGPL}
2183
2184 \DeclareRobustCommand{\FDGPL}
2185 {{\txtname{F}}\DGPL}
2186
2187 \DeclareRobustCommand{\EFDGPL}
2188 {\ensuremath{\exists}\FDGPL}
2189 \DeclareRobustCommand{\UFDGPL}
2190 {\ensuremath{\forall}\FDGPL}
2191
2192 % Alternating-Goal Plan Logic
2193 \DeclareRobustCommandx{\AGPL}[3][1=, 2=, 3=]
2194 {\PL[#1][#2][ag\argleft{,}{#3}]}
2195
2196 \DeclareRobustCommand{\EAGPL}
2197 {\ensuremath{\exists}\AGPL}
2198 \DeclareRobustCommand{\UAGPL}
2199 {\ensuremath{\forall}\AGPL}
2200
2201 \DeclareRobustCommand{\FAGPL}
2202 {{\txtname{F}}\AGPL}
2203
2204 \DeclareRobustCommand{\EFAGPL}
2205 {\ensuremath{\exists}\FAGPL}
2206 \DeclareRobustCommand{\UFAGPL}
2207 {\ensuremath{\forall}\FAGPL}
2208
2209 % Extended-Goal Plan Logic
2210 \DeclareRobustCommandx{\EGPL}[3][1=, 2=, 3=]
2211 {\PL[#1][#2][eg\argleft{,}{#3}]}
2212
2213 \DeclareRobustCommand{\EEGPL}

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2214 {\ensuremath{\exists}\EGPL}
2215 \DeclareRobustCommand{\UEGPL}
2216 {\ensuremath{\forall}\EGPL}
2217
2218 \DeclareRobustCommand{\FEGPL}
2219 {\{\textname{F}\}\EGPL}
2220
2221 \DeclareRobustCommand{\EFEGPL}
2222 {\ensuremath{\exists}\FEGPL}
2223 \DeclareRobustCommand{\UFEGPL}
2224 {\ensuremath{\forall}\FEGPL}
2225
2226 % Boolean-Goal Plan Logic
2227 \DeclareRobustCommandx{\BGPL}[3][1=, 2=, 3=]
2228 {\PL[#1][#2][bg\arglef{,}\{#3}]}
2229
2230 \DeclareRobustCommand{\EBGPL}
2231 {\ensuremath{\exists}\BGPL}
2232 \DeclareRobustCommand{\UBGPL}
2233 {\ensuremath{\forall}\BGPL}
2234
2235 \DeclareRobustCommand{\FBGPL}
2236 {\{\textname{F}\}\BGPL}
2237
2238 \DeclareRobustCommand{\EFBGPL}
2239 {\ensuremath{\exists}\FBGPL}
2240 \DeclareRobustCommand{\UFBGPL}
2241 {\ensuremath{\forall}\FBGPL}
2242
2243 % Undefined-Goal Plan Logic
2244 \DeclareRobustCommandx{\XGPL}[3][1=, 2=, 3=]
2245 {\PL[#1][#2][xg\arglef{,}\{#3}]}
2246
2247 \DeclareRobustCommand{\EXGPL}
2248 {\ensuremath{\exists}\XGPL}
2249 \DeclareRobustCommand{\UXGPL}
2250 {\ensuremath{\forall}\XGPL}
2251
2252 \DeclareRobustCommand{\FXGPL}
2253 {\{\textname{F}\}\XGPL}
2254
2255 \DeclareRobustCommand{\EFXGPL}
2256 {\ensuremath{\exists}\FXGPL}
2257 \DeclareRobustCommand{\UFXGPL}
2258 {\ensuremath{\forall}\FXGPL}

\SL ...
... 2259 % Strategy Logic
2260 \cmdtxttoparname{SL}
2261
2262 \DeclareRobustCommand{\ESL}
2263 {\ensuremath{\exists}\SL}
2264 \DeclareRobustCommand{\USL}
2265 {\ensuremath{\forall}\SL}
2266
2267 \DeclareRobustCommand{\FSL}
2268 {\{\textname{F}\}\SL}
2269
2270 \DeclareRobustCommand{\EFSL}
2271 {\ensuremath{\exists}\FSL}
2272 \DeclareRobustCommand{\UFSL}
2273 {\ensuremath{\forall}\FSL}
2274
2275 % One-Goal Strategy Logic

```

```

2276 \DeclareRobustCommandx{\OGSL}[3][1=, 2=, 3=]
2277 {\SL[#1][#2][1g\argleft{,}{#3}]}
2278
2279 \DeclareRobustCommand{\EOGSL}
2280 {\ensuremath{\exists}\OGSL}
2281 \DeclareRobustCommand{\UOGSL}
2282 {\ensuremath{\forall}\OGSL}
2283
2284 \DeclareRobustCommand{\FOGSL}
2285 {\{\textname{F}\}\OGSL}
2286
2287 \DeclareRobustCommand{\EFOGSL}
2288 {\ensuremath{\exists}\FOGSL}
2289 \DeclareRobustCommand{\UFOGSL}
2290 {\ensuremath{\forall}\FOGSL}
2291
2292 % Conjunctive-Goal Strategy Logic
2293 \DeclareRobustCommandx{\CGSL}[3][1=, 2=, 3=]
2294 {\SL[#1][#2][cg\argleft{,}{#3}]}
2295
2296 \DeclareRobustCommand{\ECGSL}
2297 {\ensuremath{\exists}\CGSL}
2298 \DeclareRobustCommand{\UCGSL}
2299 {\ensuremath{\forall}\CGSL}
2300
2301 \DeclareRobustCommand{\FCGSL}
2302 {\{\textname{F}\}\CGSL}
2303
2304 \DeclareRobustCommand{\EFCGSL}
2305 {\ensuremath{\exists}\FCGSL}
2306 \DeclareRobustCommand{\UFCGSL}
2307 {\ensuremath{\forall}\FCGSL}
2308
2309 % Disjunctive-Goal Strategy Logic
2310 \DeclareRobustCommandx{\DGPL}[3][1=, 2=, 3=]
2311 {\SL[#1][#2][dg\argleft{,}{#3}]}
2312
2313 \DeclareRobustCommand{\EDGPL}
2314 {\ensuremath{\exists}\DGPL}
2315 \DeclareRobustCommand{\UDGPL}
2316 {\ensuremath{\forall}\DGPL}
2317
2318 \DeclareRobustCommand{\FDGPL}
2319 {\{\textname{F}\}\DGPL}
2320
2321 \DeclareRobustCommand{\EFDGPL}
2322 {\ensuremath{\exists}\FDGPL}
2323 \DeclareRobustCommand{\UFDGPL}
2324 {\ensuremath{\forall}\FDGPL}
2325
2326 % Alternating-Goal Strategy Logic
2327 \DeclareRobustCommandx{\AGSL}[3][1=, 2=, 3=]
2328 {\SL[#1][#2][ag\argleft{,}{#3}]}
2329
2330 \DeclareRobustCommand{\EAGSL}
2331 {\ensuremath{\exists}\AGSL}
2332 \DeclareRobustCommand{\UAGSL}
2333 {\ensuremath{\forall}\AGSL}
2334
2335 \DeclareRobustCommand{\FAGSL}
2336 {\{\textname{F}\}\AGSL}
2337
2338 \DeclareRobustCommand{\EFAGSL}

```

```

2339 {\ensuremath{\exists}\FAGSL}
2340 \DeclareRobustCommand{\UFAGSL}
2341 {\ensuremath{\forall}\FAGSL}
2342
2343 % Extended-Goal Strategy Logic
2344 \DeclareRobustCommandx{\EGSL}[3][1=, 2=, 3=]
2345 {\SL[#1][#2][eg\arglef{,}{#3}]}
2346
2347 \DeclareRobustCommand{\EEGSL}
2348 {\ensuremath{\exists}\EGSL}
2349 \DeclareRobustCommand{\UEGSL}
2350 {\ensuremath{\forall}\EGSL}
2351
2352 \DeclareRobustCommand{\FEGSL}
2353 {\{\textname{F}\}\EGSL}
2354
2355 \DeclareRobustCommand{\EFEGSL}
2356 {\ensuremath{\exists}\FEGSL}
2357 \DeclareRobustCommand{\UFEGSL}
2358 {\ensuremath{\forall}\FEGSL}
2359
2360 % Boolean-Goal Strategy Logic
2361 \DeclareRobustCommandx{\BGSL}[3][1=, 2=, 3=]
2362 {\SL[#1][#2][bg\arglef{,}{#3}]}
2363
2364 \DeclareRobustCommand{\EBGSL}
2365 {\ensuremath{\exists}\BGSL}
2366 \DeclareRobustCommand{\UBGSL}
2367 {\ensuremath{\forall}\BGSL}
2368
2369 \DeclareRobustCommand{\FBGSL}
2370 {\{\textname{F}\}\BGSL}
2371
2372 \DeclareRobustCommand{\EFBGSL}
2373 {\ensuremath{\exists}\FBGSL}
2374 \DeclareRobustCommand{\UFBGSL}
2375 {\ensuremath{\forall}\FBGSL}
2376
2377 % Nested-Goal Strategy Logic
2378 \DeclareRobustCommandx{\NGSL}[3][1=, 2=, 3=]
2379 {\SL[#1][#2][ng\arglef{,}{#3}]}
2380
2381 \DeclareRobustCommand{\ENGSL}
2382 {\ensuremath{\exists}\NGSL}
2383 \DeclareRobustCommand{\UNGSL}
2384 {\ensuremath{\forall}\NGSL}
2385
2386 \DeclareRobustCommand{\FNGSL}
2387 {\{\textname{F}\}\NGSL}
2388
2389 \DeclareRobustCommand{\EFNGSL}
2390 {\ensuremath{\exists}\FNGSL}
2391 \DeclareRobustCommand{\UFNGSL}
2392 {\ensuremath{\forall}\FNGSL}
2393
2394 % Undefined-Goal Strategy Logic
2395 \DeclareRobustCommandx{\XGSL}[3][1=, 2=, 3=]
2396 {\SL[#1][#2][xg\arglef{,}{#3}]}
2397
2398 \DeclareRobustCommand{\EXGSL}
2399 {\ensuremath{\exists}\XGSL}
2400 \DeclareRobustCommand{\UXGSL}
2401 {\ensuremath{\forall}\XGSL}

```

```

2402
2403 \DeclareRobustCommand{\FXGSL}
2404 {\textname{F}}\XGSL}
2405
2406 \DeclareRobustCommand{\EFXGSL}
2407 {\ensuremath{\exists}\FXGSL}
2408 \DeclareRobustCommand{\UFXGSL}
2409 {\ensuremath{\forall}\FXGSL}
2410 %%** Syntax *****%

\BndSet ...
... 2411 \newcommand{\bndsym}{\flat}
2412 \newcommand{\bndset}{Bn}
2413 \cmdmthsetext{Bnd}[\bndset][\bndsym]
2414 \cmdmthsymelm{idbnd}[\bndsym_{\text{id}}]
2415 \usrmth{bnd}{-}{argfun}

\psn ...
2416 \usrmth{psn}{-}{argfun}

2417 %%** Semantics *****%

\nxt ...
2418 \usrmth{nxt}{-}{argfun}

2419 \fi
2420 %*****%
2421 %*****%
2422 %%** Macros for Automata *****%
2423 %*****%
2424 \ifaut@
2425 %%** Finite Word Automata *****%

\DFA ...
... 2426 \cmdtxtoparname{DFA}\cmdtxtoparname{NFA}\cmdtxtoparname{UFA}\cmdtxtoparname{AFA}
2427
2428 \cmdtxtoparname{DWA}\cmdtxtoparname{NWA}\cmdtxtoparname{UWA}\cmdtxtoparname{AWA}
2429
2430 \cmdtxtoparname{DFW}\cmdtxtoparname{NFW}\cmdtxtoparname{UFW}\cmdtxtoparname{AFW}
2431 \cmdtxtoparname{DWW}\cmdtxtoparname{NWW}\cmdtxtoparname{UWW}\cmdtxtoparname{AWW}
2432 \cmdtxtoparname{DBW}\cmdtxtoparname{NBW}\cmdtxtoparname{UBW}\cmdtxtoparname{ABW}
2433 \cmdtxtoparname{DCW}\cmdtxtoparname{NCW}\cmdtxtoparname{UCW}\cmdtxtoparname{ACW}
2434 \cmdtxtoparname{DPW}\cmdtxtoparname{NPW}\cmdtxtoparname{UPW}\cmdtxtoparname{APW}
2435 \cmdtxtoparname{DRW}\cmdtxtoparname{NRW}\cmdtxtoparname{URW}\cmdtxtoparname{ARW}
2436 \cmdtxtoparname{DSW}\cmdtxtoparname{NSW}\cmdtxtoparname{USW}\cmdtxtoparname{ASW}
2437 \cmdtxtoparname{DMW}\cmdtxtoparname{NMW}\cmdtxtoparname{UMW}\cmdtxtoparname{AMW}

\GFG ...
... 2438 \cmdtxtoparname{GFG}
2439
2440 \cmdtxtoparname{PD}
2441 \cmdtxtoparname{PN}
2442
2443 \cmdtxtoparname{LD}
2444 \cmdtxtoparname{LN}
2445 %%** Syntax *****%

\AutName ...
... 2446 \newcommand{\autname}{A}
2447 \usrmthlatupp{Aut}{Name}{name}[\autname]
2448 \newcommand{\autset}{Aut}
2449 \cmdmthset{Aut}[\autset]

```



```

\WAutSet ...
 2450 \newcommand{\wautset}{\WAut}
 2451 \cmdmthset{WAut}[\wautset]

\SymSet ...
 ... 2452 \newcommand{\symsym}{\sigma}
 2453 \newcommand{\symset}{\Sigma}
 2454 \cmdmthsettext{Sym}[\symset][\symsym]

\SttSet ...
 ... 2455 \providecommand{\sttsym}{q}
 2456 \providecommand{\sttset}{Q}
 2457 \cmdmthsettext{Stt}[\sttset][\sttsym]
 2458 \cmdmthset{IStt}[\sttset_{I}]
 2459 \cmdmthsymelm{istt}[\sttsym_{I}]
 2460 \cmdmthset{FStt}[\sttset_{F}]
 2461 \cmdmthsymelm{fstt}[\sttsym_{F}]

\trnFun ...
\trnRel 2462 \providecommand{\trnsym}{\delta}
 2463 \cmdmthfun{trn}[\trnsym]
 2464 \cmdmthrel{trn}[\trnsym]

 2465 %** Semantics *****%

\WrdSet ...
 ... 2466 \newcommand{\wrdsym}{w}
 2467 \newcommand{\wrdset}{Wr}
 2468 \cmdmthsettext{Wrd}[\wrdset][\wrdsym]

\Lang ...
 2469 \usrmth{Lang}{\argfun}[L]

 2470 %** Finite Tree Automata *****%

\DTA ...
 ... 2471 \cmdtxtoparname{DTA}\cmdtxtoparname{NTA}\cmdtxtoparname{UTA}\cmdtxtoparname{ATA}
 2472
 2473 \cmdtxtoparname{DFT}\cmdtxtoparname{NFT}\cmdtxtoparname{UFT}\cmdtxtoparname{AFT}
 2474 \cmdtxtoparname{DWT}\cmdtxtoparname{NWT}\cmdtxtoparname{UWT}\cmdtxtoparname{AWT}
 2475 \cmdtxtoparname{DBT}\cmdtxtoparname{NBT}\cmdtxtoparname{UBT}\cmdtxtoparname{ABT}
 2476 \cmdtxtoparname{DCT}\cmdtxtoparname{NCT}\cmdtxtoparname{UCT}\cmdtxtoparname{ACT}
 2477 \cmdtxtoparname{DPT}\cmdtxtoparname{NPT}\cmdtxtoparname{UPT}\cmdtxtoparname{APT}
 2478 \cmdtxtoparname{DRT}\cmdtxtoparname{NRT}\cmdtxtoparname{URT}\cmdtxtoparname{ART}
 2479 \cmdtxtoparname{DST}\cmdtxtoparname{NST}\cmdtxtoparname{UST}\cmdtxtoparname{AST}
 2480 \cmdtxtoparname{DMT}\cmdtxtoparname{NMT}\cmdtxtoparname{UMT}\cmdtxtoparname{AMT}
 2481 %** Syntax *****%

\TAutSet ...
 2482 \newcommand{\tautset}{\TAut}
 2483 \cmdmthset{TAut}[\tautset]

\DirSet ...
 ... 2484 \newcommand{\dirsym}{d}
 2485 \newcommand{\dirset}{\Lambda}
 2486 \cmdmthsettext{Dir}[\dirset][\dirsym]

 2487 %** Semantics *****%

\TreeSet ...
 ... 2488 \newcommand{\treesym}{T}
 2489 \newcommand{\treeset}{Tr}
 2490 \cmdmthsettext{Tree}[\treeset][\treesym]

```

```

\wot ...

2491 \usrmth{wot}{-}{argfun}

2492 \fi
2493 %*****%

2494 %*****%
2495 %** Format Tricks *****%
2496 %*****%
2497 \iffirm@

2498 \RequirePackage{multicol}

... ..

2499 %...

2500 \fi
2501 %*****%

2502 %*****%
2503 %** Figure Tricks *****%
2504 %*****%
2505 \iffig@

2506 \RequirePackage{tikz}
2507 \usetikzlibrary{calc,arrows,shapes,patterns,graphs,matrix}

2508 \tikzstyle{every node} =
2509 [draw = none, fill = none, black, thin]
2510 \tikzstyle{every edge} +=
2511 [black, thick]

2512 \tikzstyle{noall} =
2513 [draw = none, fill = none]
2514 \tikzstyle{nodraw} =
2515 [draw = none, fill = white]
2516 \tikzstyle{nofill} =
2517 [draw = black, fill = none]

2518 \ifwrpfig@
2519 % Wrapfig Package
2520 \RequirePackage{wrapfig}
2521 \fi

2522 \fi
2523 %*****%

2524 %*****%
2525 %** Table Tricks *****%
2526 %*****%
2527 \iftab@

... ..

2528 %...

2529 \fi
2530 %*****%

2531 %*****%
2532 %** Algorithm Tricks *****%
2533 %*****%
2534 \ifalg@

2535 \RequirePackage[ruled,vlined]{algorithm2e}
2536 \DontPrintSemicolon
2537 \SetInd{0.25em}{0.5em}
2538 \setlength{\algomargin}{1.25em}

\Signature ...

2539 \SetKw{Signature}{signature}

```

```

\Macro ...
\Function 2540 \SetKwFor{Macro}{macro}{}{}
\Procedure 2541 \SetKwFor{Function}{function}{}{}
 2542 \SetKwFor{Procedure}{procedure}{}{}

\Let ...
 2543 \SetKwFor{Let}{let}{in}{}

\True ...
\False 2544 \SetKw{True}{true}
 2545 \SetKw{False}{false}

\From ...
\To 2546 \SetKw{From}{from}
\DownTo 2547 \SetKw{To}{to}
 2548 \SetKw{DownTo}{downto}

\GoTo ...
\Break 2549 \SetKw{GoTo}{goto}
\Continue 2550 \SetKw{Break}{break}
 2551 \SetKw{Continue}{continue}

\Guess ...
\ExsGuess 2552 \SetKw{Guess}{guess}
\AllGuess 2553 \SetKw{ExsGuess}{$\exists$-guess}
 2554 \SetKw{AllGuess}{$\forall$-guess}

\MIf ...
\MElseIf 2555 \SetKwIF{MIf}{MElseIf}{MElse}{\#if}{\#then}{\#else \#if}{\#else}{\#endif}
\MElse
\nlr ...
 2556 \newrobustcmd{\nlr}[1]
 2557 {\addtocounter{AlgoLine}{1}%
 2558 \nlset{\arabic{AlgoLine}-\addtocounter{AlgoLine}{#1}\arabic{AlgoLine}}}

 2559 \fi
 2560 %%*****%

 2561 \endinput
 2562 \</package>

```

## 2 Change History

|       |                                                                  |       |                                                                 |
|-------|------------------------------------------------------------------|-------|-----------------------------------------------------------------|
| v0.0  | General: First public release . . . . . 1                        | v0.24 | General: Correction of fragile macros . . . . . 1               |
| v0.1  | General: Algorithm tricks . . . . . 1                            | v0.25 | General: Few additions and corrections . . . . . 1              |
| v0.10 | General: Small refinements . . . . . 1                           | v0.26 | General: Few additions . . . . . 1                              |
| v0.11 | General: Few additions and corrections . . . . . 1               | v0.27 | General: Small addition to ‘Algorithm tricks’ . . . . . 1       |
| v0.12 | General: New starred variants . . . . . 1                        | v0.28 | General: Few additions . . . . . 1                              |
| v0.13 | General: Further starred variants . . . . . 1                    | v0.29 | General: Correction of fragile macros . . . . . 1               |
| v0.14 | General: Few additions and corrections . . . . . 1               | v0.3  | General: Few problems solved . . . . . 1                        |
| v0.15 | General: Refactoring of dtx sources . . . . . 1                  | v0.30 | General: Improvements and new command variants . . . . . 1      |
| v0.16 | General: Small refinements and few additions . . . . . 1         | v0.31 | General: Small improvements . . . . . 1                         |
| v0.17 | General: Few additions . . . . . 1                               | v0.32 | General: First integration of the knowledge package . . . . . 1 |
| v0.18 | General: Few new starred variants . . . . . 1                    | v0.4  | General: Refactoring, corrections, and extensions . . . . . 1   |
| v0.19 | General: Additional starred variants . . . . . 1                 | v0.5  | General: Figure tricks . . . . . 1                              |
| v0.2  | General: Changes in ‘Auxiliary tricks’ . . . . . 1               | v0.6  | General: Small refinements . . . . . 1                          |
| v0.20 | General: New binary operators . . . . . 1                        | v0.7  | General: Refinements, corrections, and extensions . . . . . 1   |
| v0.21 | General: Refactoring of function macros . . . . . 1              | v0.8  | General: Few refinements and corrections . . . . . 1            |
| v0.22 | General: Few additions . . . . . 1                               | v0.9  | General: Small addition to ‘Algorithm tricks’ . . . . . 1       |
| v0.23 | General: New ‘Graphs’ section and small improvements . . . . . 1 |       |                                                                 |

## 3 Index

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| <a href="#">\NQPSpace(E/H/C)</a> . . . . . | <a href="#">1412</a>                                                                                                                                              | <a href="#">\possym</a> . . . . .         |                                                                                                                                                                                                  | <a href="#">\rbrace</a> . . . . .            | <a href="#">1197, 1199, 1215, 1217</a>                                                                                                                                                                                                                                                                                            |
| <a href="#">\NQPTIME(E/H/C)</a> . . . . .  | <a href="#">1411</a>                                                                                                                                              |                                           | 1479, 1481, 1482, 1483,<br>1485, 1487, 2070, 2072,<br>2073, 2074, 2076, 2078                                                                                                                     | <a href="#">\rceil</a> . . . . .             | <a href="#">1341, 1343</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\NSpace(E/H/C)</a> . . . . .   | <a href="#">1406</a>                                                                                                                                              | <a href="#">\pow</a> . . . . .            | <a href="#">1224</a>                                                                                                                                                                             | <a href="#">\rch</a> . . . . .               | <a href="#">1526</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\NTime(E/H/C)</a> . . . . .    | <a href="#">1405</a>                                                                                                                                              | <a href="#">\pre</a> . . . . .            | <a href="#">1446</a>                                                                                                                                                                             | <a href="#">\relax</a> . . . . .             | <a href="#">139</a>                                                                                                                                                                                                                                                                                                               |
| <a href="#">\num</a> . . . . .             | <a href="#">1309</a>                                                                                                                                              | <a href="#">\prfFun</a> . . . . .         | <a href="#">1519</a>                                                                                                                                                                             | <a href="#">\relset</a> . . . . .            | <a href="#">1687, 1688</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\numcc</a> . . . . .           | <a href="#">1309</a>                                                                                                                                              | <a href="#">\PrfSet</a> . . . . .         | <a href="#">1519, 2095</a>                                                                                                                                                                       | <a href="#">\RelSig</a> . . . . .            | <a href="#">1684</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\numco</a> . . . . .           | <a href="#">1309</a>                                                                                                                                              | <a href="#">\prfset</a> . . . . .         | <a href="#">1520, 1521, 2096, 2097</a>                                                                                                                                                           | <a href="#">\relsig</a> . . . . .            | <a href="#">1684, 1685</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\numoc</a> . . . . .           | <a href="#">1309</a>                                                                                                                                              | <a href="#">\prfsym</a> . . . . .         | <a href="#">1519, 1521, 2095, 2097</a>                                                                                                                                                           | <a href="#">\RelStr</a> . . . . .            | <a href="#">1698</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\numoo</a> . . . . .           | <a href="#">1309</a>                                                                                                                                              | <a href="#">\PrimaFacie</a> . . . . .     | <a href="#">1033</a>                                                                                                                                                                             | <a href="#">\relstr</a> . . . . .            | <a href="#">1698, 1699</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\nxt</a> . . . . .             | <a href="#">2418</a>                                                                                                                                              | <a href="#">\primafacie</a> . . . . .     | <a href="#">1016</a>                                                                                                                                                                             | <a href="#">\relsym</a> . . . . .            | <a href="#">1686, 1688</a>                                                                                                                                                                                                                                                                                                        |
| <b>O</b>                                   |                                                                                                                                                                   | <a href="#">\prj</a> . . . . .            | <a href="#">1244</a>                                                                                                                                                                             | <a href="#">\RequirePackage</a> . . . . .    | <a href="#">3,</a><br><a href="#">5, 6, 7, 240, 241, 242,</a><br><a href="#">248, 253, 254, 259, 264,</a><br><a href="#">279, 285, 287, 303, 309,</a><br><a href="#">311, 2498, 2506, 2520, 2535</a>                                                                                                                              |
| <a href="#">\obsFun</a> . . . . .          | <a href="#">1496</a>                                                                                                                                              | <a href="#">\Procedure</a> . . . . .      | <a href="#">2540</a>                                                                                                                                                                             | <a href="#">\resp</a> . . . . .              | <a href="#">1050</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\ObsSet</a> . . . . .          | <a href="#">1496</a>                                                                                                                                              | <a href="#">\ProcessOptions</a> . . . . . | <a href="#">139</a>                                                                                                                                                                              | <a href="#">\rfloor</a> . . . . .            | <a href="#">1335, 1337</a>                                                                                                                                                                                                                                                                                                        |
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| <a href="#">\OddSym</a> . . . . .          | <a href="#">1549</a>                                                                                                                                              | <a href="#">\prtFun</a> . . . . .         | <a href="#">1553</a>                                                                                                                                                                             | <a href="#">\Rightarrow</a> . . . . .        | <a href="#">1076, 1078</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\oddsym</a> . . . . .          | <a href="#">1551, 1552</a>                                                                                                                                        | <a href="#">\PrtSet</a> . . . . .         | <a href="#">1553</a>                                                                                                                                                                             | <a href="#">\rightarrow</a> . . . . .        | <a href="#">1071, 1609</a>                                                                                                                                                                                                                                                                                                        |
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| <a href="#">\OGPL</a> . . . . .            | <a href="#">2142, 2146, 2148, 2151</a>                                                                                                                            | <a href="#">\prtsym</a> . . . . .         | <a href="#">1553, 1555</a>                                                                                                                                                                       | <a href="#">\rmfamily</a> . . . . .          | <a href="#">406, 737</a>                                                                                                                                                                                                                                                                                                          |
| <a href="#">\OGSL</a> . . . . .            | <a href="#">2276, 2280, 2282, 2285</a>                                                                                                                            | <a href="#">\psn</a> . . . . .            | <a href="#">2416</a>                                                                                                                                                                             | <a href="#">\rng</a> . . . . .               | <a href="#">1238</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\Omega</a> . . . . .           | <a href="#">1266</a>                                                                                                                                              | <a href="#">\PSpace(E/H/C)</a> . . . . .  | <a href="#">1410</a>                                                                                                                                                                             | <a href="#">\Role</a> . . . . .              | <a href="#">1043</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\omega</a> . . . . .           | <a href="#">1265</a>                                                                                                                                              | <a href="#">\pthFun</a> . . . . .         | <a href="#">1442</a>                                                                                                                                                                             | <a href="#">\role</a> . . . . .              | <a href="#">1041</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\Omicron</a> . . . . .         | <a href="#">1270</a>                                                                                                                                              | <a href="#">\PthSet</a> . . . . .         | <a href="#">1442, 1886</a>                                                                                                                                                                       | <a href="#">\rrbracket</a> . . . . .         | <a href="#">1100, 1102</a>                                                                                                                                                                                                                                                                                                        |
| <a href="#">\omicron</a> . . . . .         | <a href="#">147, 1269</a>                                                                                                                                         | <a href="#">\pthset</a> . . . . .         | <a href="#">1443, 1444, 1887, 1888</a>                                                                                                                                                           | <a href="#">\rst</a> . . . . .               | <a href="#">1246</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\oplus</a> . . . . .           | <a href="#">1568</a>                                                                                                                                              | <a href="#">\pthsym</a> . . . . .         | <a href="#">1442, 1444, 1886, 1888</a>                                                                                                                                                           | <a href="#">\rVert</a> . . . . .             | <a href="#">1329, 1331</a>                                                                                                                                                                                                                                                                                                        |
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| <a href="#">\oppsym</a> . . . . .          | <a href="#">1475, 1476</a>                                                                                                                                        | <a href="#">\PTL</a> . . . . .            | <a href="#">1925</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\Opr</a> . . . . .             | <a href="#">1859</a>                                                                                                                                              | <a href="#">\pto</a> . . . . .            | <a href="#">1254</a>                                                                                                                                                                             | <b>S</b>                                     |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\out</a> . . . . .             | <a href="#">1524</a>                                                                                                                                              | <b>Q</b>                                  |                                                                                                                                                                                                  | <a href="#">\S</a> . . . . .                 | <a href="#">1955</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\overline</a> . . . . .        | <a href="#">1109, 1123</a>                                                                                                                                        | <a href="#">\QAE</a> . . . . .            | <a href="#">1625</a>                                                                                                                                                                             | <a href="#">\SATG</a> . . . . .              | <a href="#">1455</a>                                                                                                                                                                                                                                                                                                              |
| <b>P</b>                                   |                                                                                                                                                                   | <a href="#">\QAFMC</a> . . . . .          | <a href="#">1915</a>                                                                                                                                                                             | <a href="#">\SaveDoubleAcute</a> . . . . .   | <a href="#">1954</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\P</a> . . . . .               | <a href="#">1953</a>                                                                                                                                              | <a href="#">\QATL</a> . . . . .           | <a href="#">2019</a>                                                                                                                                                                             | <a href="#">\SavePilcrow</a> . . . . .       | <a href="#">1953</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\PackageWarning</a> . . . . .  | <a href="#">135</a>                                                                                                                                               | <a href="#">\QATLP</a> . . . . .          | <a href="#">2034</a>                                                                                                                                                                             | <a href="#">\SaveSectionSymbol</a> . . . . . | <a href="#">1955</a>                                                                                                                                                                                                                                                                                                              |
| <a href="#">\PDL</a> . . . . .             | <a href="#">1959</a>                                                                                                                                              | <a href="#">\QATLS</a> . . . . .          | <a href="#">2049</a>                                                                                                                                                                             | <a href="#">\scshape</a> . . . . .           | <a href="#">724, 737</a>                                                                                                                                                                                                                                                                                                          |
| <a href="#">\Percontra</a> . . . . .       | <a href="#">1032</a>                                                                                                                                              | <a href="#">\QBF</a> . . . . .            | <a href="#">1584</a>                                                                                                                                                                             | <a href="#">\seqofcmd</a> . . . . .          | <a href="#">197, 216, 220</a>                                                                                                                                                                                                                                                                                                     |
| <a href="#">\percontra</a> . . . . .       | <a href="#">1015</a>                                                                                                                                              | <a href="#">\QCTL</a> . . . . .           | <a href="#">1970</a>                                                                                                                                                                             | <a href="#">\seqofgrklet</a> . . . . .       | <a href="#">223, 683</a>                                                                                                                                                                                                                                                                                                          |
| <a href="#">\PH</a> . . . . .              | <a href="#">1416</a>                                                                                                                                              | <a href="#">\QCTL</a> . . . . .           | <a href="#">1970</a>                                                                                                                                                                             | <a href="#">\seqofgrklow</a> . . . . .       | <a href="#">215, 224, 227, 679, 799, 813</a>                                                                                                                                                                                                                                                                                      |
| <a href="#">\Pi</a> . . . . .              | <a href="#">1423, 1424, 1727</a>                                                                                                                                  | <a href="#">\QCTLP</a> . . . . .          | <a href="#">1985</a>                                                                                                                                                                             | <a href="#">\seqofgrkupp</a> . . . . .       | <a href="#">219, 224, 229, 681</a>                                                                                                                                                                                                                                                                                                |
| <a href="#">\pi</a> . . . . .              | <a href="#">1442, 1508, 1886, 2106</a>                                                                                                                            | <a href="#">\QCTL</a> . . . . .           | <a href="#">1985</a>                                                                                                                                                                             | <a href="#">\seqoflatlet</a> . . . . .       | <a href="#">212, 677, 799, 813</a>                                                                                                                                                                                                                                                                                                |
| <a href="#">\PL</a> . . . . .              | <a href="#">1773, 2125</a>                                                                                                                                        | <a href="#">\QCTLS</a> . . . . .          | <a href="#">2000</a>                                                                                                                                                                             | <a href="#">\seqoflatlow</a> . . . . .       | <a href="#">208, 213, 227, 673</a>                                                                                                                                                                                                                                                                                                |
| <a href="#">\playFun</a> . . . . .         | <a href="#">1508</a>                                                                                                                                              | <a href="#">\QEA</a> . . . . .            | <a href="#">1625</a>                                                                                                                                                                             | <a href="#">\seqoflatupp</a> . . . . .       | <a href="#">210,</a><br><a href="#">213, 229, 675, 757, 771, 785</a>                                                                                                                                                                                                                                                              |
| <a href="#">\PlaySet</a> . . . . .         | <a href="#">1508, 2106</a>                                                                                                                                        | <a href="#">\QLTL</a> . . . . .           | <a href="#">1940</a>                                                                                                                                                                             | <a href="#">\seqoflet</a> . . . . .          | <a href="#">230, 689, 827, 847, 861,</a><br><a href="#">875, 889, 935, 949, 964, 978</a>                                                                                                                                                                                                                                          |
| <a href="#">\playset</a> . . . . .         | <a href="#">1509, 1510, 2107, 2108</a>                                                                                                                            | <a href="#">\QMC</a> . . . . .            | <a href="#">1899</a>                                                                                                                                                                             | <a href="#">\seqoflow</a> . . . . .          | <a href="#">226, 231, 685</a>                                                                                                                                                                                                                                                                                                     |
| <a href="#">\playsym</a> . . . . .         | <a href="#">1508, 1510, 2106, 2108</a>                                                                                                                            | <a href="#">\QML</a> . . . . .            | <a href="#">1852</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\PlnSet</a> . . . . .          | <a href="#">2110</a>                                                                                                                                              | <a href="#">\Qnt</a> . . . . .            | <a href="#">1622</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\plnset</a> . . . . .          | <a href="#">2111, 2112, 2113, 2115</a>                                                                                                                            | <a href="#">\QntSet</a> . . . . .         | <a href="#">1627</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
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| <a href="#">\plrFun</a> . . . . .          | <a href="#">1488, 1489</a>                                                                                                                                        | <a href="#">\QPSpace(E/H/C)</a> . . . . . | <a href="#">1412</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\PlrSym</a> . . . . .          | <a href="#">1473, 1484, 1485,</a><br><a href="#">1503, 1504, 1515, 1516,</a><br><a href="#">2075, 2076, 2101, 2102,</a><br><a href="#">2113, 2114, 2120, 2121</a> | <a href="#">\QPTIME(E/H/C)</a> . . . . .  | <a href="#">1411</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">\plrsym</a> . . . . .          | <a href="#">1473, 1474</a>                                                                                                                                        | <a href="#">\QPTL</a> . . . . .           | <a href="#">1929</a>                                                                                                                                                                             |                                              |                                                                                                                                                                                                                                                                                                                                   |
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